

Additional Results and Reproduction Details

Pin Ni, Francesca Medda, Ramin Okhrati

This companion accompanies *When a High Score Is an Illusion: Certifying Genuine versus Repackaged Forecasting Skill*. The main paper contains every core theoretical statement and proof. This document preserves the expanded comparison protocols, complete model tables, additional bounds and provenance qualifications. Results from distinct fitting rules remain separate. The code release provides aggregate evidence and executable public or synthetic analyses at <https://github.com/MaxwellNi/certifying-forecasting-skill>.

1 Corrections and interpretation of the stored forecasts

The historical row labelled TimesFM-2.5 was produced by a forecasting network whose adapter did not load the intended checkpoint. The table now calls it an *untrained forecasting network*; its stored predictions and numerical audit results are unchanged. A separate rerun used the appropriate pretrained architecture, loading checks and contexts ending at each forecast origin. These later aggregates describe different forecasts. Historical availability of the pretrained weights has not been established.

For 29 of the 30 historical forecasters, excess-return training targets used inconsistent risk-free-rate units. Subtracting a common monthly shift preserves within-month ranks mathematically for frozen outcomes. It does not preserve the model that would result from correct retraining; finite-precision rounding can also change ties. The restricted case remains an audit of stored outputs.

The retail price-and-calendar example has standard-fold statistic 7.68, ordered-middle statistic -0.82 , and ordered-all statistic -1.21 . Croston SBA has -8.59 , 6.11 , and -1.04 , respectively. Seven models are retained in each of the first two configurations, but their identities differ. Middle-fold evaluation changes support as well as nuisance fitting, so this comparison alone does not identify the causal share of feedback bias. An additional matched-middle comparison fixes evaluation rows: the price model changes from 6.41 to -0.82 and Croston SBA from -19.43 to 6.11 when only fitting direction changes; family retentions change from five to seven. This establishes sensitivity to the nuisance-fitting rule on that sample, not a population calibration result. The ratings example in the main paper supplies the accuracy-versus-association illustration without relying on stability of the retail model's retention.

A previously reported solver discrepancy bound of 3.2×10^{-10} has a located aggregate verification record: its largest recorded discrepancy is below that bound. It concerns replication of frozen audit calculations, not correctness of forecast training or statistical calibration.

2 Detailed matched-panel changes

Same forecasts, different audit decisions Complementary-fold nominal screens retain identical sets under both exponents and splines. Directional sensitivities use $\beta = 2$ and the original scale; two-way-cluster checks use standard fits. Neither jointly validates both changes.

On all folds, complementary fitting retains 10/11, 7/10, 6/10 and 0/10 models in electricity, retail, ratings and portfolios, with 1, 2, 4 and 1 abstentions.

Figure 1 uses identical middle folds. Retail price/calendar changes from 6.41 to -0.82 and Croston SBA from -19.43 to 6.11 ; three models enter and one leaves the retained set. Their mean products change from 0.01014 to -0.00158 and from -0.01988 to 0.00552 , respectively, on the same 68,000 rows over 17 weeks. At each resolution, let r_X, r_Y be complementary residuals and Δ_X, Δ_Y the changes in fitted means. Then

$$\Delta \bar{R} = -\overline{r_X \Delta_Y} - \overline{r_Y \Delta_X} + \overline{\Delta_X \Delta_Y}. \quad (1)$$

The same extrapolation weights give $-0.00382, -0.00449, -0.00342$ for price/calendar and $0.01242, 0.00970, 0.00327$ for Croston. Each term reinforces the reversal. These are fit changes, not causal feedback identified by Proposition 4.

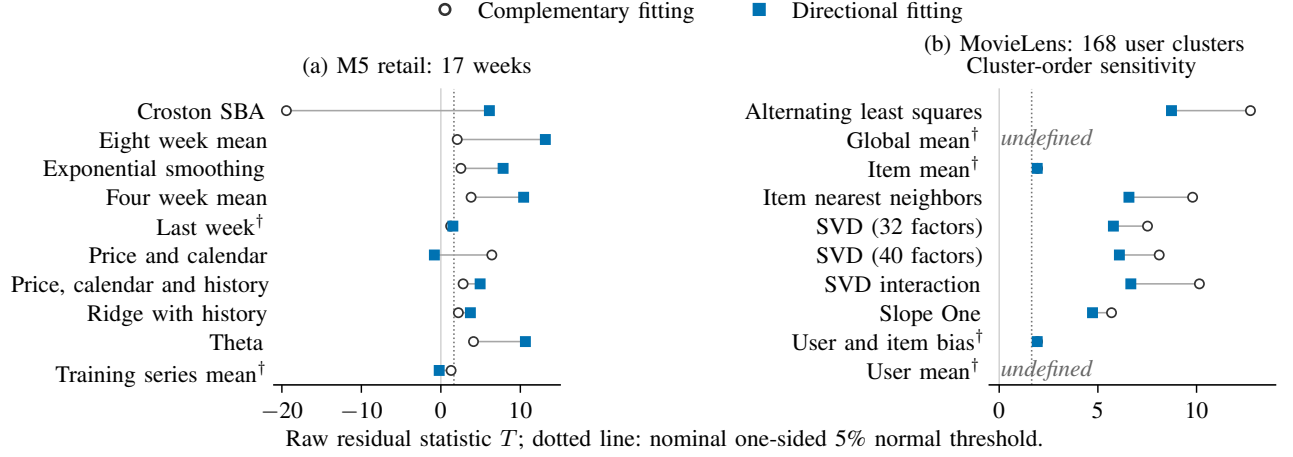


Figure 1: Fitting sensitivity on identical middle-fold observations, ranks, bins and uncertainty rules ($\beta = 2$). Lines join raw statistics; the dotted line is the nominal 5% normal threshold before BY. Daggers mark full-cohort abstentions; two pairs are undefined. Ratings use ordered user clusters. These are screening statistics, not calibrated discoveries.

Accuracy and residual association The ratings item-mean baseline has mean absolute error 0.693 and abstains; alternating least squares has error 0.719 and is retained under all four fold/support choices. Thus association and prediction accuracy differ. The electricity copy has raw $T = 28.72$ with ten-bin adjustment and 4.94 with splines before guarding. The same copy has $T = -24.85$ and -5.36 under the two exponents. These degenerate scores are abstained and do not calibrate nondegenerate nulls.

Rental-demand comparison The earlier Seoul study [1] freezes 16 baseline/candidate pairs and a 2,016-hour confirmation window. Ridge augmentation gives 22,431 raw-count MSE gain and equals ungated selection. The seasonal candidate passes the copy guard, but its eight-week t_7 value 0.00668 becomes 0.1094 under BY, causing 299,790 MSE opportunity loss. For Ridge, gain over convex blending is 5,624; the weekly interval [833, 10,414] excludes zero whereas the fortnight interval $[-898, 12,145]$ includes it. These temporal approximations and all strategies remain in the companion. Augmentation gain and gating gain are separate quantities.

Does the adjustment capture the baseline? An omitted entity–baseline interaction gives 300/300 nominal positives under every additive adjustment. In a near-copy rank null with 25 periods, 80 entities and noise 0.01, degrees-of-freedom correction and t_{24} give 982/1,000 rejections for 32 bins, 46 for linear-plus-spline adjustment and 48 for splines. The companion retains the full grid and scalar comparisons. Their different targets and sizes preclude a general calibration or power ranking.

Why the sign of learning error matters For two equally likely categories with midrank means (0.35, 0.65) in both channels, fits $f = (0.5, 0.8)$ and $g = (0.2, 0.5)$ give $b_H = -0.0225$. At $\vartheta = 0.02$, absolute subtraction approaches -0.025 as validation and evaluation grow, whereas signed subtraction approaches 0.02. Rank construction fixes the marginal mean $1/2$, not these category means.

Resolution-screen implementation The score is $s_i = \sum_q w_q R_i^{(q)}$, where

$$\sum_q w_q = 1, \quad \sum_q w_q q^{-\beta} = 0.$$

A bounded-weight expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ leaves $O(q_{\min}^{-2\beta})$ bias, which must be negligible relative to the standard error along with fitting error. Smooth uniform controls give $\beta = 2$; Gaussian quantile bins have order $1/(q \log q)$ instead. Both exponents and splines are prespecified screens, not selected by significance.

Scoring cost Single-model audit time rises from 0.16 seconds near 31,000 rows to 14.2 seconds at one million. Sorting matches exact pair calculations, including ties. The release supplies runtime settings for this $O(N \log N)$ implementation; these timings exclude source forecast training.

Rows outside the directional centering conditions Table 1 reports the implemented fitting support. Fallback uses a pooled complementary fit when one directional training side is empty. Unseen-entity rows lack an entity fit on the relevant training side. These counts are per channel and must not be added without their rowwise intersection. Every model in a task shares the displayed counts. Ratings order user clusters rather than time, and even its middle-fold rows include unseen entities. Thus the directional centering theorem is not a joint calibration argument for these rank-adjusted screens.

Table 1: Directional fitting support by task. Shares are percentages of evaluation rows.

Task	Folds	Rows	Forecast fallback	Outcome fallback	Forecast unseen	Outcome unseen
Electricity	all	34,992	20.03	19.89	0.00	0.00
Electricity	middle	21,024	0.00	0.00	0.00	0.00
Portfolios	all	27,272	20.55	19.40	0.00	0.08
Portfolios	middle	16,377	0.00	0.00	0.00	0.07
Ratings	all	12,088	17.04	23.30	22.81	19.32
Ratings	middle	7,212	0.00	0.00	33.33	29.26
Retail	all	112,000	21.43	17.86	0.00	0.00
Retail	middle	68,000	0.00	0.00	0.00	0.00

The source count table and its deterministic summary script are in `results/directional_comparison/`; all underlying model results remain unchanged.

3 Complete-family audit procedure

The main paper presents the target-and-information table. The executable procedure retains the following steps for every fixed family:

1. Fix forecasts, outcomes, target, sampling units, fitting rule, analysis scale and level.
2. For every model, record outcome-free redundancy and evaluation support.
3. Select the applicable information row and construct its residual score and preselected bound or statistic.
4. Record the null and assumptions for theorem-supported inputs; label other inputs as nominal screens.
5. For p -value rules, set $p^* = 1$ for recorded abstention, otherwise $p^* = p$.
6. Apply BY to the complete p -value family, including abstentions. For fixed-level bounds on K candidates, use level α/K for Bonferroni familywise control instead.
7. Return abstain, retain or not retained, with target, validity status and all raw values.

4 Panel scale and redundancy details

4.1 Adjustment and the uncertainty scale

At resolution q , cross-fitted regressions on the represented controls leave residuals $e_X^{(q)}, e_Y^{(q)}$. The panel implementation uses additive entity effects and q baseline bins. Period enters ranking and clustering. This additive approximation need not equal the full joint conditional mean. Let $R_i^{(q)} = e_{X,i}^{(q)} e_{Y,i}^{(q)}$. The i.i.d. generalized covariance measure uses $T(q) = \sqrt{n} \bar{R}^{(q)} / \widehat{\text{sd}}(R^{(q)})$. For cluster inference and products s_i , let $L_g = \sum_{i \in g} s_i$, $n_g = |g|$, and

$$\widehat{\text{se}}_{\text{cl}}^2 = n^{-2} \sum_{g=1}^M (L_g - n_g \bar{s})^2, \quad T_{\text{cl}} = \bar{s} / \widehat{\text{se}}_{\text{cl}}. \quad (2)$$

Ordered screens use Newey–West variance with a stated Bartlett lag [2]; lag zero gives Eq. (2). Cluster GCM inference needs a cluster CLT, consistent variance and negligible fitted-product error [3, 4]; fixed lags and additive fits alone do not establish them.

1. Check observed redundancy	2. Measure residual association	3. Evaluate the complete family
Fix the family, baseline and peer design. Record copies, shared or reversed orderings, and constant ranks.	Specify fold direction and controls. Compare adjustment and references; check bias and the uncertainty scale.	Retain every raw statistic. Set guarded decision values to one. Apply BY to all models.
ABSTAIN: observed redundancy or a degenerate statistic.	RETAIN: retained by the screen, with validity conditional on its inputs.	NOT RETAINED: below the screen; predictive value remains undecided.

Figure 2: **From forecasts to an interpretable audit.** Keep the baseline, target and sampling design fixed. Report observed redundancy, residual association and complete-family decisions. FDR control requires valid input p -values.

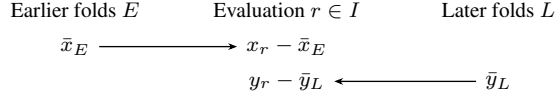


Figure 3: **Directional fitting for an audit.** Fit the score mean earlier and the outcome mean later, then multiply the two residuals. Evaluation rows require both sides. Forecast training is unchanged.

4.2 Redundancy and resolution sensitivity

For unit-variance X , let $\rho^2 = R^2(X | Z)$. The Cauchy-Schwarz inequality gives

$$|\theta_\infty| \leq \sqrt{1 - \rho^2} \|Y - \mathbb{E}[Y | Z]\|_2. \quad (3)$$

High overlap limits covariance, not studentized power. Our outcome-free abstention rule records exact copies, matching or reversed cluster orders with ties, and constant ranks. Setting $p^* = 1$ on abstention preserves valid marginal inputs because $p^* \geq p$; it does not calibrate invalid ones.

Resolution sensitivity On a fixed grid $\mathcal{Q}$, let $s_i = \sum_q w_q R_i^{(q)}$ with $\sum_q w_q = 1$ and $\sum_q w_q q^{-\beta} = 0$. An expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ then leaves $O(q_{\min}^{-2\beta})$ bias for bounded weights. Inference requires this remainder and fitting error below the standard error. Smooth uniform controls give $\beta = 2$; Gaussian quantile bins instead have order $1/(q \log q)$. The companion proves both cases. We compare both fixed exponents with splines as screens, without choosing by significance.

Connection to classical rank correlation For continuous margins, $\Gamma = (3\tau - \rho_S)/12$, where τ and ρ_S are population Kendall and Spearman correlations. This is Hoeffding's classical decomposition [5, 6]. The conditional audit separates that term from nuisance error and quantifies its amplification relative to sampling uncertainty.

For continuous margins, the population rank identity follows from $\mathbb{E}[a_{12}b_{12}] = (1 + \tau)/4$ and $\mathbb{E}[F_V(V)F_W(W)] = (3 + \rho_S)/12$.

5 Expanded figures and case protocols

A numerical audit For electricity DLinear with complementary all-fold fitting and exponent two, the mean product is 0.016425, its 14-lag HAC standard error is 0.001874 and $T = 8.763$ over 729 days. The nominal $p = 9.52 \times 10^{-19}$ enters the complete 11-model family; DLinear is retained. These are the recorded screen's calculations, rather than a calibration check.

For ordered periods, let $A_t = L_t - n_t \bar{s}$ on the complete calendar of M_{cal} periods, with $A_t = 0$ for a missing period. At specified lag ℓ , the heteroskedasticity and autocorrelation consistent (HAC) estimator of Newey and West [2] is

$$\begin{aligned} \widehat{\text{se}}_{\text{HAC}}^2 &= n^{-2} \left[\sum_{t=1}^{M_{\text{cal}}} A_t^2 \right. \\ &\quad \left. + 2 \sum_{h=1}^{\min(\ell, M_{\text{cal}}-1)} \left(1 - \frac{h}{\ell+1}\right) \sum_{t=h+1}^{M_{\text{cal}}} A_t A_{t-h} \right]. \end{aligned} \quad (4)$$

Lag zero recovers Eq. (2), without a degrees-of-freedom multiplier. Fixed lags define the reported screens.

Table 2: Complete-family screens at 0.05, holding original copy flags fixed. All/middle denotes evaluation folds; complementary/directional denotes nuisance fitting. MovieLens orders user clusters, not time. Counts do not establish calibrated discoveries.

Task	Models	Complementary		Directional		Abstain
		All	Middle	All	Middle	
Electricity	11	10	10	10	10	1
M5	10	7	5	6	7	2
MovieLens-25M	10	6	6	6	6	4
OSAP forecasts	10	0	0	0	0	1

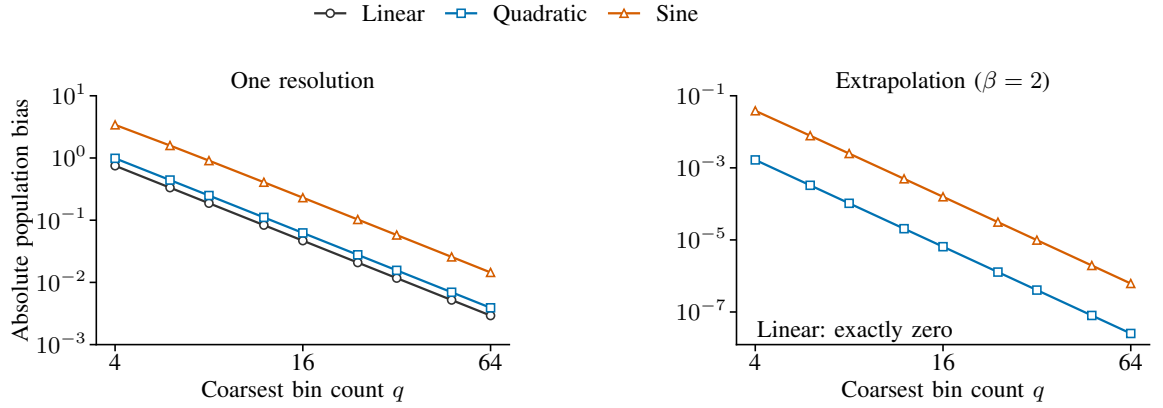


Figure 4: **Exact population approximation bias.** For $X = f(U) + \epsilon$, $Y = f(U) + \eta$ and uniform U , use $f(U) = 12U$, $12U^2$, or $6 \sin(2\pi U)$. The right panel combines bins $q\{1, 2, 4\}$ with weights $(-1/6, 1/2, 2/3)$. The linear remainder is zero; the other remainders persist. These are oracle identities, not fitted simulation means.

Separate financial targets A two-sided factor-alpha audit using Fama and French factors and momentum [7, 8], with 12 HAC lags, retains 122 of 212 published signals. This tests another target and is not a fifth independent dataset. In the restricted forecast family, the single standard-fold nominal positive changes from $T = 2.45$ (BY-adjusted $p = 0.85$) to 1.29 with ordered middle folds; no model is retained. Among sixteen strongly negative standard statistics, median absolute magnitude falls from 3.34 to 1.08. These comparisons diagnose nuisance sensitivity rather than establish absence of predictive information.

Separate simulations on two observed incidence patterns use entity-and-period clustering. Standard all-fold fitting has largest absolute mean 3.092, versus 0.251 for directional middle-fold fitting on the larger pattern. This contrast also changes evaluation support. At lag-one outcome correlation 0.9, another stress test gives an ordered-versus-independent-fit mean contrast up to 6.47 standard units, outside the innovation assumption.

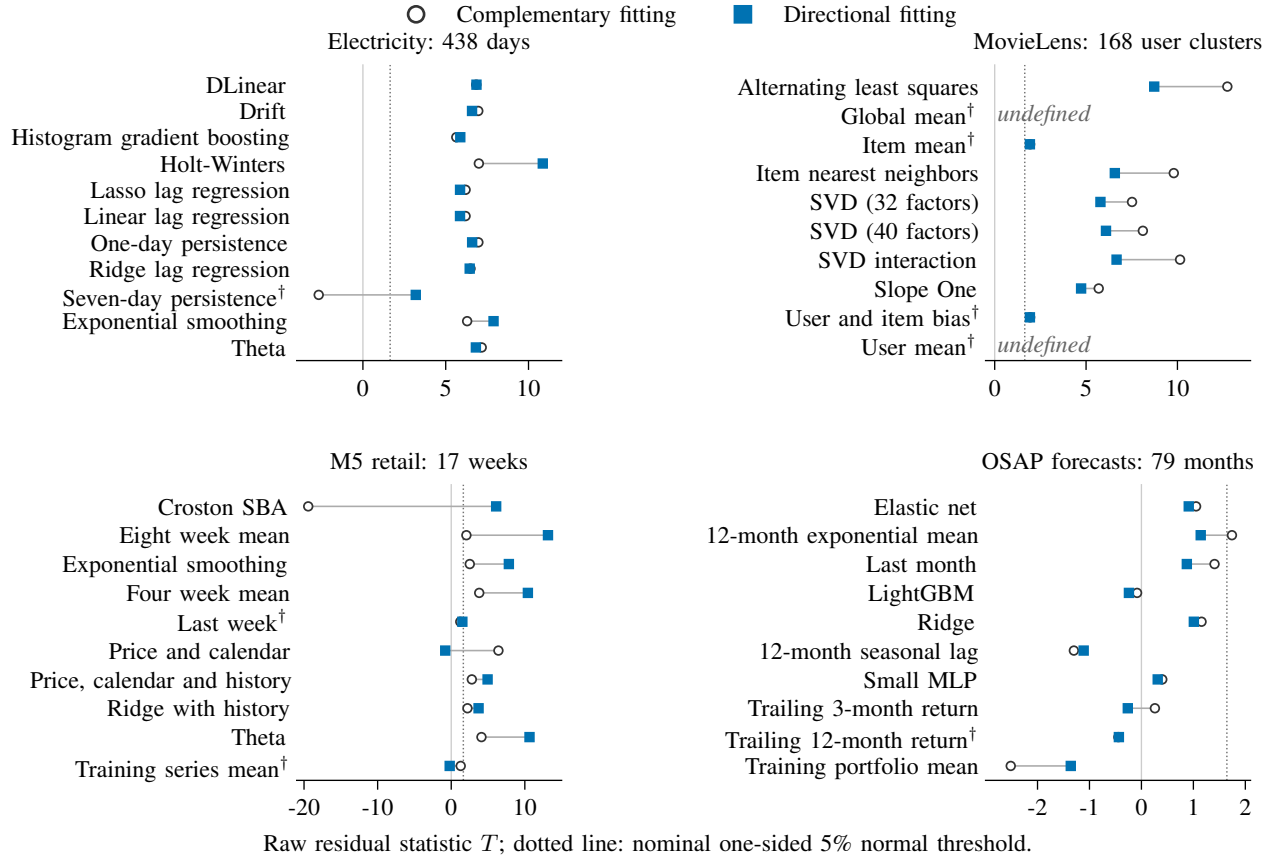


Figure 5: **Same forecasts and observations, different nuisance training.** All 41 models use the middle three folds, identical ranks and bins, $\beta = 2$ and the original uncertainty rule. Lines join raw statistics; the dotted line is the nominal 5% normal threshold. Daggers mark fixed full-cohort abstentions; two pairs are undefined. MovieLens orders user clusters. These are sensitivity comparisons, not calibrated discoveries.

Table 3: Scalar comparison, $n = 400$, 300 paired draws per cell. The first four rates test independent nulls; power is the minimum over four covariance alternatives. Zero covariance with dependence is an alternative for KCI, a null for the covariance tests.

Test	false-positive rate				min.	zero
	linear	smooth	quad.	t_5	power	cov.
GCM (8 bins)	0.220	0.070	0.987	0.090	0.980	0.060
GCM (32 bins)	0.140	0.097	0.590	0.120	0.957	0.097
GCM (splines)	0.100	0.070	0.113	0.093	0.947	0.050
Extrap. $\beta = 1$	0.120	0.093	0.210	0.107	0.943	0.093
Extrap. $\beta = 2$	0.140	0.090	0.613	0.107	0.967	0.077
KCI (gamma)	0.070	0.047	0.047	0.070	0.727	1.000

6 Expanded standard-fit comparison protocols

The following comparisons use the standard nuisance folds unless explicitly stated otherwise. They complement the distinct reference-sampling and feedback experiments in the main paper. Complete cell-level results, including unsuccessful settings, remain in the release.

6.1 Matched comparison on independent observations

All methods receive the same $n = 400$ observations per replication. Canonical KCI [9] uses `causal-learn` 0.1.4.8, Gaussian kernels, empirical bandwidths, regularization 0.001, and a gamma null approximation. Independent matrix calculations verify its statistic and gamma moments. GCM variants share two balanced folds and i.i.d. studentization. Adjustment uses eight or 32 bins, cubic splines with eight training quantile knots and ridge penalty 0.001, or the ladder $\{8, 12, 16, 24, 32\}$. The original comparison fixes $\beta = 1$; a separately recorded extension evaluates $\beta = 2$ on identical draws. Bins, knots and coefficients use training folds; KCI uses its native full sample.

Let Z, U, e_X, e_Y be independent standard normal variables. The common baseline is $g(Z) = 0.8Z, 0.8 \tanh(1.5Z)$, or $0.8(Z^2 - 1)$. The heavy-tail case uses the smooth baseline and independent variance-one t_5 errors. Under the null, $X = g(Z) + e_X$ and $Y = g(Z) + e_Y$. The positive-covariance alternative adds $0.5U$ to both, giving conditional covariance 0.25. A further alternative uses the smooth baseline and $Y = g(Z) + (e_X^2 - 1)/\sqrt{2} + e_Y$. Its conditional covariance with X is zero, but it is conditionally dependent.

Each of the nine cells uses 300 paired replications. Raw variables retain independent sampling. U-centered partial distance correlation [10], verified in the package, targets a different null.

6.2 Calibration and power

Table 3 and Fig. 6 show incomplete bias correction. At $\beta = 1$, extrapolation’s quadratic-null rate is $63/300 = 0.210$, with Wilson 95% interval $[0.168, 0.260]$. At $\beta = 2$, it rises to $184/300 = 0.613$, with interval $[0.557, 0.667]$. All four $\beta = 2$ null intervals lie above 0.05. Gaussian quantile bins need not satisfy the uniform-control expansion proved below; neither exponent is justified here by Proposition 1. KCI’s four rates range from 0.047 to 0.070; their individual intervals contain 0.05. These are pointwise intervals.

At $\beta = 1$, minimum power is 0.943, versus 0.727 for KCI. Unequal null rejection rates prevent ranking these raw powers at matched size. Splines improve both worst null rate and minimum power over $\beta = 1$ extrapolation. Under dependence with zero covariance, KCI rejects 300/300 and $\beta = 1$ extrapolation 28/300: the two tests address different nulls.

Comparison with supplied null simulations A retrospective secondary analysis assigns the first 150 null draws per regime to calibration and the remaining 150 to validation. Each method’s Monte Carlo value is $p_{MC} = (1 + \sum_{b=1}^{150} \mathbf{1}\{p_b \leq p_0\})/151$, where p_b and p_0 are the calibration and evaluation raw p -values. Exchangeability of the 151 null ranks, including ties, gives marginal rejection at most $7/151$ at level 0.05; rates conditional on a realized bank can differ. On the 300 independent positive draws, $\beta = 1$ detects 0.910/0.937 in the linear/smooth cases versus KCI’s 0.727/0.753; under quadratic controls, splines give 0.933 versus 0.853 for $\beta = 1$. The package retains all 48 cells and 60 paired contrasts, using exact two-sided binomial tests of discordant decisions and Holm adjustment. The supplied null law is extra information, not a real-panel calibration.

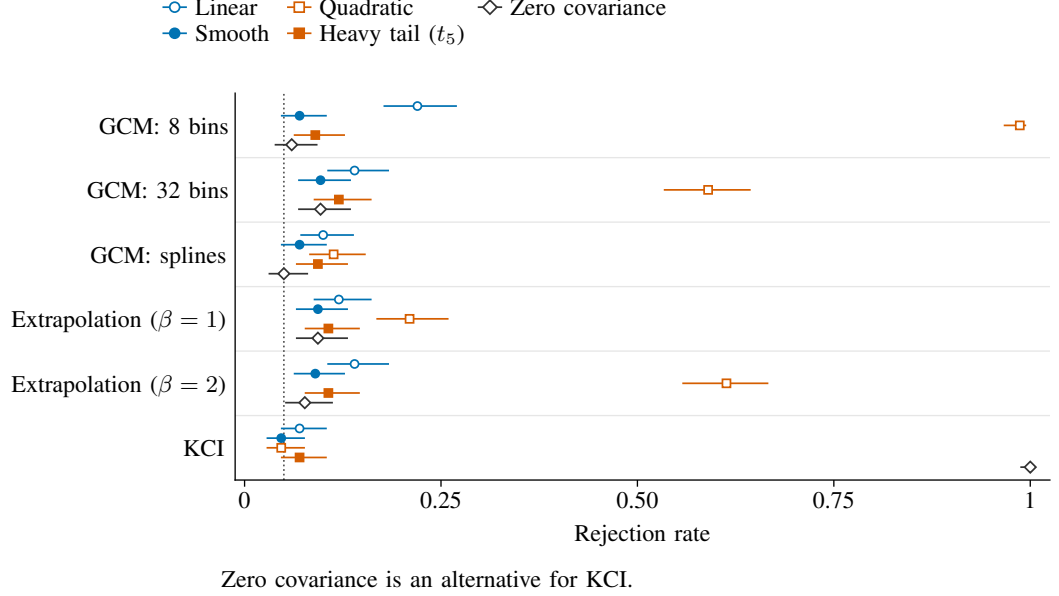


Figure 6: **Matched scalar comparison.** Each method shows the four independent nulls and the zero-covariance dependence case from Table 3; intervals are pointwise 95% Wilson intervals and the line marks 0.05. The dependence case is an alternative for KCI. All methods use the same data within a replication. The extrapolated statistic retains approximation error, particularly under the quadratic baseline.

Table 4: Rough controls, 300 paired draws. K, S, O : kink, step, oscillation; 0: independent null; +: positive covariance. K_d has dependence with zero covariance, a null for GCM and alternative for KCI. Entries are rejection proportions.

Method	K_0	S_0	O_0	K_+	S_+	O_+	K_d
GCM, 8 bins	.180	.230	.810	.993	.993	1.000	.110
GCM, 32 bins	.120	.140	.217	.963	.983	.993	.097
Spline GCM	.070	.177	.157	.957	.983	.967	.080
Extrap., $\beta = 1$	.093	.120	.147	.947	.967	.980	.090
KCI	.057	.097	.037	.733	.863	.670	1.000

6.3 Rough controls and short panels

A further recorded scalar comparison fixes three controls before the run: $g(Z) = 0.8(|Z| - \sqrt{2/\pi})$, $0.8 \text{sign}(Z)$, and $0.8\sqrt{2} \sin(3Z)$. These give a kink, a jump and rapid oscillation in 2,100 datasets (Table 4). Extrapolation’s worst null rate is 0.147, versus 0.177 for splines and 0.097 for KCI. No regime meets the joint criterion of lower null rejection than splines, disjoint pointwise 95% intervals and power within 0.05.

The panel study uses 80 entities, $M = 25, 50, 100$ independent periods, five whole-period folds, and 300 paired draws per data-generating setting. The independent-score null is $Y = a_i + 0.6B + 0.3 \sin(B) + \eta$, with independent normal B, X, η and fixed $a_i \in [-0.6, 0.6]$. A high-overlap null sets $X = B + 0.1\epsilon$, $Y = B + 0.1\eta$, where each period’s B is a permuted fixed 80-point grid plus a common shift. Given a focal baseline rank, the peer grid is fixed; exchangeable independent error arrays retain the conditional rank null. This argument does not cover arbitrary random peer grids. The positive control uses $Y = B + 0.1(0.6\epsilon + 0.8\eta)$.

At $M = 100$, $\beta = 2$ reduces the high-overlap shift: mean T changes from -13.52 to -0.30 in the released summaries; Table 5 gives rejection counts. Its independent-null rate is $11/300$, with a 95% Wilson interval $[0.021, 0.064]$ containing 0.05. At $M = 25$, however, both $\beta = 2$ null intervals lie above 0.05. At $M = 25$, for $\beta = 1$ in the independent-null case, the reported standard error is only 0.856 times the empirical standard deviation of the estimated mean. Both approximation bias and too few clusters can distort the test.

A local spline GCM uses all 2,700 simulated panels with identical ranks, controls, folds and cluster scales. Training quantiles define eight cubic-spline knots; duplicate knots are removed, and ridge penalty 0.001 applies only to spline coefficients, with unpenalized entity effects. At $M = 25$, its two null rates are $24/300$ and $30/300$, whose intervals also lie entirely above 0.05. At $M = 100$, both are $10/300$. All three methods reject every positive control, so these alternatives cannot establish a

Table 5: Rejections among 300 paired panels with identical ranks, entities, folds and cluster scale. All 33 settings and their intervals are provided in the package.

Condition	M	$\beta = 1$	$\beta = 2$	Spline
Independent null	25	25	25	24
Independent null	50	18	17	17
Independent null	100	12	11	10
High overlap null	25	0	23	30
High overlap null	50	0	15	17
High overlap null	100	0	7	10
Positive control	25	300	300	300
Positive control	50	300	300	300
Positive control	100	300	300	300

power advantage.

On all 41 public forecasts, splines use identical predictions, samples, folds, HAC lags and complete-family BY. All three guarded retained sets agree; Table 2 gives the extrapolation counts and the package gives the spline counts. Before guarding, splines reject the electricity copy ($T = 4.94$) and ratings copy ($T = 3.20$).

All 33 settings have 300 replications: 27 combinations of three conditions, three cluster counts and $\beta \in \{0.5, 1, 2\}$ on the primary ladder, plus six settings at $M = 100, \beta = 1$ using $\{4, 6, 8, 12, 16\}$ or $\{16, 24, 32, 48, 64\}$. At $M = 100$ and $\beta = 1$, the coarse ladder rejects 0/300 positive-control panels, whereas the primary $\{8, 12, 16, 24, 32\}$ and fine ladders each reject 300/300.

The smooth scalar study uses Fig. 4’s functions, known uniform bins and bivariate standard normal errors with correlation zero or 0.2. Three sample sizes and 300 paired draws give 5,400 datasets. Twofold histogram fits on $q_{\min}\{1, 2, 4\}$ compare coarsest, finest, and both extrapolated statistics: 21,600 calculations. With $q_{\min} = \lceil n^{1/5} \rceil$ and $\beta = 2$, linear and quadratic null rates range from 0.030 to 0.073. For the sine example they are 0.157, 0.087, and 0.120 at $n = 2,000, 8,000$, and $32,000$. Population calculations confirm a substantial remainder despite the correct leading exponent.

6.4 Public-data forecast protocol

Four public datasets support five tasks: electricity and retail demand, ratings, return forecasts and factor alpha. Model families and temporal rules precede fitting. Panel audits use the five-resolution ladder with both exponents; the presentation order follows the panel checks. Ranks retain ties. Five contiguous blocks exclude whole evaluation clusters from each joint fit of additive entity and baseline-bin effects. Undefined ranks, negligible residuals or standard error at most 10^{-10} in standardized-rank product units yield $p^* = 1$, as do guarded copies. The floor prevents studentizing solver noise. Every model remains in BY; ten-bin adjustment and interleaved folds are sensitivities.

UCI Electricity [11] uses the first 48 meters in the verified provider archive, excluding incomplete boundary days. Eleven models forecast next-day mean demand with annual cutoffs in 2013 and 2014: 34,992 predictions per model over 729 days. The baseline is seven-day seasonal persistence; ranks are daily, with 14 HAC lags. Two-meter perturbation tests find no future-target influence on earlier predictions for all eleven models.

For M5 [12], a training-only rule selects 4,000 product-store series on a complete weekly grid. There are 992,000 training and 112,000 evaluation series-weeks, the latter spanning 28 complete weeks. The final partial week is excluded. Price features use previous weeks; normalization and imputation use training data. Histories update after each week, so the task is rolling one-week prediction. Ten models share last week’s sales as baseline, within-week ranks, and two weekly HAC lags. The small cluster count limits inference.

MovieLens-25M [13] uses ratings before January 1, 2017 for training and 2017 for evaluation. Training activity alone selects 6,000 users; 1,368,078 training ratings support ten models with 12,088 held-out ratings from 279 users. Items are entities, users are clusters, and the baseline is the training item mean. Within-user ranks assign zero to 18 singleton users. There is no HAC ordering across users; their clustering does not account automatically for shared-item dependence.

For Open Source Asset Pricing (OSAP) [14], all 173,302 long-short returns match the authors’ public 2025.10 release. Ten models use annual training cutoffs with monthly history updates. The target is the next calendar month’s return; the baseline compounds the preceding 12 months through the forecast origin. Evaluation covers 27,272 portfolio-months, 209 portfolios and 132 origins from January 2013 to December 2023, with targets February 2013 to January 2024. Three published portfolios have no admissible forecast rows. Ranks and folds are monthly, with 12 HAC lags. This evaluates published return series.

A separate OSAP task tests each long-short series’ intercept against zero in a Fama and French five-factor regression with

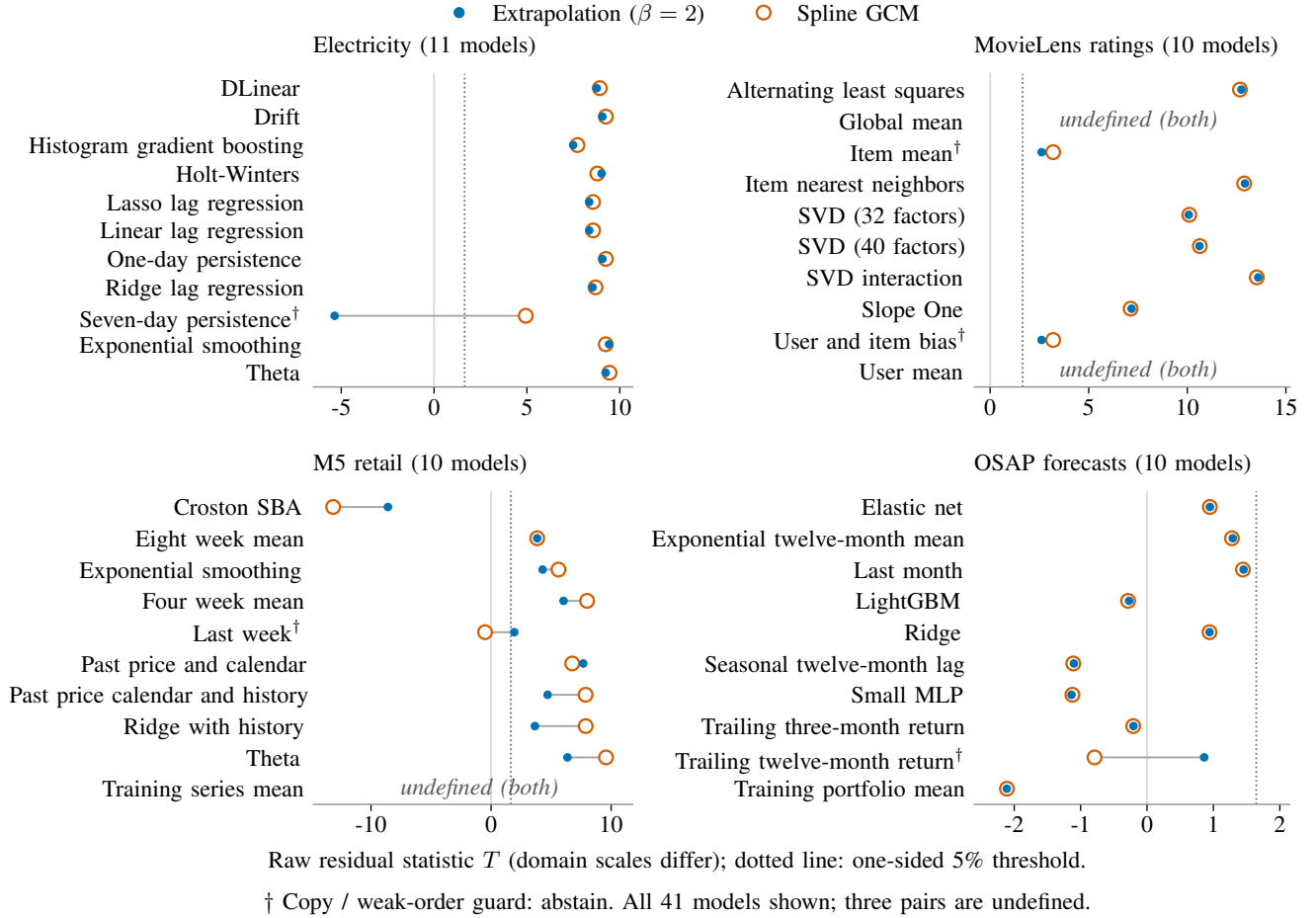


Figure 7: **Same forecasts, different adjustments.** All 41 public models, with 38 finite pairs and three undefined pairs. Points show raw extrapolated ($\beta = 2$) and spline statistics before the copy rule; the vertical line is the one-sided 5% normal threshold. Daggers identify finite copy or weak-order guards. Larger statistics do not establish better calibration or power. Full scores, MAEs and family-adjusted values accompany the reproduction package.

momentum [7,8]. Calendar-based Newey and West covariance [2] uses 12 monthly lags and two-sided normal-tail p -values. BY uses all 212 signals; six- and 24-lag estimates are sensitivities. This shares a data source with the forecast task, so it is not a fifth independent dataset.

6.5 Retrospective audit of a restricted case

We also audit stored predictions from thirty models in a UK insider-trading study. The data remain private. Each retained record has an observed next-calendar-month outcome and all twelve preceding calendar-month returns, excluding the forecast month, for the momentum control. This corrects incomplete histories and next-observed-month substitutions; the rule was fixed before recomputation after seeing historical results. Supports differ: 2,613 to 31,045 firm-months over 72 or 132 months. Available stored predictions are averaged within firm-month; source counts and target rounding differ, so this is not a common-sample model ranking.

This audit retains five whole-month folds, additive firm and momentum-bin effects, the same ladder, $\beta = 1$ primary and $\beta = 2$ sensitivity. Equation (2) uses months without HAC or a copy guard, differing from the public protocol. Historical full-panel imputation makes this a retrospective output audit. Only model-level aggregates are distributed.

A separate common-support reconstruction retrains the adversarial residual-ranking MLP, Ridge and histogram gradient boosting with training-only preprocessing and matured labels. Across all 55 fits, 18,036 observations and 132 common months, mean monthly Spearman correlations are MLP 0.00275, Ridge 0.05667, boosting 0.03794 and one-month momentum 0.02314. The MLP has no descriptive advantage; no significance comparison is claimed. The package records every seed aggregate and temporal rule.

7 Estimated means, family inference and near-copy controls

7.1 Estimated conditional means for tied ranks

The main paper’s Corollary 1 gives a studentized limit with independent estimated means. The public generator `learned_reference_study.py` evaluates a finite-control instance. Let $Z \sim \text{Bernoulli}(1/2)$ and, conditionally independently, $V, W \mid Z \sim \text{Bernoulli}(0.2 + 0.6Z)$. Marginal midranks take values $1/4$ and $3/4$; their conditional means are $0.35 + 0.3Z$. The target is zero, $\Gamma = 9/400$, and the first-projection variance is $1/625$. These are exact values, checked by finite-state enumeration.

Training uses its own leave-one-out empirical midranks and two category means. It is independent of evaluation, but its fitted functions are shared across evaluation groups. Training sizes are 32, 512 and 8,192; exact means provide a diagnostic fourth arm. Group sizes are 16 and 64, with 25, 100 and 400 independent resampled groups. The 1,000 replications share evaluation primitives across arms and nested group sizes. There are 48 cells and 48,000 records. Population fitting-error products and L^2 bounds are evaluated from the known synthetic law; these diagnostics are not given to the test and are not confidence bounds available under an unknown law.

Across the six sampling configurations, distinct-reference rates are 0.041–0.055 with exact means, 0.136–0.527 with 32 training observations, 0.046–0.079 with 512, and 0.041–0.057 with 8,192. For 512 training observations, two of six pointwise Wilson intervals exclude 0.05, both with 400 groups. For 8,192, none does. The experiment supports the role of training accuracy, not a finite-sample universal threshold or a power comparison. The main captioned peer study retains the separate fixed-peer counterexamples.

7.2 Complete raw-score model families

The generator `independent_entity_family.py` tests the complete raw-score branch rather than only a marginal 0.05 rejection frequency. It uses $G = 25, 100, 400$ independent Gaussian entity trajectories, 20 observations, five contiguous folds and the same 12 middle observations per entity. Forecasts use 0.2 times the previous five outcome innovations plus unit-variance forecast noise. Forecast noises have pairwise correlation 0.5 across models, and all models share outcomes. Entity intercepts and all noises are independent across entities.

Families have 10 or 11 models. All-null families and families with three injected alternatives of coefficients 0.03, 0.06 or 0.10 share the same primitives. Each of 1,000 evaluation families runs the fitting, equal-entity studentization, numerical abstention and complete-family BY steps. Current-outcome injections are positive controls, not prospective forecasts. The full 216 cells retain failed methods and all hypotheses.

A separate 5,000-draw null bank is independent of evaluation. For each method and entity count, plus-one upper-tail ranks give Monte Carlo values with ties included. The same bank calibrates exchangeable model marginals. The bank’s minimum value $1/5001$ resolves the first BY thresholds for $K = 10, 11$. It supplies the known simulated null law, which is extra information. Reported evaluation-family Monte Carlo intervals are conditional on the realized bank; they do not integrate uncertainty over new banks. Pointwise Hoeffding intervals additionally bound finite-evaluation FDR uncertainty; a zero observed error rate does not imply zero risk. Normal, t_{G-1} and supplied-null references are all retained. The raw entity scale already uses sample standard deviation with divisor $G - 1$; applying a second degrees-of-freedom multiplier would be incorrect.

For $G = 400, K = 11$, the all-null FDR for complementary, gapped and directional fitting is 0, 0.928 and 0.011 with the normal reference. It is 0, 0.926 and 0.011 with t_{G-1} , and 0.008, 0.015 and 0.009 with independent supplied-null calibration. Across all six size/family combinations, directional normal-reference all-null FDR is 0.011–0.020. These simulations support the stated independent-entity branch, not a ranked, binned or serially dependent panel guarantee.

At coefficient 0.06, $G = 400, K = 11$, calibrated complementary and directional powers are 0.8443 and 0.6660. The paired difference, directional minus complementary, is -0.1783 , with pointwise Monte Carlo interval $[-0.1976, -0.1591]$. Both use independent null calibration at the same nominal level; their realized FDRs are not forced to be equal. Directional centering therefore provides an analytic alternative to an external null bank, not power dominance after supplying that additional information.

7.3 Near-copy representation and independent evaluation

The generator `near_copy_adjustment.py` uses 25 independent periods, 80 entities, a permuted fixed baseline grid and independent homogeneous Gaussian noises. Conditional on a focal baseline rank, the competing baseline multiset is fixed. The two full noise arrays are independent at zero correlation, so the two conditional rank distributions factorize. This supplies an exact nondegenerate rank null for positive score noise. Exact copies form a separate zero-product-variance boundary.

Comparators share five complementary folds, ranks, observations and the same adjustment information: 32 bins, both extrapolation exponents, eight-knot cubic splines, and splines with an explicitly unpenalized linear term. Exact conditional

rank means are a diagnostic arm. The historical period-cluster statistic uses divisor M ; the small-sample variant uses $M - 1$ and t_{24} . Unlike the equal-entity study above, this changes the scale as well as the reference distribution.

For noise 0.01, 32 bins reject 984/1,000 evaluation null panels. The spline and linear-plus-spline normal counts are 57 and 54; their small-sample counts are 48 and 46. Exponent one rejects zero but has mean statistic -13.49 , so zero rejection alone is not evidence of centering. The new experiment is distinct from the historical fixed restricted-pattern stress, which gave 93/100 nominal positives for bins and 11/100 for splines and did not regenerate the full rank-null design.

Independent 1,000-draw calibration and 1,000-draw evaluation banks also compare weak injected correlations. The original 0, 0.3, 0.6 grid saturated its positive controls. The recorded exploratory extension adds 0.03, 0.06, 0.10 on the same primitives and retains every original cell. At noise 0.01 and correlation 0.06, supplied-null-calibrated powers are 0.125, 0.069, 0.157, 0.335, 0.344 and 0.338 for bins, exponent one, exponent two, splines, linear-plus-splines and exact means, respectively. The small linear-term difference is local and exploratory; it establishes no broad superiority over splines. All counts, paired differences and calibration inputs accompany the executable scripts.

7.4 A matched classical temporal moment

A further comparison uses the forward orthogonal deviation of Arellano and Bover [15], with a recursively centered predictable instrument. It compares a moment construction, not the complete panel GMM estimator. The method was specified after the preceding results were reviewed, with its formula and primary comparisons fixed before this additional run.

For the same 20 observed periods and evaluation rows $t = 5, \dots, 16$, set $c_t = \sqrt{(20 - t)/(21 - t)}$ and define

$$S_{gk}^{\text{FOD}} = \frac{\sum_{t=5}^{16} c_t (X_{gkt} - \bar{X}_{gk, < t})(Y_{gt} - \bar{Y}_{g, > t})}{\sum_{t=5}^{16} c_t}.$$

The instrument subtracts all preceding observed predictions, and the outcome subtracts all subsequent observed outcomes. Under the stated innovation conditions, the instrument is orthogonal to current and future outcome innovations, so this score has exact null mean zero. Independent entities permit the same sample-standard-deviation studentization. The positive design-only denominator leaves the statistic unchanged from averaging the weighted products, while making the mean equal to the constant innovation-injection coefficient τ . The blocked directional score has the same mean τ . Under time-varying effects, the two target averages have different weights; their equivalence is specific to the constant-effect comparison. More explicitly, replacing the constant injection by $X_t = X_t^0 + \tau_t \varepsilon_t$ with independent unit-variance innovations and fixed τ_t gives expected directional mean $12^{-1} \sum_{t=5}^{16} \tau_t$ and classical mean $\sum_{t=5}^{16} c_t \tau_t / \sum_{t=5}^{16} c_t$. The earlier-centered forecast contains no current or future innovation, except for its declared current injection, so expansion leaves only τ_t in each product. Unequal c_t establish the different weights algebraically; no efficiency advantage is inferred from this identity.

All original primitive draws, evaluation rows, entity weights and full $K = 10, 11$ families are reused. The new method receives its own 5,000 calibration statistics on the same independent null draws, followed by 1,000 evaluation families. Every original method's means, standard errors, statistics, probabilities and BY decisions replay identically. The four-method study has 288 cells, including all three reference laws, three entity counts and all four signal levels. It extends the preceding study; repeated cells are not independent confirmation.

At $G = 400, K = 11$, the classical moment's all-null FDR is 0.012 with the normal reference, 0.011 with t_{399} , and 0.015 with supplied-null calibration. The corresponding directional values are 0.011, 0.011 and 0.009. For $\tau = 0.06$, normal-reference power is 0.7517 versus 0.6577; calibrated power is 0.7523 versus 0.6660. The calibrated paired power difference is 0.0863, with pointwise Monte Carlo interval $[0.0721, 0.1006]$ and Bonferroni-adjusted normal Monte Carlo interval $[0.0637, 0.1089]$ over the 27 primary contrasts. The latter is an approximate simultaneous Monte Carlo interval, conditional on the calibration bank, not a finite-sample statistical guarantee. Pointwise Hoeffding FDR bounds accompany the normal and Wilson summaries.

The classical moment has higher power than the blocked directional score in this design. Calibrated complementary fitting is still more powerful (0.8443), but requires the externally supplied null law to correct its biased reference. These comparisons separate exact centering from statistical efficiency and do not establish power dominance for the proposed blocked construction. The generator, input compatibility checks, all saved arrays, paired intervals and literal-product tests are released.

8 Standard cluster GCM conditions

The next result applies the standard GCM argument at the cluster scale [3, 4].

Theorem 1 (Conditional validity). *Under (A1) to (A3) below, with nuisance approximation bias negligible relative to the standard error, the statistic T_{cl} in Eq. (2) converges in distribution to $\mathcal{N}(0, 1)$ under conditional independence. The one-sided level- α test is consistent for fixed alternatives with $\theta_\infty > 0$ and a standard error tending to zero.*

The proof also covers a pointwise zero target with oracle mean $o(a_n)$. For a fixed negative target, a consistent mean and vanishing standard error give rejection probability tending to zero. Neither statement is uniform over drifting nulls. Training excludes complete evaluation clusters.

Assumptions For the oracle product combination, let s_i^* use population nuisance regressions, $L_g^* = \sum_{i \in g} s_i^*$, and $\mathbb{E}[L_g^*] = n_g \mu_n$, treating cluster sizes as fixed. Define $v_n^2 = \sum_g \text{Var}(L_g^*)$ and $a_n = v_n/n$.

(A1) *Cluster sampling and central limit theorem.* The oracle cluster sums are independent across clusters, $M \rightarrow \infty$, $\max_g n_g/n \rightarrow 0$, and $a_n \rightarrow 0$. For some $\nu \in (0, 2]$, $\sum_g \mathbb{E}|L_g^* - n_g \mu_n|^{2+\nu}/v_n^{2+\nu} \rightarrow 0$.

(A2) *Negligible nuisance remainder.* With $\Delta_g = L_g - L_g^*$, require $\sum_g \Delta_g = o_p(v_n)$ and $\sum_g \Delta_g^2 = o_p(v_n^2)$. In an i.i.d. setting, the usual product-rate condition $\|\hat{g}_X - g_X\|_2 \|\hat{g}_Y - g_Y\|_2 = o_p(n^{-1/2})$ controls the quadratic remainder, while weighted prediction-error and moment bounds control the linear terms. Growing clusters require those bounds at scale a_n .

(A3) *Non-degenerate product scale.* $0 < v_n^2 < \infty$ and the cluster-product moments in (A1) exist. In the i.i.d. specialization invoking $\sqrt{n}$ rates, additionally assume $v_n^2/n \rightarrow \sigma^2 \in (0, \infty)$.

Relation to the implementation Nuisances refer to the declared score. Cross-fitting and HAC do not alone prove (A2) and (A1).

Proof of Theorem 1. Expanding fitted products gives $R - R^* = -\Delta_X e_Y^* - e_X^* \Delta_Y + \Delta_X \Delta_Y$, where $\Delta = \hat{g} - g$. Assumption (A2) controls these terms at the cluster scale. By (A1), the centered oracle sum divided by v_n tends to $\mathcal{N}(0, 1)$ and its sum of squared centered cluster contributions, divided by v_n^2 , tends in probability to one. Empirical centering changes this ratio by $o_p(1)$ because $\bar{s} - \mu_n = O_p(a_n)$ and $\sum_g n_g^2/n^2 \leq \max_g n_g/n \rightarrow 0$. Both parts of (A2) and the Cauchy-Schwarz inequality then permit fitted products. Thus $\widehat{\text{se}}_{\text{cl}}/a_n \rightarrow_p 1$. A null mean $o(a_n)$ gives the normal limit; a positive limiting mean and vanishing scale give consistency.

9 Conditional resolution theory

The main paper records the fixed-grid score as a sensitivity analysis. This section retains the full conditional expansion and its counterexamples. Assumptions (A1)–(A3) are stated in the preceding section; (A4) is given below.

Additional assumption (A4) *Resolution bias expansion.* The population mean obeys $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ with a common c , a specified $\beta > 0$, and a remainder bound uniform over the ladder. For an expanding ladder take fixed m , $q_j/q_{\min} \rightarrow a_j$ with distinct positive a_j , so the intercept weights have bounded absolute sum. Inference also requires the stated bias-to-standard-error rate. This is an assumption on the residual rank covariance, not a consequence of assigning percentiles.

9.1 A conditional resolution correction

Assume the population numerator has the expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ with a specified exponent $\beta > 0$. On a resolution grid $\mathcal{Q} = \{q_1, \dots, q_m\}$ of distinct resolutions, let w be the intercept weights from regressing a vector of product means on $\{1, q^{-\beta}\}$. They satisfy $\sum_j w_j = 1$ and $\sum_j w_j q_j^{-\beta} = 0$. Define

$$s_i = \sum_{j=1}^m w_j R_i^{(q_j)}, \quad T_{\text{ex}} = \bar{s}/\widehat{\text{se}}(s). \quad (5)$$

Product means are combined before studentization; their cross-resolution covariance enters the variance. Use Eq. (2) for independent clusters or the declared HAC scale for a screen.

Theorem 2 (Leading-bias cancellation). *Under (A4), with bounded $\sum_j |w_j|$, the population mean of the oracle combination s^* is $\theta_\infty + b_n$, where $b_n = O(q_{\min}^{-2\beta})$. If (A1) to (A3) hold for this combination and $b_n/a_n \rightarrow 0$, where a_n is its oracle standard error, then $T_{\text{ex}} \Rightarrow \mathcal{N}(0, 1)$ under conditional independence. For an i.i.d. non-degenerate scale, a sufficient bias condition is $\sqrt{n} q_{\min}^{-2\beta} \rightarrow 0$. For fixed $\theta_\infty > 0$, $b_n \rightarrow 0$ and $a_n \rightarrow 0$ give consistency.*

The following calculations derive $\beta = 2$ for uniform scalar controls with C^3 means and the rate window $n^{1/8} \ll q_{\min} \leq q_{\max} \ll n^{1/2}$. Gaussian quantile bins instead have bias of order $1/(q \log q)$; bounded ranks can also violate the q^{-2} expansion (Proposition 2). Lemma 1 quantifies exponent error. We consequently treat fixed-grid extrapolation as a conditional correction and report both exponents alongside splines.

Proof of leading-bias cancellation. For extrapolation set $u_j = q_j^{-\beta}$ and $X_Q = [\mathbf{1} \ u]$. The intercept weights are

$$w = X_Q (X_Q^\top X_Q)^{-1} (1, 0)^\top.$$

They have $\mathbf{1}^\top w = 1$, $u^\top w = 0$ and bounded absolute sum. The combined mean is therefore $\theta_\infty + O(q_{\min}^{-2\beta})$. Applying the main paper's GCM limit to this combination gives the claimed limit when $b_n/a_n \rightarrow 0$. At a fixed ladder, the admissible null expansion $cq^{-\beta} + dq^{-2\beta}$ instead leaves $d \sum_j w_j q_j^{-2\beta}$, generally nonzero. Separate per-resolution validity does not supply the combination's assumptions.

Proposition 1 (A smooth scalar case). *Let $U \sim \text{Uniform}[0, 1]$, $X = f(U) + \epsilon$, $Y = g(U) + \eta$, with square-integrable conditionally mean-zero errors and $f, g \in C^3[0, 1]$. For the population means f_q, g_q on q equal-width bins,*

$$\begin{aligned} \mathbb{E}[(X - f_q(U))(Y - g_q(U))] \\ = \theta + \frac{1}{12q^2} \int_0^1 f'(u)g'(u) du + O(q^{-4}), \end{aligned} \quad (6)$$

where $\theta = \mathbb{E}[\epsilon\eta]$. Thus (A4) holds with $\beta = 2$ in this setting.

Proof. Write $h = 1/q$ and expand about each bin midpoint c . After subtracting bin means, the leading terms are $f'(c)V + \frac{1}{2}f''(c)(V^2 - \mathbb{E}V^2)$ and the corresponding expression for g , with a uniform $O(h^3)$ remainder and $V \sim \text{Uniform}[-h/2, h/2]$. Odd moments vanish, giving bin covariance $f'(c)g'(c)h^2/12 + O(h^4)$. The midpoint approximation to the integral of $f'g'$ has error $O(h^2)$. Averaging the bin covariances proves the expansion.

For the rate window, use i.i.d. observations, known uniform bins, deterministic proportional twofold splits and training-fold histogram means. Assume bounded conditional fourth moments and $0 < \text{Var}(\epsilon\eta) < \infty$. Histogram fitting error is $O_p(\sqrt{q_{\max}/n})$ in L^2 . The linear terms are conditionally centered; the quadratic mean is $O_p(q_{\max}/n)$. Thus $q_{\max} = o(\sqrt{n})$ makes fitting error negligible at root- n scale, while bounded fourth moments give variance consistency. Combining this with $\sqrt{n}q_{\min}^{-4} \rightarrow 0$ gives the stated nonempty window. The sine results show finite-sample error despite this rate window; this calculation does not establish the panel assumptions.

Gaussian quantile bins need not have the assumed leading power This failure already occurs with oracle adjustment, without fitting error. Let $Z \sim \mathcal{N}(0, 1)$, let $I_q(Z)$ index its $q \geq 2$ equal-probability bins, and write $z_j = \Phi^{-1}(j/q)$, with $\phi(\pm\infty) = 0$. For $X = aZ + \epsilon$, $Y = aZ + \eta$, with Z, ϵ, η mutually independent and both errors square-integrable and mean-zero, the binned residual covariance is $a^2 D_q$, where

$$D_q = 1 - q \sum_{j=1}^q [\phi(z_{j-1}) - \phi(z_j)]^2 \asymp \frac{1}{q \log q}. \quad (7)$$

Thus neither $c/q + O(q^{-2})$ nor $c/q^2 + O(q^{-4})$ holds for $a \neq 0$. This is an application of scalar quantization [16], not a new quantization principle. To prove the order, put $t = z_{q-1}$, $\lambda = \phi(t)/(1 - \Phi(t))$ and $v_t = 1 + t\lambda - \lambda^2$. The two tail bins contribute $2v_t/q$. Interior conditional variances are at most one quarter of bin width squared; the Cauchy-Schwarz inequality therefore gives

$$\frac{2v_t}{q} \leq D_q \leq \frac{2v_t}{q} + \frac{1}{4q^2} \int_{-t}^t \frac{dz}{\phi(z)}. \quad (8)$$

The normal-tail expansion gives $v_t \sim t^{-2}$, $t^2 \sim 2 \log q$, $\phi(t) \sim t/q$, and the integral is asymptotic to $2/(t\phi(t))$. Hence $1 \leq \liminf q \log(q) D_q \leq \limsup q \log(q) D_q \leq 5/4$.

Proposition 2 (A boundary obstruction for population ranks). *Let $Z, \epsilon_X, \epsilon_Y$ be independent standard normals and $X = \rho_X Z + \sqrt{1 - \rho_X^2} \epsilon_X$, $Y = \rho_Y Z + \sqrt{1 - \rho_Y^2} \epsilon_Y$, with $0 < \rho_X, \rho_Y < 1$. Set $V = \Phi(X)$, $W = \Phi(Y)$, $U = \Phi(Z)$, $a = \rho_X / \sqrt{2 - \rho_X^2}$ and $b = \rho_Y / \sqrt{2 - \rho_Y^2}$. For the oracle residual covariance B_q after q equal-width bins of U ,*

$$a^2 + b^2 \leq 1 \implies q^2 B_q \longrightarrow \infty. \quad (9)$$

Thus bounded population ranks do not guarantee a finite leading q^{-2} coefficient, even though $X \perp Y \mid Z$.

Proof. The conditional means are $f(u) = \Phi(a\Phi^{-1}(u))$ and $g(u) = \Phi(b\Phi^{-1}(u))$, obtained by integrating over an independent standard normal. Every bin covariance is nonnegative because both means increase. On any fixed $[\delta, 1 - \delta]$, their derivatives are bounded; the midpoint expansion, retaining only fully interior bins, gives $\liminf q^2 B_q \geq \frac{1}{12} \int_{\delta}^{1-\delta} f' g'$. With $z = \Phi^{-1}(u)$,

$$f'(u)g'(u) du = \frac{ab}{\sqrt{2\pi}} e^{-(a^2+b^2-1)z^2/2} dz. \quad (10)$$

This integral diverges as $\delta \downarrow 0$ when $a^2 + b^2 \leq 1$, proving the claim. For equal correlations the condition is $\rho_X = \rho_Y \leq \sqrt{2/3}$. A finite derivative integral in the other regime alone does not establish the $O(q^{-4})$ remainder in (A4).

Misspecified exponents For distinct $q_1, \dots, q_m > 0$, $m \geq 2$, let $w(\beta)$ be the ordinary least-squares (OLS) intercept weights on $[1, q^{-\beta}]$, and $q_+ = \max_j q_j$. Define the remaining leading bias in finest-resolution units by

$$\varrho(\beta, \beta_*) = q_+^{\beta_*} \sum_j w_j(\beta) q_j^{-\beta_*}. \quad (11)$$

Lemma 1 (Direction of exponent error). *For positive exponents, $\varrho = 0$ exactly when $\beta = \beta_*$; otherwise $\text{sign}(\varrho) = \text{sign}(\beta - \beta_*)$. The positive ratio need not be at most one.*

Proof. Put $u_j = q_j^{-\beta}$ and $p = \beta_*/\beta$. The intercept of u^p has positive denominator $m \sum_j u_j^2 - (\sum_j u_j)^2$ and numerator

$$\sum_{i < j} u_i u_j (u_j - u_i) (u_i^{p-1} - u_j^{p-1}). \quad (12)$$

Every term has the stated sign; all vanish when $p = 1$.

On $\{8, 12, 16, 24, 32\}$, $\varrho(2, 0.5) = 1.066$ and $\varrho(2, 1) = 1.046$: remaining leading bias can exceed that of the finest bin.

For $\mu(q) = \theta + cq^{-\beta_*} + r(q)$, the intercept bias is $cq_+^{-\beta_*} \varrho + \sum_j w_j r(q_j)$. Underestimating the exponent makes only the leading term conservative for a positive one-sided test when $c > 0$; for $c < 0$ the sign reverses. The smooth scalar coefficient is $c = \int f' g' / 12$, not necessarily positive. Remainder, nuisance and sampling errors still require control. With $\Sigma = \text{Cov}((\bar{R}^{(q_j)})_{j=1}^m)$, the intercept variance is $w^\top \Sigma w$; $\|w\|_2^2$ compares variance only when $\Sigma \propto I$. Estimated tied ranks, raw-scale smoothness or Lipschitz continuity alone do not establish the required signed expansion.

10 Observable category budgets and a general covariance boundary

The main paper's reference theorem uses all three observation roles in the order-three U-statistic. A concrete dependent boundary has constant controls, V uniform on $\{0, 1, 2, 3\}$ and $W = (1, 3, 0, 2)_V$.

The focal residual product is $3/64$ for $V = 0, 3$ and $-3/64$ for $V = 1, 2$. Each reference-role term is $1/64$ for $V = 0, 3$ and $-1/64$ otherwise. Hence the target and reuse coefficient are zero, but the full variance is $25/4096$, compared with focal variance $9/4096$. This is a counterexample to using focal variance under an arbitrary covariance null, not an error in the released group denominator. Across 3,000 independent trials at $M = 400, N = 64$, the group test rejects 143 times and the deliberately focal-only test 482 times. Exact count enumeration, generic triple kernels and the full projection are independently checked in the release. For a local mixture with $W = V$ of weight $c/\sqrt{MN}$, the target is $(5/64)c/\sqrt{MN}$ and the limiting statistic is $\mathcal{N}(c, 1)$. The local sequence keeps the full projection variance away from zero.

The initial absolute-range categorical certificate replaces an externally supplied error allowance with an observable validation construction. Each validation pair contains an independent focal and reference row; both observations are charged. Conditional-mean confidence rectangles and simultaneous binomial probability limits share the failure budget $\delta = 10^{-4}$. The small linear program in the main paper accounts for rare and unobserved cells. This uses classical Hoeffding and binomial confidence arguments; those inequalities are not new contributions.

The study has two or eight categories, balanced or with one probability 0.005. Given category c , both Bernoulli margins have success probability p_c and covariance $\gamma p_c(1 - p_c)$. The two-category probabilities are (0.2, 0.8); the eight probabilities are equally spaced from 0.2 to 0.8. The midrank target is $\gamma \sum_c \Pr(Z = c) p_c(1 - p_c)/4$. Training sizes are 32, 512, 8,192; validation uses 512 or 8,192 pairs; $M = 100$ or 400, $N = 64$, and $\gamma = 0, 0.25, 0.5$. Each of 108 configurations has 1,000 replications. Every original maximum-cell result and its probability-weighted follow-up is retained. The weighting design followed the original rare-cell failure and reused its random primitives; it is not a new independent confirmation.

Table 6: Finite-certificate positive decisions out of 1,000. All use 8,192 training observations, 8,192 validation pairs and 400 groups of 64 peers: 50,176 total observations.

Category law	γ	Target	Maximum cell	Probability weighted
Two balanced	0	0	0	0
Two balanced	0.25	0.01000	65	54
Two balanced	0.50	0.02000	1,000	1,000
Eight balanced	0	0	0	0
Eight balanced	0.25	0.01321	0	0
Eight balanced	0.50	0.02643	1,000	1,000
Eight, rare cell	0	0	0	0
Eight, rare cell	0.25	0.01366	0	0
Eight, rare cell	0.50	0.02731	0	1,000

No nuisance-envelope or lower-bound coverage failure occurred in the recorded grids. This is not a zero-risk claim. Weak-signal power remains limited. Because weighting reallocates the failure budget, it need not improve each realization: 30 paired decisions favor only maximum-cell, whereas 3,182 favor only weighting. Generic covariance-null theory and fixed-family BY validity are mathematical results; this experiment measures single-certificate behavior rather than a new full-family benchmark.

11 Exact decomposition of the retail fit changes

For the unchanged 68,000 middle-fold rows over 17 weeks, define complementary residuals r_X, r_Y and directional-minus-complementary fitted mean changes Δ_X, Δ_Y . At each resolution, $\Delta \bar{R} = -r_X \Delta_Y - r_Y \Delta_X + \Delta_X \Delta_Y$. Combining each term with the unchanged exponent-two intercept weights preserves this identity. All three terms reinforce each change at

Table 7: Empirical mean-score change, decomposed by fitted-mean shifts.

Model	Outcome fit	Forecast fit	Joint term	Total
Price/calendar	-0.00381653	-0.00448701	-0.00341526	-0.01171880
Croston SBA	0.01242305	0.00970073	0.00327125	0.02539503

every tested resolution. The largest contribution is the forecast-mean shift for price/calendar and the outcome-mean shift for Croston. The full 15×15 cross-resolution/term HAC covariance is released; original means and standard errors reproduce. These are empirical fitting components, not identified causal shares of the linear population feedback model.

12 Locked public rental-demand confirmation

The Seoul Bike Sharing Demand data [1] contain 8,760 hourly observations from December 2017 through November 2018. The provider’s unchanged source and attribution are included under CC BY 4.0. The protocol was frozen locally before download, and models, adjustments, coefficients and choices were frozen before confirmation outcomes were parsed. This is a historical reanalysis of a newly obtained source, not an external registration, third-party replication or new transport domain for the project.

Seven-day embargoes separate the stages. Inputs are past counts at lags 1, 2, 3, 24, 25, 48, 168, past-only daily/weekly means and calendar harmonics. No contemporary weather, operational status or outcome innovation is used. The previous hourly count is assumed available; publication latency and historical provider vintages are not reconstructed. Confirmation outcomes enter only later lag features in sequential one-hour forecasting, never a model refit or selection. Perturbing all current/future counts leaves all forecasts at or before each checked origin unchanged.

Purpose	Inclusive dates	Hours
Forecast training	2017-12-08 to 2018-04-27	3,384
Nuisance fitting	2018-05-05 to 2018-06-29	1,344
Audit and selection	2018-07-07 to 2018-08-31	1,344
Confirmation	2018-09-08 to 2018-11-30	2,016

Table 8: Locked Seoul confirmation: raw rental-count MSE. All choices were fixed before the final 12 weeks. A gate admits only BY-retained candidates.

Forecast rule	Seasonal control	Ridge control
Adjusted baseline	333,914	53,554
Conventional convex blend	40,231	36,747
Ungated residual augmentation	34,125	31,123
Audit-gated augmentation	333,914	31,123

Two baseline controls, previous-day counts and Ridge, each have eight candidates: previous hour/day/week, Ridge, histogram gradient boosting, reflected Ridge, an exact baseline copy and a noise-perturbed baseline. The adjustment uses an intercept, linear baseline term and five-knot cubic splines, fitted only in the nuisance window. These are estimated projections, not true conditional means. Conventional convex blending, ungated residual augmentation and BY-gated augmentation each select by the audit-window MSE, with the adjusted baseline always available. The 16 pairs remain in the family. The gate uses eight weekly means, lag-one Bartlett HAC, an 8/7 correction and nominal t_7 values; none of these choices proves real-data calibration.

On Ridge, gated and ungated augmentation are identical. Their confirmation gain over the adjusted baseline is 22,431.43 squared counts. Their gain over conventional blending is 5,623.72, but weekly and fortnight intervals give different evidence about its sign. The direct boosting forecast has MSE 40,231.11; its difference from the augmentation is inconclusive under both references. On the seasonal baseline, gating blocks the useful augmentation and loses 299,789.52 squared counts relative to ungated selection. Both branches remain; none was chosen after confirmation. Intervals are pointwise temporal approximations, not exact or simultaneous coverage. All 74 risk rows, 82 paired comparisons, 16 gain identities and 888 weekly aggregates are released.

The exact empirical gain of fixed fits and coefficient t is $2t\hat{\theta}_{\text{confirm}} - t^2\hat{v}_{\text{confirm}}$. For Ridge, $t = 0.786989$, whereas the ex-post empirical optimum is 0.942380. The difference in squared coefficients explains 626.95 squared counts relative to that empirical optimum. It is neither the unknown population optimum nor a coefficient used for prediction. The projected gain identity measures what the learned combination achieved; it does not establish a general benefit from gating.

13 Rectangular validation: sufficient sample cost

Use the rectangular allowances defined in the main paper, with its range R , independent-triple count J , and separate validation and evaluation error allocations. This result is retained for that construction only. For the rectangular choice $\hat{B} = \hat{B}_+$, the observable width $W_B = \hat{B}_+ - \hat{B}_-$ supports sample planning. On its validation event, the range-only bound has conditional power at least $1 - \eta$, $0 < \eta < 1$, if

$$\vartheta > W_B + \frac{R}{\sqrt{2J}} \left(\sqrt{\log \frac{1}{\alpha - \delta}} + \sqrt{\log \frac{1}{\eta}} \right). \quad (13)$$

This sufficient condition uses the joint rectangular event, not the aggregate bound combined with the old lower allowance. It is not an optimal detection boundary. Count training and both members of every validation pair in the total budget.

Proof. On the rectangular validation event, $b_H \geq \hat{B}_-$ and $\mathbb{E}\bar{D} = \vartheta + b_H$. Conditional on validation, Hoeffding’s lower-tail bound gives

$$\bar{D} \geq \vartheta + b_H - R\sqrt{\log(1/\eta)/(2J)}$$

with probability at least $1 - \eta$. Subtract $\hat{B}_+$ and the range-only radius. The resulting lower bound is positive under Eq. (13). The rectangular event jointly controls both allowances; an independently constructed aggregate upper bound and this lower allowance require the appropriate union allocation.

14 Signed allowances, variance and observation cost

The refined reference certificate in the main paper separates a signed upper learning-bias limit from a sampling allowance. The original absolute-range construction and every previously released result remain available. This section specifies the new comparisons and their limits.

Signed interval optimization For each validation confidence rectangle, evaluating the bilinear product at all four corners gives its exact minimum and maximum. The probability constraints are unchanged from the absolute construction. An exchange argument proves the greedy optimizer: moving available mass from a smaller coefficient to a larger one cannot decrease the objective. Negative coefficients still receive mass when needed to satisfy total mass one. For identical intervals and caps, each signed upper coefficient is no larger than its absolute-error product, hence $\hat{B}_+ \leq \hat{B}_{\text{abs}}$ pointwise. No additional confidence failure is incurred. This comparison does not license changing the validation size or error allocation after viewing the evaluation results.

For a concrete persistent-error example, take one category and Bernoulli margins of $1/2$ with joint probabilities $P_{00} = P_{11} = 0.29$, $P_{01} = P_{10} = 0.21$. Marginal midranks are $1/4$ and $3/4$, so $\vartheta = 0.01$. Fixed fits $f = 0.65$, $g = 0.35$ give $b_H = -0.0225$. As validation and evaluation sizes grow, the absolute lower bound tends to -0.035 while the signed bound tends to 0.01 . The point is loss from subtracting an unnecessary absolute value under persistent misspecification, not a new generic concentration inequality or superiority over classical signed inference.

Variance from independent triples Partition evaluation rows by their predeclared order within each group. For each consecutive triple, average all six permutations of focal, forecast-reference and outcome-reference roles. Ignore the final $N \bmod 3$ rows only for this variance estimate; the full statistic retains them. The J symmetric kernels are independent under the common-law design. Their ordinary sample variance uses denominator $J - 1$. The proof in the main paper applies the variance confidence bound and the full-U concentration event by a union bound; it does not assume independence between this sample variance and the full mean. The factor two in the inverted variance-based probability is essential. Taking the minimum of separately valid range and variance probabilities without an error allocation would be a different, unsupported procedure.

The Bernstein constant For the latter constant, the Bernstein bound is $\log \mathbb{E} e^{\lambda(\bar{D} - \mathbb{E}[\bar{D}|H])} \leq v\lambda^2/[2(1 - c\lambda)]$, with $v = \sigma^2/J$, $c = R/(3J)$. Setting $t = \sqrt{2x/v}$ and $\lambda = t/(1 + ct)$ gives exponent x at $\sqrt{2vx} + cx$; the zero-variance case follows directly.

Comparators The independent-triple comparator uses the *mean of these triples*, with radius

$$s\sqrt{2x/J} + \frac{7Rx}{3(J-1)}, \quad x = \log\{2/(\alpha - \delta)\},$$

from the classical empirical Bernstein bound [17]. Applying this radius directly to the full U-statistic is not justified by that theorem. The pooled comparator computes the full order-three U-statistic over all MN peers and measures its kernel variance with $\lfloor MN/3 \rfloor$ triples. Pooling is legitimate here because every group follows the same law. Both comparators receive the same signed learning allowance, fitted means, validation pairs and evaluation rows. They are strong alternatives, not methods assumed to have lower power.

Fixed simulation grid The two-category law has equal category masses and Bernoulli success probabilities $(0.2, 0.8)$. The eight-category law has masses $(0.005, 0.995/7, \dots, 0.995/7)$ and success probabilities equally spaced from 0.2 to 0.8 . Conditional covariance is $\gamma p_c(1 - p_c)$, with $\gamma \in \{-0.125, 0, 0.125, 0.25, 0.5\}$. The target is $\gamma \sum_c \Pr(Z = c)p_c(1 - p_c)/4$. Training uses 8,192 observations and empirical-midrank cell fits. Validation has 512 or 8,192 disjoint pairs; evaluation has $M \in \{25, 100, 400, 1600\}$ groups of $N = 64$ peers. The two-category 8,192-pair setting additionally shifts the fitted means by $+0.15$ and -0.15 , clipped to $[0, 1]$. Five constructions give 500 cells with 1,000 paired replications per cell, at $\alpha = 0.05$, $\delta = 10^{-4}$. These are single-certificate comparisons. A separate helper chooses a predeclared family-size-aware validation budget; it does not retune this study's error allocation or turn it into a new family experiment.

The study protocol and executable hashes were saved before the full run. A two-replication computational pilot checked the same declared grid. After adding a defensive check to the independent-triple API, the entire 500,000-row computation was repeated and both result files were byte-identical. This is a reproducibility check, not a second statistical confirmation. The public protocol preserves the original and replay links.

Table 9: Positive decisions out of 1,000, with estimated means and 8,192 validation pairs. Every row counts all training and evaluation observations and both members of each validation pair.

Law	ϑ	Total rows	Absolute range	Signed range	Variance full U	Triples Bernstein	Pooled variance
Two	0.010	30,976	0	0	1,000	1,000	1,000
Two	0.010	50,176	58	78	1,000	1,000	1,000
Two	0.005	30,976	0	0	6	24	6
Two	0.005	50,176	0	0	1,000	1,000	1,000
Eight	0.01366	30,976	0	0	435	433	461
Eight	0.01366	50,176	0	0	1,000	1,000	1,000
Eight	0.00683	50,176	0	0	0	0	0

All 200 nonpositive cells have zero observed rejections, and all 500 cells have zero observed lower-bound coverage failures. Neither means zero risk; for a single 0/1000 binomial cell the one-sided 95% upper limit is approximately 0.003. All 512-pair validation settings remain undetected. Rare categories and weak signals therefore still expose a learning-budget bottleneck. The new full-U variance bound adds 11,565 positive decisions relative to the absolute-range construction across the paired grid, but the benefit is largely shared by classical comparators. Relative to independent triples, its net gain is only 38 decisions; relative to pooled variance it has 51 fewer. These grid sums are descriptive, not population-weighted performance measures.

Sample planning and family resolution For a desired effect $\vartheta_0 > W_B$, solving the main paper’s sufficient power condition gives a strict lower requirement on the number of triples:

$$J > \frac{R^2}{2(\vartheta_0 - W_B)^2} \left(\sqrt{\log \frac{1}{\alpha - \delta}} + \sqrt{\log \frac{1}{\eta}} \right)^2.$$

Choose evaluation size after independent validation and before evaluating new rows. The guarantee is conditional on the simultaneous validation event, whose failure probability is at most δ . No finite count is certified by this condition when $\vartheta_0 \leq W_B$; collecting only more evaluation rows cannot remove that sufficient-condition bottleneck. This is not a necessary lower bound or an optimal allocation theorem. For a fixed K -model BY family, setting $\delta = \rho\alpha/(KH_K)$ with $0 < \rho < 1$ keeps the probability floor below the first-rank threshold. With fixed $\delta = 10^{-4}$ and $K = 100$, that threshold is about 9.64×10^{-5} . One isolated strong signal cannot pass the first rank, although later step-up ranks can still reject.

15 A finite-population certificate on a public archive

The source is UCI ElectricityLoadDiagrams20112014 [11], under CC BY 4.0. We keep the first 96 source meters and nominal daily means of the provider’s 96 quarter-hour readings. A single histogram gradient boosting model trains on 67,392 rows before 2013 and remains fixed over 730 days in 2013–2014. Lagged observations update only as they become available. The source’s zeros are retained, including those before some meters became active. Categories combine target weekend with whether the previous-week forecast exceeds its pre-2013 training median. That frozen median is zero. Category counts in the 70,080-row archive are 883, 49,229, 350 and 19,618.

Conditional on the fixed archive, independent replacement draws in the training, validation and evaluation streams have a common finite-population law. Repeated row identities do not create dependence between independently drawn indices. This establishes the sampling assumptions literally for the finite-archive target, irrespective of temporal dependence in the source. It does not establish independence of chronological observations or inference about a future population. The source had previously been used in the project; this is not an unseen source or external preregistration.

The three candidates are boosting, the previous-week forecast and a category-code copy. The protocol fixes all seven method comparisons, 4,096 mean-fitting rows, 8,192 validation pairs and 12,288 evaluation rows before computing the audit. The primary pooled-U variance certificate has boosting lower bound 0.070503 and seasonal lower bound 0.069418. Exact census targets, computed after writing every certificate, are 0.078219 and 0.077414. All seven methods retain both forecasts; the category-copy control is not retained. The common BY-adjusted values for the two positives are 0.000275.

The 32,768 queries access 26,140 unique rows. Forecast preparation already accesses the complete source archive and takes 8.39 seconds in the recorded environment. Census targets for all three candidates take about 0.039 seconds, versus 0.328 seconds for one pooled boosting U-statistic; its independent triples take about 0.00046 seconds. Thus no end-to-end access

or runtime advantage is established. The coarse categories leave substantial continuous-baseline information, as the positive seasonal candidate illustrates. The result demonstrates a valid finite-archive certificate, not additional forecast value beyond the full seasonal baseline. The legal derived archive, draw indices, frozen model, protocol and all bounds are included. A portable source-data replay reproduces the forecasts and numerical results within the recorded numerical tolerances; timings naturally vary.

16 Why the rental-demand gate blocked a useful candidate

The seasonal-control boosting candidate was not flagged as a copy. Its raw-count residual mean was 125,102.410, with lag-one weekly HAC standard error 38,057.967 over eight selection weeks, giving $T = 3.28715$ and nominal t_7 probability 0.00667738. It ranked third in the complete 16-pair family, whose third threshold is 0.00277307. The monotone BY-adjusted value is 0.10940387, attained via the next rank's 0.0080903 probability. The eight weekly means are all positive but uneven. The HAC standard error is only 3.41% above the independent-week standard error; limited weekly replication and the family threshold are the main numerical bottlenecks. No rank transformation is used in this study.

The boosting candidate under Ridge control has raw probability 0.0000344919 and adjusted probability 0.00186572, and is selected by both gated and ungated rules. Four useful seasonal alternatives are blocked; two harmful Ridge alternatives are also blocked but would not be selected by the ungated rule. Hence the gate provides no realized Ridge benefit and incurs 299,789.52 MSE opportunity loss under the seasonal control. Removing guarded hypotheses, splitting the family or replacing the tail law after observing this result would create a different procedure. Such arithmetic sensitivities are labelled retrospective in the release; none is presented as a newly confirmed gate.

17 Additional finite bound and weighting identities

17.1 A finite-sample certificate with a bias allowance

For the finite guarantee, fit nuisances on independent training clusters. Let H contain the training and design information. For evaluation cluster g , let S_g be its mean combined residual product. Prespecified nonnegative weights λ_g sum to one. The target $\theta_H = \sum_g \lambda_g \mathbb{E}[\langle X_g - m_{Xg}, Y_g - m_{Yg} \rangle / n_g \mid H]$ is the weighted oracle residual covariance, where $m_{Xg} = \mathbb{E}[X_g \mid C_g, H]$, m_{Yg} likewise, and C_g is the full evaluation-control array of cluster g .

Proposition 3 (Bias-aware finite certificate). *Conditional on H , suppose the evaluation clusters are independent and $S_g \in [\ell_g, u_g]$ almost surely. All weights and bounds are H -measurable. Write $\mu_H = \sum_g \lambda_g \mathbb{E}[S_g \mid H]$ and $V_H = \sum_g \lambda_g^2 (u_g - \ell_g)^2$. If an H -measurable allowance B satisfies $\Pr(\mu_H - \theta_H \leq B) \geq 1 - \delta$, for $0 \leq \delta < \alpha$, then*

$$L_\alpha = \sum_g \lambda_g S_g - B - \sqrt{\frac{1}{2} V_H \log(1/(\alpha - \delta))} \quad (14)$$

satisfies $\Pr(\theta_H \geq L_\alpha) \geq 1 - \alpha$. Thus deciding positively only when $L_\alpha > 0$ has type-I error at most α under $\theta_H \leq 0$ almost surely.

Proof. Conditional Hoeffding concentration [18] bounds the upper tail of $\sum_g \lambda_g S_g - \mu_H$ by $\exp(-2x^2/V_H)$. Allocate $\alpha - \delta$ to this tail and δ to failure of the bias allowance, then apply a union bound. If $V_H = 0$, the weighted score equals its conditional mean almost surely.

With $\lambda_g = n_g/n$ the target is pooled-row; equal weights give an average-cluster target. A flat resolution curve or a good held-out fit does not supply B , and total prediction risk in place of conditional-mean error gives a valid but powerless allowance. The real-data normal screens do not apply this finite theorem.

The weighting tradeoff Bias cancellation can increase variance. With $u_j = q_j^{-\beta}$ and $X_Q = [\mathbf{1} \ u]$, the classical constrained optimum for $\Sigma \succ 0$ is $w_G = \Sigma^{-1} X_Q (X_Q^\top \Sigma^{-1} X_Q)^{-1} (1, 0)^\top$. For any feasible w , $X_Q^\top (w - w_G) = 0$, so

$$w^\top \Sigma w = w_G^\top \Sigma w_G + (w - w_G)^\top \Sigma (w - w_G). \quad (15)$$

An additional second-order constraint costs variance strictly only if $(u^{\odot 2})^\top w_G \neq 0$. On our ladder at $\beta = 1$, it raises the minimum squared weight norm by a factor of 6.44, a variance ratio only for $\Sigma \propto I$. We use fixed intercept weights.

The cost of a second bias constraint This calculation is classical constrained least squares, consistent with generalized bias reduction [19]. For $m \geq 3$ distinct positive resolutions and fixed $\beta > 0$, write $u_j = q_j^{-\beta}$, $h_j = u_j^2$, $C = [\mathbf{1} \ u]$, $w_0 = C(C^\top C)^{-1}(1, 0)^\top$, and $r = [I - C(C^\top C)^{-1}C^\top]h$. Let $\kappa_0 = h^\top w_0$. Among weights satisfying $C^\top w = (1, 0)^\top$ and $h^\top w = \kappa$, the unique minimum-norm solution and its squared norm are

$$\begin{aligned} w_*(\kappa) &= w_0 + \frac{\kappa - \kappa_0}{r^\top r} r, \\ V(\kappa) &= \|w_0\|^2 + \frac{(\kappa - \kappa_0)^2}{r^\top r}. \end{aligned} \quad (16)$$

Indeed, feasible perturbations are orthogonal to w_0 ; their projection onto r supplies the additional constraint. For $\mathcal{Q} = \{8, 12, 16, 24, 32\}$ and $\beta = 1$, $\kappa_0 = -5845/1179648$, $r^\top r = 7777/1811939328$, $V(\kappa_0) = 269/256$, and $V(0) = 22567/3333$. Thus full second-order cancellation raises the minimum squared norm by $V(0)/V(\kappa_0) = 6.44356$; the endpoint directions have cosine 0.39395. This is a variance ratio only for isotropic cross-resolution covariance. For general $\Sigma \succ 0$, the constrained GLS optimum is $w_G = \Sigma^{-1}C(C^\top \Sigma^{-1}C)^{-1}(1, 0)^\top$; adding $h^\top w = 0$ costs variance strictly only if $h^\top w_G \neq 0$. Neither result implies a false-positive floor. Changing β changes the constraint class and is not this fixed- β frontier. For the intercept weights $w(\beta)$ on $[1, q^{-\beta}]$, define $\varrho(\beta, \beta_*) = q_+^{\beta_*} \sum_j w_j q_j^{-\beta_*}$. The preceding resolution analysis proves its sign is that of $\beta - \beta_*$ when nonzero.

Table 10: Remaining leading bias in finest-resolution units on $\{8, 12, 16, 24, 32\}$. Rows give the assumed exponent; columns give the true exponent. Values are deterministic, rounded to three decimals.

β	β_*					$\ w\ _2$
	0.5	1	1.5	2	3	
0.5	0	-2.119	-6.483	-15.336	-68.823	1.874
1	0.714	0	-1.611	-5.074	-27.228	1.025
1.5	0.950	0.700	0	-1.679	-13.468	0.774
2	1.066	1.046	0.796	0	-6.656	0.664

17.2 Separating rank-reference bias from nuisance error

Ranks introduce a second bias when both channels reuse the same finite reference sample. To isolate it, condition on independent training data H and take $N \geq 3$ i.i.d. triples (V_i, W_i, Z_i) per independent group, with a common law. Let F_V, F_W be population marginal mid-distributions, $m_V(Z) = \mathbb{E}[F_V(V) \mid Z, H]$, and m_W analogous. This result targets $\vartheta = \mathbb{E}[(F_V(V) - m_V)(F_W(W) - m_W) \mid H]$, which differs from the standardized panel-rank target. Set $a_{ij} = \mathbf{1}\{V_j < V_i\} + \frac{1}{2}\mathbf{1}\{V_j = V_i\}$, define b_{ij} similarly, and let $k = N - 1$, $U_i = k^{-1} \sum_{j \neq i} a_{ij}$, $V'_i = k^{-1} \sum_{j \neq i} b_{ij}$. For H -measurable nuisance fits $f, g \in [0, 1]$, define

$$\begin{aligned} C_i &= (U_i - f(Z_i))(V'_i - g(Z_i)), \\ D_i &= C_i + \frac{U_i V'_i - k^{-1} \sum_{j \neq i} a_{ij} b_{ij}}{k - 1}. \end{aligned} \quad (17)$$

Proposition 4 (Reference-reuse decomposition). *Conditional on H , write $\Gamma = \mathbb{E}[a_{12}b_{12} - F_V(V_1)F_W(W_1) \mid H]$ and $b_H = \mathbb{E}[(m_V - f)(m_W - g) \mid H]$. Then, including ties,*

$$\begin{aligned} \mathbb{E}[C_i \mid H] &= \vartheta + b_H + \frac{\Gamma}{N - 1}, \\ \mathbb{E}[D_i \mid H] &= \vartheta + b_H. \end{aligned} \quad (18)$$

Distinct reference indices remove the $\Gamma/(N - 1)$ term exactly; either exact nuisance removes b_H . The following argument gives the proof and a finite confidence bound using classical U-statistic concentration [5, 18]. Independent groups alone are insufficient: dependent peers can leave positive bias even with exact nuisances. This correction is therefore a separate construction under i.i.d. peer sampling, not a repair for arbitrary dependent panels.

Distinct-reference proof and bound For distinct i, j, l , reference independence conditional on the focal row and H gives $\mathbb{E}[a_{ij}b_{il} \mid V_i, W_i, Z_i, H] = F_V(V_i)F_W(W_i)$. The score D_i averages $(a_{ij} - f(Z_i))(b_{il} - g(Z_i))$ over these ordered pairs.

Conditional centering removes both linear nuisance-error terms, leaving $\vartheta + b_H$. The $N - 1$ shared pairs among $(N - 1)^2$ pairs in C_i add $\Gamma/(N - 1)$. This proves Eq. (18), including ties.

For M equal independent groups, let $\bar{D}$ average their group means. With an H -measurable $B \geq b_H$ almost surely,

$$L_\alpha = \bar{D} - B - \sqrt{\frac{2\log(1/\alpha)}{M\lfloor N/3 \rfloor}} \quad (19)$$

has conditional coverage at least $1 - \alpha$ for ϑ . Indeed, the symmetrized degree-three kernel lies in $[-1, 1]$. Averaging over permutations into disjoint triples recovers each group mean. Exponential convexity and Hoeffding's lemma bound its centered MGF by $\exp\{t^2/(2\lfloor N/3 \rfloor)\}$; group independence gives the stated radius. If the training envelope fails with probability δ , replace α by $\alpha - \delta > 0$ for unconditional coverage.

Group independence alone is insufficient. Let Z be a fair bit, $V = Z \text{ xor } E_V$, $W = Z \text{ xor } E_W$, with independent flips of probabilities $1/4$ and $1/3$. Duplicate one triple within each group. Then $\vartheta = 0$, and exact nuisances are $m_V(z) = 3/8 + z/4$, $m_W(z) = 5/12 + z/6$. All comparisons equal $1/2$, giving $D_i = (1/2 - m_V(Z))(1/2 - m_W(Z)) = 1/96 > 0$. Thus dependent peers invalidate the reference argument even with exact fits.

Theorem 3 (Oracle sampling-ratio consequence). *Consider $M \rightarrow \infty$ independent groups, each containing $N \rightarrow \infty$ i.i.d. rows from the same fixed law. Suppose $V \perp W \mid Z$ and both nuisances in Eq. (17) are exact: $f = m_V$, $g = m_W$. Set*

$$r_V = F_V(V) - m_V(Z), \quad r_W = F_W(W) - m_W(Z), \quad \sigma^2 = \mathbb{E}[r_V^2 r_W^2] > 0.$$

Let C_g, D_g be the group averages of C_i, D_i , respectively. For $A = C, D$, write $\bar{A} = M^{-1} \sum_g A_g$ and $\hat{v}_A = (M - 1)^{-1} \sum_g (A_g - \bar{A})^2$. Define $T_A = \sqrt{M} \bar{A} / \sqrt{\hat{v}_A}$ for $A = C, D$. Then

$$\begin{aligned} \sqrt{MN} \bar{D} &\Rightarrow \mathcal{N}(0, \sigma^2), \\ \sqrt{MN} \{\bar{C} - \Gamma/(N - 1)\} &\Rightarrow \mathcal{N}(0, \sigma^2), \\ N \hat{v}_D &\rightarrow_p \sigma^2, \quad N \hat{v}_C \rightarrow_p \sigma^2. \end{aligned} \quad (20)$$

Thus T_D converges to $\mathcal{N}(0, 1)$. If $M/N \rightarrow c \in [0, \infty)$, then $T_C \Rightarrow \mathcal{N}(\Gamma\sqrt{c}/\sigma, 1)$. If $\Gamma > 0$ and $M/N \rightarrow \infty$, then $T_C \rightarrow_p +\infty$. The limiting one-sided naive rejection probability is therefore $1 - \Phi(z_{1-\alpha} - \Gamma\sqrt{c}/\sigma)$ for finite c , and one in the divergent positive-bias case.

Proof. Symmetrize the ordered-triple kernel $(a_{ij} - m_V(Z_i))(b_{il} - m_W(Z_i))$ over its six role assignments. Under the raw conditional null its first Hoeffding projection is $h_1(O) = r_V r_W / 3$. The focal role contributes $r_V r_W$. For the reference- V role, the focal W residual has conditional mean zero given its controls and focal V , so that role contributes zero. The reference- W role is analogous. The U-statistic decomposition gives

$$D_g = N^{-1} \sum_i r_{V,gi} r_{W,gi} + R_g, \quad \mathbb{E} R_g = 0, \quad \mathbb{E} R_g^2 = O(N^{-2}).$$

Group independence makes the variance of $\sqrt{MN} \bar{R}$ of order N^{-1} . The MN leading bounded i.i.d. terms obey the ordinary CLT.

Let J_g average $a_{ij} b_{ij}$ over ordered pairs $i \neq j$, and let Q_g average $a_{ij} b_{il}$ over ordered distinct triples. The definitions give $C_g - D_g = (J_g - Q_g)/(N - 1)$ and $\mathbb{E}(J_g - Q_g) = \Gamma$. Bounded kernels of degrees two and three imply $\text{Var}(J_g - Q_g) = O(N^{-1})$. Consequently the variance of

$$\frac{\sqrt{MN}}{N - 1} \left\{ M^{-1} \sum_g (J_g - Q_g) - \Gamma \right\}$$

is $O(N^{-2})$, proving the centered shared-reference CLT.

For either statistic, the centered fourth moment is $O(N^{-2})$. To see this, represent the bounded U-statistic as an average over permutations of disjoint triples, then apply convexity to the fourth power. Each permutation average contains $\lfloor N/3 \rfloor$ independent bounded centered terms. The shared-reference statistic has the same bound by embedding its degree-two component in a degree-three kernel with uniformly bounded coefficients. Hence the variance of the scaled cluster second-moment average is $O(M^{-1})$; replacing its true mean by the sample mean costs $O_p(M^{-1})$ after multiplication by N . Together with the variance limits from the decompositions above, this proves consistency of $N \hat{v}_A$.

Slutsky's theorem gives the finite- c limits. In the divergent case, $T_C / [\Gamma\sqrt{MN}/\{(N - 1)\sigma\}] \rightarrow_p 1$. This ratio argument does not require a centered normal approximation at a diverging studentized shift.

Both exact conditional means and nonzero first-projection variance are essential to this statement. Honest training alone is insufficient. It gives no exact small- M calibration and does not apply to dependent or within-period standardized panel ranks. More independent groups can make reference reuse detectable even when the population residual covariance is zero; distinct references remove that specific error.

18 A comparison with supplied null simulations

The main scalar comparison uses each method’s normal or gamma reference. Their different null rejection rates preclude a direct power ranking. Here we ask a narrower question: how do the six fixed statistics compare when each receives an independent bank from the exact same null generator? This is a retrospective secondary analysis of existing outputs. The split, all methods and all comparisons were fixed before the secondary calculation; they were not fixed before the original simulation results became available.

For each of the four scalar regimes, replications 0 through 149 under the null form a calibration bank of $B = 150$ fully observed datasets, each with $n = 400$. Null replications 150 through 299 and all 300 positive-covariance replications are evaluation data. The generating seeds and sample hashes are disjoint across these roles. All six methods receive the same observations within each replication. The extra zero-covariance dependence design is excluded because it has different null membership for KCI and covariance tests.

Write p_b for the native reference value on bank dataset b and p_0 for that on an evaluation dataset. Use

$$p_{MC} = \frac{1 + \sum_{b=1}^{150} \mathbf{1}\{p_b \leq p_0\}}{151}. \quad (21)$$

Under exchangeable bank and new-null datasets and a fixed score function, the rank argument gives $\Pr(p_{MC} \leq 0.05) \leq 7/151 = 0.04636$. To see this, an exchangeable continuous score has a uniform rank among 151 values; the inclusive comparison makes ties conservative. This is a classical Monte Carlo reference [20], not a new learned calibration method. It requires a supplied null generating law and does not apply to the observational panels in the main paper.

The guarantee averages over the bank as well as the evaluation dataset. After one bank is fixed, its conditional rejection probability can differ from $7/151$. The 150 validation indicators are independent conditional on that bank, but share calibration randomness marginally. Accordingly, all Wilson intervals in Fig. 8 describe the conditional rate for the realized bank; they do not validate the marginal rank theorem.

Table 11: Oracle-bank comparison. Each entry is the rejection count divided by its denominator: 150 validation nulls or 300 positive-covariance alternatives. The same bank is used within each method and regime.

Method	Linear		Smooth		Quadratic		Heavy tails	
	Null	Power	Null	Power	Null	Power	Null	Power
GCM, 8 bins	7/150	278/300	7/150	292/300	4/150	179/300	3/150	289/300
GCM, 32 bins	12/150	274/300	9/150	282/300	6/150	241/300	6/150	278/300
GCM, splines	8/150	275/300	6/150	280/300	8/150	280/300	4/150	279/300
Extrapolation, $\beta = 1$	11/150	273/300	10/150	281/300	12/150	256/300	6/150	278/300
Extrapolation, $\beta = 2$	11/150	283/300	10/150	287/300	8/150	246/300	5/150	286/300
KCI, gamma	13/150	218/300	11/150	226/300	4/150	231/300	6/150	268/300

At exponent one, extrapolation detects 273/300 and 281/300 linear and smooth alternatives, compared with KCI’s 218/300 and 226/300. In the quadratic design, splines detect 280/300, compared with 256/300 for exponent one and 246/300 for exponent two. All 15 method pairs in all four positive-effect regimes are compared with two-sided exact discordance tests, followed by Holm correction over 60 comparisons. The two exponent-one gains over KCI and the quadratic spline gain over exponent one remain significant, with adjusted values 2.53×10^{-11} , 4.93×10^{-10} and 0.00765. These comparisons are conditional on the realized calibration banks. All 10,800 evaluated method rows, 48 cells and 60 contrasts are retained.

The result identifies a conditional power tradeoff under supplied null information; it does not identify a uniformly preferable adjustment. The cost includes 150 complete oracle null datasets per regime, and the secondary analysis supplies no evidence of actual-panel validity or independent confirmation on new observations.

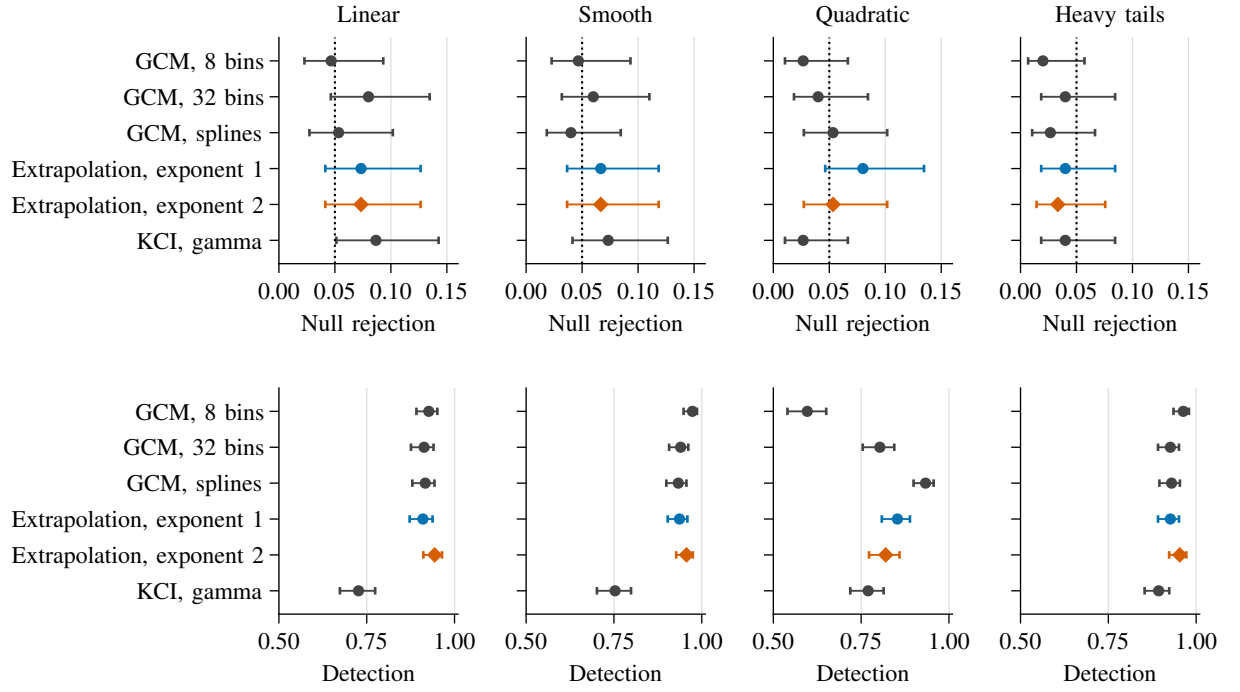


Figure 8: **Calibration and detection with a supplied null bank.** All 48 method, regime and effect cells are shown. Points and intervals are the observed rates and conditional-bank 95% Wilson intervals. The dotted line in the null panels marks 0.05. Colors distinguish the two extrapolation exponents; all methods share the same displayed scales. The calibration datasets are additional oracle information, unavailable for direct use on the real panels.

19 Legacy sensitivity experiments and comparator scope

We reproduce four earlier simulation drivers with their original settings, retaining every decision and every unfavorable result. These experiments use empirical within-period ranks against random peers and row-random nuisance folds. Their zero latent added signal does not establish conditional independence of the ranked quantities given the focal baseline rank and entity. We therefore report rejection under zero or positive latent added signal, rather than calibrated Type-I error or size-corrected power. The separate matched-null studies in the main paper address a different design.

The synthetic additive nuisance fit jointly estimates firm, period and momentum-bin effects within each training fold. It is distinct from the historical sequential real-panel adjustment. Every simulated panel is balanced, so the clustered SE’s mean-total centering equals size centering. These two implementation checks do not establish independent folds, negligible nuisance error or valid ranked-panel normal-reference inference.

19.1 Adaptive exponent and complete ablation

The adaptive heuristic fits a log slope to absolute successive differences of the five evaluated product means, clips the exponent to $[0.5, 2]$, and tests weights selected using those same means. It omits the varying adjacent-resolution factor in a pure-power difference and treats the selected weights as fixed in its SE. A noiseless spectrum $\theta + cq^{-1}$ already yields $\hat{\beta} = 1.180886$ on this ladder. In contrast, the geometric triple 8, 16, 32 identifies a positive pure-power order for $c \neq 0$ from the ratio of successive differences. Failure of this heuristic does not prove that the order cannot be estimated.

Table 12 gives all four regimes, with fifty paired draws per signal condition. The average estimated order is computed on the zero-signal draws. Independent draws within each cell give the pointwise Wilson intervals in Figure 9; they are Monte Carlo intervals, not guarantees for the scientific null.

Table 12: Adaptive-exponent replay. Entries are rejected draws/total draws. Zero and positive denote latent signal coefficients 0 and 0.15.

Regime	Zero signal		Positive signal		Mean $\hat{\beta}$
	Fixed	Adaptive	Fixed	Adaptive	
Smooth	3/50	4/50	50/50	50/50	1.303450
Nonlinear	7/50	48/50	50/50	50/50	1.960841
Larger panel	5/50	8/50	50/50	50/50	1.834468
Heavy tails	5/50	4/50	48/50	47/50	1.109617

Changing the assumed order changes the leading-bias constraint. On the five-resolution ladder, squared weight norms fall from 1.050781 at $\beta = 1$ to 0.449668 at 1.96 and 0.441348 at 2. At 2 the weights no longer annihilate the q^{-1} term. This is not the fixed-order second-order-annihilation frontier; the observed rejection change does not identify a unique causal mechanism.

The complete ablation uses overlap 0.8, four regimes and nine settings. Each signal cell has 150 draws except the larger panel’s 80; the two signal conditions use disjoint seed ranges. All settings share the same data draws within each condition. There is no fold-scheme ablation. The short ladder is $\{8, 16, 32\}$ and the long ladder is $\{8, 12, 16, 20, 24, 28, 32, 40\}$. Other settings use $\{8, 12, 16, 24, 32\}$, equal-mass bins, twelve fitting sweeps and exponent one unless the row changes that component. Table 13 retains the long ladder as well as all exponent changes.

Table 13: Complete ablation. Each cell gives zero-signal / positive-signal rejection rates. Larger-panel rates use 80 draws; other rates use 150.

Setting	Smooth	Nonlinear	Larger panel	Heavy tails
Default	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933
Short ladder	0.127 / 1.000	0.160 / 0.987	0.100 / 1.000	0.087 / 0.933
Long ladder	0.113 / 1.000	0.220 / 0.987	0.113 / 1.000	0.093 / 0.960
$\beta = 0.5$	0.047 / 0.960	0.000 / 0.000	0.050 / 1.000	0.033 / 0.847
$\beta = 1.5$	0.147 / 1.000	0.833 / 1.000	0.125 / 1.000	0.147 / 0.960
$\beta = 2$	0.180 / 1.000	0.973 / 1.000	0.163 / 1.000	0.147 / 0.987
Equal-width bins	0.113 / 1.000	0.153 / 0.987	0.113 / 1.000	0.087 / 0.940
4 fit sweeps	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933
24 fit sweeps	0.113 / 1.000	0.147 / 0.987	0.113 / 1.000	0.093 / 0.933

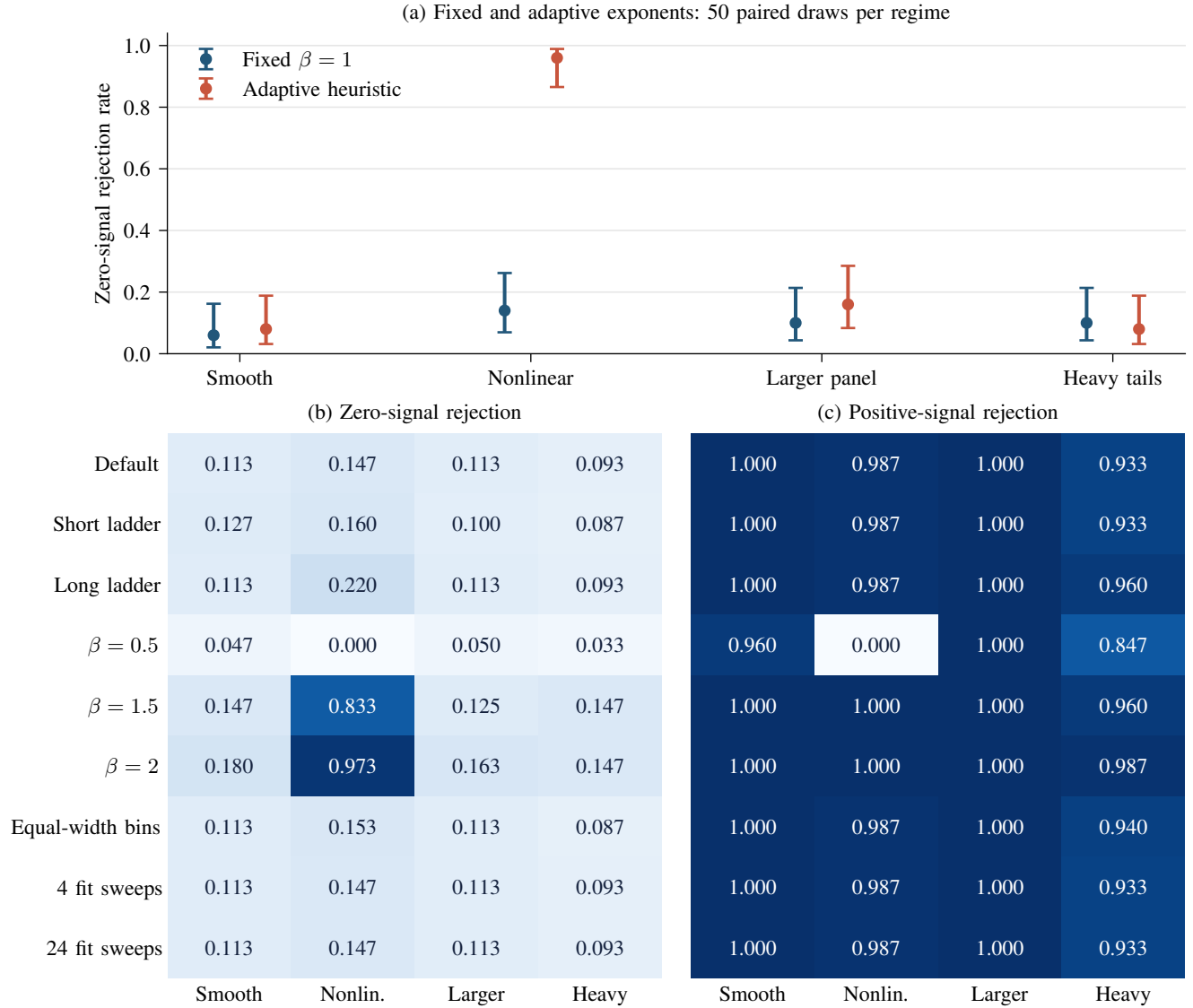


Figure 9: Complete legacy sensitivity, with no settings dropped. Panel (a) shows pointwise 95% Wilson intervals over 50 draws per method/regime. Panels (b,c) show all ablation rates, with 150 draws except 80 in the larger panel. These random-peer-ranked experiments do not validate the ranked conditional null. Full integer counts, denominators and cellwise intervals are supplied in the machine-readable tables.

19.2 Historical grid and stress designs

The earlier grid has eight implementations, five overlaps $\{0.5, 0.65, 0.8, 0.9, 0.95\}$ and four regimes: 160 method/design pairs, each with two signal conditions. Each condition uses 150 draws, except 80 for the larger panel. The legacy criterion requires zero-signal rejection at most 0.18 and positive-signal rejection at least 0.90; it is a descriptive tolerance rule, not nominal 0.05 calibration. A stricter historical prose criterion used less than 0.15 and more than 0.95. Table 14 makes the two criteria explicit.

The comparator names describe the implemented variants. Coarse/fine GCM use IID product SEs, while extrapolation uses month clustering. The linear variant fits a second-stage rank-momentum slope on the full sample. Sign weighting and binned projection learn from residuals that already depend on outer evaluation labels; deterministic target perturbations confirm this feedback. Local permutation conditions on month and momentum bins but omits firm. The grid therefore does not establish performance of the published WGCM, PCM or local-permutation procedures under their respective assumptions. The separately mislabeled RFF/distance routines are absent from this eight-implementation grid.

Table 14: Legacy grid summary over twenty designs per implementation. Extrema are descriptive; failure counts refer only to the stated tolerance rules and do not measure scientific-null validity.

Implementation	Maximum	Minimum	Failed designs/20	
	Zero signal	Positive signal	.18/.90	.15/.95
Cluster extrapolation	0.147	0.953	0/20	0/20
Coarse GCM	1.000	0.993	20/20	20/20
Fine GCM	1.000	0.973	16/20	19/20
Rank-linear projection	1.000	1.000	20/20	20/20
Naive permutation	1.000	0.993	20/20	20/20
Legacy sign weighting	0.920	0.507	15/20	20/20
Legacy binned projection	1.000	0.520	18/20	20/20
Month/bin permutation	0.633	0.847	10/20	15/20

Stress designs use 600 firms/60 periods, 120/100 with serial latent processes, 120/98 after a three-period rolling outcome sum, or 120/24. The rolling sum includes the current period, so it is not a strict forward forecasting target. A common additive period shock disappears under within-period ranking; these constructed processes do not isolate a unique cause of miscalibration. All positive-signal losses remain in Table 15.

Table 15: All stress conditions. Rates with pointwise 95% Wilson Monte Carlo intervals; these are not simultaneous intervals or ranked-null guarantees.

Design	Draws	Zero signal	Positive signal
Many fixed effects	100	0.030 [0.010, 0.085]	1.000 [0.963, 1.000]
Serial processes	200	0.085 [0.054, 0.132]	0.990 [0.964, 0.997]
Overlapping outcomes	200	0.105 [0.070, 0.155]	0.750 [0.686, 0.805]
Twenty-four periods	200	0.175 [0.129, 0.234]	0.760 [0.696, 0.814]

These replays recover reproducible sensitivity measurements without restoring claims of an unavoidable error floor, unique sensitivity to the exponent, or superiority over faithful native comparator implementations. The accompanying machine-readable tables retain all 416 rate cells, integer counts, exact denominators and pointwise intervals, including all 320 grid rates. No settings are changed after observing the replay.

20 What an exact reference can establish

The following benchmark separates three questions: whether an estimator preserves its covariance target, whether its null distribution is valid, and whether learning improves detection. Its reference distribution is classical Gaussian regression geometry [21, 22]. It does not establish a new distribution-free certificate for the ranked observational panels in the main paper.

20.1 Preserving the target before testing it

Condition on full controls and independent training information. Let clusters be independent, with arbitrary dependence within each cluster. Write $x = X - h_X$, assume $E[Y] = \Phi\gamma$ for a declared full-column-rank design Φ , and allow $E[x]$ to be unrestricted. The fixed-design target is

$$\theta = \sum_g \text{tr}\{D_g \text{Cov}(Y_g, X_g)\}, \quad D_g = I_{n_g}/n. \quad (22)$$

Here $n = \sum_g n_g$. Equal cluster weights would define a different target.

Proposition 5 (Block preservation and feasibility). *A design-only bilinear estimator $x^\top AY$ is unbiased for Eq. (22), for every score mean, γ and feasible within-cluster cross-covariance, if and only if $A\Phi = 0$ and $A_{gg} = D_g$ for every cluster. Such a matrix exists if and only if deleting each cluster preserves the column rank of Φ .*

Proof. Its expectation is $E[x]^\top A\Phi\gamma + \sum_g \text{tr}\{A_{gg} \text{Cov}(Y_g, X_g)\}$. Varying the score mean, γ and each cross-covariance gives both necessary conditions; substitution gives sufficiency. The row-block equation is $A_{g,-g}\Phi_{-g} = -D_g\Phi_g$. If $\Phi_{-g}v = 0$, invertibility of D_g forces $\Phi_g v = 0$, hence $v = 0$. Conversely, full rank permits

$$A_{gg} = D_g, \quad A_{g,-g} = -D_g\Phi_g(\Phi_{-g}^\top\Phi_{-g})^{-1}\Phi_{-g}^\top. \quad (23)$$

This is the cluster leave-out regression construction of established leave-out estimation [23], specialized to the covariance target. A period-specific dummy loses all support when its period is deleted, so unrestricted period means fail the criterion. A pseudoinverse does not remove that obstruction. Cancelling only $\Phi^\top A\Phi$ or normalizing only $\text{tr}(A)$ is insufficient. For example, with $\Phi = [1, (-1, 0, 2)^\top]$, $v = (2, -3, 1)^\top$ and $A = vv^\top/14$, both $\Phi^\top A\Phi = 0$ and $\text{tr}(A) = 1$ hold. Independent rows with cross-covariances $(-1, 1, 0)$ have average covariance zero but $E[X^\top AY] = 5/14$.

20.2 The role of a learned score mean

Under full-channel null independence, let $\delta = E[X] - h_X$, $\Sigma_X = \text{Var}(X)$, $\Sigma_Y = \text{Var}(Y)$ and $K = A\Sigma_Y A^\top$. Since $A\Phi = 0$, conditioning first on X gives

$$\text{Var}\{(X - h_X)^\top AY\} = \text{tr}(K\Sigma_X) + \delta^\top K\delta. \quad (24)$$

Learning helps this variance when it reduces the second term. Ordinary prediction mean squared error need not order errors in the metric K . If $E[Y] = \Phi\gamma + r$, the remaining bias is exactly $\delta^\top Ar$. Neither independent training nor a small training loss certifies that unobserved remainder. Empirical Gram corrections also have direct higher-order influence-function predecessors [24]; we do not claim that replacing a power law with a Gram matrix is itself new.

20.3 A finite reference with known covariance shape

Now use a separate model, which may include serial dependence:

$$Y \mid X, Z, H \sim \mathcal{N}(\Phi\gamma, \sigma^2 V), \quad V = LL^\top \succ 0. \quad (25)$$

The mean span and covariance shape V are supplied correctly before evaluation outcomes; only γ and the positive scale σ are unknown. H denotes independent training. Let P project orthogonally away from $L^{-1}\Phi$, let $d = \text{rank}(P) \geq 2$, and set $r = PL^{-1}Y$. For an outcome-independent raw direction $a = a(X, Z, H)$, put $u = PL^\top a / \|PL^\top a\|$ when this direction is nonzero.

Proposition 6 (Classical Gaussian reference). *Under Eq. (25),*

$$T_a = \frac{u^\top r}{\sqrt{\{\|r\|^2 - (u^\top r)^2\}/(d-1)}} \sim t_{d-1}. \quad (26)$$

Proof. In an orthonormal basis of the residual space whose first vector is u , the first Gaussian coordinate is independent of the other $d-1$ coordinates. Their squared norm, divided by σ^2 , is chi-squared with $d-1$ degrees of freedom. The nuisance mean has been projected out.

For Eq. (23), use $a = A^\top(X - h_X)$. The identity $\Phi^\top a = 0$ means the reference projection does not alter its numerator. Shared leave-out training otherwise makes cluster summands dependent; their sample standard deviation is not an exact reference. Known serial V validates Eq. (26), but it does not extend the independent-cluster covariance-target identity.

An estimated or incorrect V , omitted mean directions, arbitrary non-Gaussian errors and directions selected using evaluation outcomes are not covered. Directly ranking Y does not preserve this Gaussian model. The null is the stated conditional no-additional-mean model, not every distribution with zero signed covariance. These assumptions are true by construction in the following simulation, not verified for real forecasts.

20.4 A finite comparison with the classical regression direction

For $Y = \Phi\gamma + \tau X + \sigma L\epsilon$, $\tau \geq 0$, let $s = PL^{-1}X \neq 0$. The generalized least-squares (GLS) direction is $u_* = s/\|s\|$. Among tests of the form Eq. (26) with the same Φ, V and one-sided level $\alpha < 1/2$, this direction has at least as much conditional power as any other outcome-independent direction.

To prove this, a general unit direction u gives independent numerator and denominator components

$$N_u \sim \mathcal{N}(\tau u^\top s / \sigma, 1), \quad D_u \sim \chi_{d-1}^2(\tau^2\{\|s\|^2 - (u^\top s)^2\}/\sigma^2). \quad (27)$$

The aligned direction maximizes the numerator mean by Cauchy-Schwarz and sets denominator noncentrality to zero. A noncentral chi-squared variable is stochastically larger than its central counterpart, as seen from its Poisson mixture. The positive critical value makes rejection increasing in the numerator and decreasing in the independent denominator. This proves the comparison. It is a ceiling within this specified classical model and test class, not general optimality. An off-axis alternative has a noncentral denominator, so the ordinary noncentral- t power formula applies only to the aligned case.

20.5 Fixed simulation and all reference violations

Four fixed designs use $M \in \{12, 36\}$ periods, six units per period and serial correlation $\rho \in \{0, 0.6\}$. For entity index $i = 0, \dots, 5$, let z be a fixed standard Gaussian draw plus $0.25(i - 2.5)$, and let Φ contain six entity indicators, z and z^2 . The known covariance is the Kronecker product of the AR(1) correlation matrix and a six-unit equicorrelation matrix with correlation 0.4. For independent standard Gaussian innovations, generate

$$X = \Phi\gamma_X + L\epsilon_X, \quad Y = \Phi\gamma_Y + \tau L\epsilon_X + L\epsilon_Y, \quad X_0 = \Phi\gamma_X + L\epsilon_0. \quad (28)$$

Entity coefficients range linearly from -1 to 1 for X and from 0.5 to -0.5 for Y ; coefficients on (z, z^2) are $(2, 3)$ and $(0.3, 0.8)$ respectively. Effects are $\tau \in \{0, 0.15, 0.35\}$. The learned mean fits X_0 on the declared design. All methods have access to the same pilot scores and evaluation outcomes; no pilot outcome labels are supplied and no information-cost advantage is claimed.

Five scalar-adjustment directions use training-fold bin means at 8, 12, 16, 24, 32, both recorded extrapolation exponents, or the recorded cubic spline. These fit z only, so their uncalibrated normal/HAC results are adjustment probes, not replays of the main paper's entity-adjusted rank-panel procedure. Their Gaussian-reference versions all use the same full declared mean span and covariance, as do both leave-cluster directions and GLS. Thus calibrated direction comparisons have common model information.

Each fixed design has 1,000 independent innovation draws. Pairing three effects gives 12,000 distinct panels from 4,000 primitive draws. Each panel is also evaluated with the quadratic mean omitted and with dependence ignored. Those two reference violations are explicitly outside the theorem. All 468 method cells and 468,000 results are retained. This is one paired study, not 468 independent replications of generalization.

The 32 correctly specified finite-reference null rates range from 3.5% to 6.3%; all simultaneous Bonferroni 99% binomial intervals include 5%. This is an implementation diagnostic supporting the already specified theorem, not evidence that its premises hold elsewhere. The independent verification reconstructs every reported statistic and probability from the frozen primitives using separate matrix solves and reference formulas. The four primary paired learning comparisons use $\tau = 0.15$

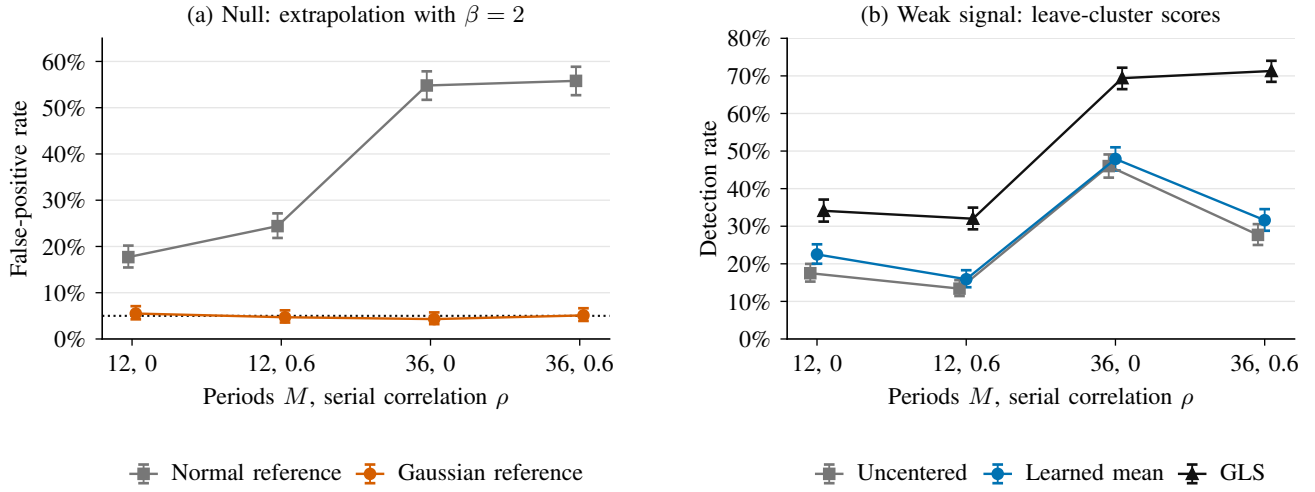


Figure 10: Four fixed designs under the correctly specified Gaussian model. Left: the scalar extrapolation direction with its original normal reference or the finite Gaussian reference. The dotted line is 5%. Right: weak-signal detection with uncentered or independently learned leave-cluster scores and GLS. Bars are marginal Wilson 95% intervals, not simultaneous intervals. Learning improves the chosen score but does not reach GLS. No real-panel calibration is inferred.

Table 16: Correct-reference weak-signal detection, $\tau = 0.15$; 1,000 paired draws per fixed design. Extrapolation uses $\beta = 2$. All entries use the same finite Gaussian reference. The full results also include nulls, stronger signals and deliberately incorrect references.

M	ρ	Extrapolation	Spline	Uncentered	Learned	GLS
12	0.0	.145	.138	.175	.225	.341
12	0.6	.110	.112	.134	.159	.320
36	0.0	.232	.292	.460	.479	.694
36	0.6	.164	.210	.277	.316	.713

and a one-sided exact binomial test of discordant decisions, with Holm correction across all four designs. Adjusted values are 0.000109, 0.00407, 0.01356 and 0.00371, respectively. They establish a gain over the uncentered leave-cluster direction in these models, not superiority to GLS. When dependence is ignored, finite-reference rejection rates across the outside-model null probes range from 5.1% to 19.7%. Omitting the quadratic mean term makes both learned leave-cluster and GLS directions reject in all 1,000 draws of each design. The restricted reference null is then false, despite conditional independence under the richer generator at $\tau = 0$. These results expose the consequences of asserting the wrong reference model.

21 Restricted-case results and their limits

These are retrospective audits of thirty stored prediction outputs in a UK insider-trading study. Each retained observation has the next calendar month’s return and a complete previous twelve-calendar-month return history, excluding the forecast month. The availability rule was fixed before recomputation, after examining the old results. It corrects source-data alignment; it is not a new prospective confirmation study.

The audit uses five contiguous whole-month folds. Adjustment is additive in firm and momentum-bin effects, at resolutions 8, 12, 16, 24, 32. The primary exponent is one, with two as sensitivity. Month clustering uses no between-month HAC and no copy or overlap guard. These settings differ from the guarded public screens. Normal-reference and BY calculations do not establish calibration.

Models differ in evaluation support, with 72 or 132 month clusters. Stored records, including repeated forecast records, are averaged within firm-month; source seed counts are one, four or five. Not every row has five predictions. Outcome encodings and stored precision differ; although subtracting a common monthly benchmark preserves ranks mathematically, float32 rounding can alter ties. This is not a common-sample or identical-target-array model ranking.

Historical feature preparation included firm-level forward fill followed by full-panel mean imputation. Removing incomplete audit records does not retrain forecasters, establish release times of all inputs, or verify foundation-model pretraining dates. These results therefore concern retrospective stored outputs, not deployable forecasting gains or implementable profits.

Both exponents retain zero models after BY. The adversarial residual-ranking MLP is the only nominal positive, with $T_1 = 2.453289$, $p_1 = 0.00707783$, and adjusted $p_1 = 0.848275$. For exponent two, $T_2 = 2.483673$, $p_2 = 0.00650175$, and adjusted $p_2 = 0.779232$. Its 31,045 observations span 1,852 firms and 132 months. Non-rejection establishes neither independence nor absence of predictive information. Historical full-support and outcome-only corrected aggregates are also supplied; each has one nominal positive and zero BY retentions.

Only model-level statistics, counts and methodological limitations are included. The underlying observations, entity identifiers, dates, predictions, residuals, monthly totals, training data and private loaders remain undistributed.

Table 17: All thirty restricted-case models on calendar-complete evaluation records. n is the number of firm-months; M counts months. Adjusted values use the entire thirty-model family, separately for each exponent. Every adjusted value exceeds 0.05.

Model	n	M	T_1	BY p_1	T_2	BY p_2
Adversarial residual-ranking MLP	31045	132	2.453	0.848	2.484	0.779
Chronos-Bolt-tiny	21428	132	0.561	1.000	0.591	1.000
Chronos-Bolt-small	21428	132	0.607	1.000	0.640	1.000
Chronos-Bolt-base	21428	132	0.452	1.000	0.485	1.000
Chronos-Bolt-base (extended coverage)	23881	132	-0.034	1.000	0.001	1.000
Moirai-1.0-R-base	21428	132	-3.542	1.000	-3.546	1.000
Moirai-1.0-R-large	21428	132	-3.372	1.000	-3.358	1.000
Untrained forecasting network	21428	132	-3.033	1.000	-2.988	1.000
WPMixer	8922	132	-2.324	1.000	-2.355	1.000
MambaSimple	8922	132	-2.219	1.000	-2.231	1.000
MSGNet	8922	132	-1.967	1.000	-1.991	1.000
DLinear	14770	132	0.107	1.000	0.101	1.000
FiLM	2613	132	-2.644	1.000	-2.671	1.000
TiDE	2613	132	-0.650	1.000	-0.619	1.000
SegRNN	2613	132	-0.625	1.000	-0.664	1.000
PatchTST	8113	72	-3.230	1.000	-3.227	1.000
iTransformer	8113	72	-4.002	1.000	-4.024	1.000
TimesNet	8113	72	-1.221	1.000	-1.232	1.000
TimeMixer	8113	72	-3.694	1.000	-3.682	1.000
TimeXer	8113	72	-3.990	1.000	-3.987	1.000
TFT	15699	132	-0.594	1.000	-0.577	1.000
NHITS	8690	132	-1.107	1.000	-1.134	1.000
DTAF	8113	72	-4.602	1.000	-4.582	1.000
DUET	16014	132	-2.185	1.000	-2.184	1.000
FreqCycle	8113	72	-3.787	1.000	-3.779	1.000
TimeBridge	16014	132	-0.955	1.000	-0.943	1.000
FT-Transformer	31045	132	-6.863	1.000	-6.863	1.000
TabM	31045	132	-3.315	1.000	-3.316	1.000
ImplicitDecoding	8113	72	-3.023	1.000	-3.017	1.000
TimeFilter	16014	132	-1.538	1.000	-1.541	1.000

22 A common-support temporal reconstruction comparison

The historical thirty-model table above is not a common-sample comparison and does not repair training-time information use. We therefore conduct a separate reconstruction experiment with the actual adversarial residual-ranking MLP, Ridge regression, histogram gradient boosting and an unfit momentum control. It retains 18,036 identical evaluation records and 132 common months. No month is excluded for a constant score or insufficient comparison support. This is a descriptive retrospective comparison, not a fresh prospective holdout.

Eleven expanding annual folds use the previously specified evaluation years. At each fit, training origins and next-calendar-month target maturity must both precede or equal the cutoff. Missing labels are excluded, never filled. All models use the same twelve available features: preceding one-, three-, six- and twelve-month returns, preceding twelve-month volatility, the transform $\log(1 + \text{count})$ for all, buy and sell disclosures, buy share, mean and maximum disclosure delay, and same-day disclosure share. The exact features are reconstructed independently from the restricted source records. Finite training medians fill features, with zero only for an entirely missing training column; means and population standard deviations are then fitted on training rows. Zero training scales become one.

The neural model uses its existing raw-rank and residual-rank squared losses, gradient reversal for issuer and month heads, and training-group stability penalty. The loss weights are 0.25, 1, 0.1 and 0.1 respectively; the additional worst-group and conditional-value-at-risk terms are disabled. Predictions combine the raw and residual heads with weights 0.25 and 1. The network has two hidden layers of width 128 with layer normalization and GELU activations, dropout 0.1, batch size 1,024, AdamW learning rate 0.001, weight decay 0.0001 and gradient clipping at five. All 20 epochs are run for seeds 11, 23 and 47 in each fold. The primary neural prediction is the fixed arithmetic mean of these seeds. There is no evaluation-based early stopping or parameter selection.

Ridge uses penalty ten. Histogram gradient boosting uses squared-error loss, learning rate 0.05, 200 iterations, at most fifteen leaves, at least thirty training samples per leaf, regularization one and 255 bins, without early stopping. Both fit the same within-training-month rank target as the raw neural head. The unfit control uses preceding one-month return. Fixed settings do not imply exhaustively optimized comparators.

All 55 fits finish: 33 neural, eleven Ridge and eleven boosting fits. The primary metric gives each common month equal weight and then averages its Spearman correlation. Pooled correlation answers a different question. Table 18 retains both metrics and every neural seed. The custom objective does not yield a descriptive advantage in this comparison. No significance test, confidence interval, model selection or finite certificate is inferred from these correlations.

Table 18: Temporal reconstruction on identical support. Correlations are descriptive. The three-seed mean is fixed before fitting; individual seeds are retained. All models share 18,036 records and 132 monthly evaluations.

Model	Mean monthly	Pooled	Monthly SD
Adversarial residual-ranking MLP, mean	.002746	-.000346	.090433
MLP, seed 11	.002577	.013670	.090463
MLP, seed 23	-.000632	-.014784	.102442
MLP, seed 47	.003744	-.002502	.090673
Ridge regression	.056668	.040327	.129204
Histogram gradient boosting	.037945	.022672	.117709
One-month momentum	.023142	-.004441	.122953

Independent checks rebuild every aggregate correlation, verify all training cutoffs and transformations, and reproduce all 33 neural and eleven Ridge checkpoints exactly. Boosting predictions and metrics are independently checked, but those models are not refitted in that verification. No raw observations, date rows, entity identifiers, forecasts, fitted transformation values, checkpoints or loaders are distributed; only model-level aggregates and protocol descriptions are supplied.

Historical availability remains unresolved. Transaction uniqueness is filtered across the current full extract; identifiers and adjusted prices also come from current source fields. Population selection is consequently retrospective despite verified training-only transformations and label maturity. The twelve-feature experiment neither repairs every historical model nor verifies vendor vintages, input release times, a new-security population or implementable profits.

Table 19: Feedback checks on two fixed observation patterns. Outcomes are simulated under the conditional null. Both statistics use entity and period clusters; the ordered version evaluates middle folds. Entries are the largest absolute mean and largest positive rejection rate across the stated cells, each with 200 replications. These two maxima need not occur in the same cell.

Design	Standard folds		Ordered folds	
	mean	Rate	mean	Rate
31,045 rows, six cells	3.092	0.890	0.251	0.070
8,113 rows, six cells	1.686	0.505	0.303	0.095
One-month lookback, one cell	2.237	0.735	0.242	0.090

23 New design checks and reproduction map

An executable independent-entity benchmark The script `equal_entity_directional.py` uses entirely synthetic trajectories, deterministic folds and equal entity weights to examine the raw-score corollary in the main paper. It includes complementary, directional and simple gapped-complementary means on the same middle rows. The gapped rule excludes a lookback’s width on either side of each evaluation fold; it is not a complete implementation of a previously published estimator. There are 500 paired replications in each of 84 cells. Entity counts are 25, 100 and 400; trajectory lengths 20 and 40; lookbacks one and five. Positive controls inject 0.1 or 0.3 times the current outcome innovation into the score and are not prospective forecasts. With 400 entities, 20 observations and lookback five, complementary, gapped and directional mean statistics are -2.677 , 3.148 and -0.002 , respectively; positive rejection counts are 0, 458 and 27 out of 500. Directional null rates across the complete grid range from 0.054 to 0.088, so finite-sample calibration is not exact. The package includes all cell summaries and replication-level synthetic statistics, exact expected means, standard-error comparisons and intervals.

The directory `results/design_validation/` contains the peer-sampling, feedback, interaction, near-copy, conditional-power, clustering, runtime, bootstrap and pretrained-rerun aggregates. Its README describes the design and unit of every table. The MDE values are measured increases in mean monthly rank correlation; the injected coefficient in the conditional-power experiment is a different quantity. Null quantiles fitted on the same simulated null replications do not constitute independent calibration.

The synthetic script `panel_design_checks.py` verifies the exact feedback identities for general weights and contiguous folds, the closed form for finite lookback, a serial-dependence counterexample, and fixed- versus resampled-peer expectations including ties. The directional audit script additionally reports complementary and directional fits on both all folds and matched middle folds. It retains original bin assignments and cohort-level copy flags; empty training sides and entities absent from a training side are reported explicitly. User-cluster ordering in the ratings task is not time ordering.

For continuous margins, the rank-reference term also has the classical representation $\Gamma = (3\tau - \rho_S)/12$. In a tie-free group of size N , let J average $a_{ij}b_{ij}$ over ordered distinct pairs and Q average $a_{ij}b_{i\ell}$ over ordered distinct triples. Then $J = (1 + \hat{\tau})/4$, $Q = (3 + \hat{\rho}_3)/12$, and

$$C_g - D_g = \frac{3\hat{\tau} - \hat{\rho}_3}{12(N-1)}, \quad \hat{\rho}_S = \frac{(N-2)\hat{\rho}_3 + 3\hat{\tau}}{N+1}.$$

Here $\hat{\rho}_3 = 12Q - 3$ is the degree-three U-statistic version of Spearman’s rho. This classical decomposition is attributed to the rank-statistic literature [5, 6]; it is not claimed as a new rank-correlation identity.

24 Paired factorial attribution of certificate efficiency

This section completes the missing absolute-allowance plus variance-bound comparison on the already inspected synthetic streams. It is a *post-exposure ablation*, not an independent new confirmation. The additional analysis protocol fixes all original settings and the reported paired contrasts before the added results are computed. The original study and every previously published result remain available.

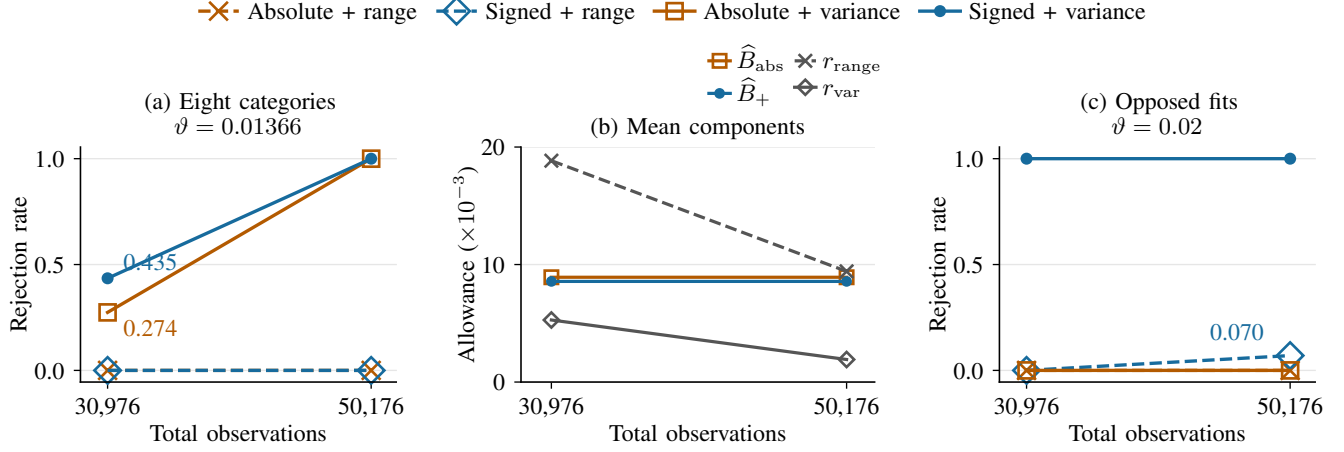


Figure 11: Complete factorial comparison. Four constructions share observations and 1,000 repetitions: (a) eight-category detection; (b) mean learning and sampling allowances; (c) detection with opposite fitted-mean shifts in two categories. Training uses 8,192 rows, validation 8,192 pairs, and 64 peers per group. Both pair members count toward cost; lines join only tested budgets.

Target and common information. Writing R_X, R_Y for population marginal midranks, including ties, the common target is

$$\vartheta = \mathbb{E}[(R_X - \mathbb{E}[R_X | C])(R_Y - \mathbb{E}[R_Y | C])].$$

Conditional on independent fitted category means f_c, g_c , the same order-three full-U center $\hat{U}$ has expectation $\vartheta + b_H$, where $b_H = \sum_c \pi_c(\mu_{Xc} - f_c)(\mu_{Yc} - g_c)$. The four lower bounds are

$$L_{s,t} = \hat{U} - \hat{B}_s - r_t, \quad s \in \{\text{abs}, +\}, \quad t \in \{\text{range}, \text{var}\}.$$

They share all training, validation and evaluation draws, the fitted kernel range, and the full-U center. The two variance constructions also share the same sample variance of disjoint symmetric triples and the same $J = M \lfloor N/3 \rfloor$. Only the original absolute allowance is substituted for the signed allowance in the newly completed cell. We abbreviate the four constructions AR, SR, AV and SV in that order.

The absolute comparator is exactly the original one. On the common validation rectangles, put

$$e_c = \max_{x \in \{L_{Xc}, U_{Xc}\}} |x - f_c| \max_{y \in \{L_{Yc}, U_{Yc}\}} |y - g_c|.$$

Then $\hat{B}_{\text{abs}} = \max_{\pi \in \mathcal{P}} \sum_c \pi_c e_c$, where $\mathcal{P}$ is the original capped probability simplex. The signed allowance instead uses the largest signed corner product, with the identical rectangles, probability caps and allocation of δ . Empirical category weights, a worst-cell maximum and an absolute value of the summed product would be different comparators. The range and variance radii retain their original constants and error allocation; the variance inversion's factor two is retained. No assertion is made that these two sampling bounds are pointwise ordered for arbitrary inputs.

Complete grid and observation cost. All 100 original settings and 1,000 paired replications remain: the two- and eight-category laws, all five signals including zero and negative signals, $M \in \{25, 100, 400, 1600\}$, $N = 64$, both original validation budgets for estimated fits, and all original two-category opposed-shift settings. There are 400,000 method rows and 400 method-setting cells. The four constructions use 8,192 training rows, and their common charged cost is $8,192 + 2n_{\text{val}} + 64M$.

The displayed budgets with $n_{\text{val}} = 8,192$ are 30,976 and 50,176. The original 40 nonpositive-target settings give 160 factorial cells; all have zero observed rejections. All 400 cells have zero observed lower-bound coverage failures. These observations do not imply zero risk.

Table 20: Completed paired factorial counts out of 1,000. All displayed cases use 8,192 validation pairs.

Law / fit	ϑ	Cost	AR	SR	AV	SV
Two / estimated	0.010	50,176	58	78	1,000	1,000
Eight / estimated	0.013655	30,976	0	0	274	435
Eight / estimated	0.013655	50,176	0	0	1,000	1,000
Two / opposed	0.020	30,976	0	0	0	1,000
Two / opposed	0.020	50,176	0	70	0	1,000

Paired intervals and attribution. For decisions A, B , let N_+ count $B = 1, A = 0$ and N_- count $B = 0, A = 1$. Each is marginally binomial, although the two counts are dependent. Two two-sided 97.5% Clopper–Pearson intervals for the discordant probabilities jointly cover with probability at least 95% by the union bound. Subtracting opposite endpoints therefore gives a conservative exact 95% interval for the paired probability difference. These are pointwise intervals for each fixed setting and contrast; there is no simultaneous claim for the 600 intervals in the artifact. They are neither independent-proportion intervals nor McNemar tests presented as confidence intervals.

Table 21: All six paired differences in the two highlighted estimated-fit settings. Entries are differences in rejection probability and conservative exact pointwise 95% intervals.

After minus before	Two, cost 50,176	Eight, cost 30,976
SR–AR	0.020 [0.007, 0.032]	0.000 [−0.004, 0.004]
AV–AR	0.942 [0.919, 0.957]	0.274 [0.238, 0.307]
SV–AR	0.942 [0.919, 0.957]	0.435 [0.395, 0.471]
AV–SR	0.922 [0.897, 0.940]	0.274 [0.238, 0.307]
SV–SR	0.922 [0.897, 0.940]	0.435 [0.395, 0.471]
SV–AV	0.000 [−0.004, 0.004]	0.161 [0.131, 0.189]

Thus the two-category increase from 58 to 1,000 is largely a sampling-bound improvement. The nonsaturated eight-category case isolates a further signed-allowance benefit under the variance bound: 161 favorable and zero unfavorable discordant replications, with 274 common rejections. Its paired increment is 0.161, with interval [0.131379, 0.188805]. At the lower opposed-fit budget, changing only the learning allowance under the range bound or only the sampling bound under the absolute allowance yields no additional decisions, whereas combining both yields 1,000; the SV–AV interval is [0.991255, 1].

Every continuous lower-bound difference decomposes exactly:

$$L_{+, \text{var}} - L_{\text{abs}, \text{range}} = (\hat{B}_{\text{abs}} - \hat{B}_+) + (r_{\text{range}} - r_{\text{var}}).$$

For the nonsaturated eight-category case the mean contributions are approximately 0.000354 from learning and 0.013557 from sampling. Mean absolute/signed allowances are 0.008918/0.008564 at both budgets; mean range/variance radii fall from 0.018838/0.005281 to 0.009419/0.001918. The training and validation draws are unchanged between these budgets. A small allowance difference can change decisions near the threshold, while thresholding itself creates interactions.

The paired interaction is $D = (SV - AV) - (SR - AR)$. Pointwise signed-allowance monotonicity implies $D \in \{-1, 0, 1\}$, so the same discordant-probability interval applies. For the two highlighted cases its values are respectively -0.020 with interval $[-0.032376, -0.007010]$, and 0.161 with interval $[0.131379, 0.188805]$. Order-averaged learning and sampling contributions have supports $[0, 1]$ and $[-1, 1]$. The artifact uses pointwise Hoeffding intervals with half-widths $\sqrt{\log(40)/(2n)}$ and $\sqrt{2\log(40)/n}$, respectively, and truncates to these known supports. For the eight-category case the contributions are 0.0805 [0.037553, 0.123447] and 0.3545 [0.268606, 0.440394]. Across the entire grid, the two factor orders give $1,348 + 10,217 = 8,340 + 3,225 = 11,565$ extra decisions. These totals are descriptive because different settings share draws; they have no pooled-binomial interval or unique causal allocation.

Persistent multicategory errors. For the actual two-category $\vartheta = 0.02$ opposed-shift setting, the true category means of both rank variables are (0.35, 0.65). In original training replication zero, the deployed shifted fits, rounded to eight decimal

places, are

$$f \simeq (0.49982599, 0.80113030), \quad g \simeq (0.20166229, 0.49928231).$$

The unrounded fits give $b_H = -0.0225014272$ and $\mathbb{E}\hat{U} = -0.0025014272$. These are one random training replication, not fixed fits imposed on all replications. A concise algebraic illustration fixes $f = (0.5, 0.8)$ and $g = (0.2, 0.5)$, for which both category product errors equal -0.0225 . At $\vartheta = 0.02$, vanishing uncertainty makes the signed lower bound tend to 0.02 but the absolute lower bound to -0.025 . In general the absolute limit loses $2 \sum_c \pi_c \{-(\mu_{Xc} - f_c)(\mu_{Yc} - g_c)\}_+$. This explains a persistent-error distinction; it does not establish superiority over fit-free conditional estimators.

Stronger classical comparisons and evidence scope. The artifact retains all 32 existing matched-comparator settings, 300 paired replications, eight methods and 128 specified contrasts. At the weaker two-category $\vartheta = 0.005$ setting with cost 30,976, SV, independent Bernstein, independent betting, pooled-U and pooled betting reject 1/10/263/2/273 out of 300; AV also rejects 1. Independent betting minus SV is 0.873333, with paired interval $[0.809435, 0.913159]$. The denominator differs from the 1,000-replication factorial table. Test-only betting has no plotted lower-confidence-bound radius. The classical Bernstein and betting constructions are not new methods, and these data refute a general dominance interpretation of the full-U variance certificate.

The portable artifact at `results/certificate_factorial` includes all 400,000 rows, the original 500,000-row inputs, all 48,000 matched-comparator input rows, exported scientific protocols and historical hashes, the pre-execution serialization-correction trace, and independent verification of all rows, all paired intervals, and selected original primitives. Its reproducible compact figure uses only settings declared before these added results. The publication rendering changes notation and leading zeros, without changing the data. The older public study and its broader grid remain available.

25 Public hourly forecast selection and chronological confirmation

This study compares association certificates with separately measured forecast losses on two public tasks. The forecasting rules, controls, inference methods, budgets and primary comparisons were fixed before source retrieval. The complete results include baseline fallbacks and unsuccessful primary comparisons. Independent sampling from each fixed selection archive supports its stated finite-archive association target; it does not make the original time series independent or certify a later forecast-loss target.

25.1 Sources, preparation and chronology

Appliances Energy Prediction, contributed by Luis Candanedo (2017), contains 19,735 ten-minute records [25]; its original ZIP is <https://archive.ics.uci.edu/static/public/374/appliances+energy+prediction.zip> (DOI <https://doi.org/10.24432/C5VC8G>). Metro Interstate Traffic Volume, contributed by John Hogue (2019), contains 48,204 records [26]; The ZIP is available at <https://archive.ics.uci.edu/static/public/492/metro+interstate+traffic+volume.zip> (DOI <https://doi.org/10.24432/C5X60B>). Both source datasets and the distributed derived hourly series, numeric forecast arrays and result tables are under CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>. Redistribution retains author attribution and records aggregation, reindexing, feature construction, clipping and model predictions as changes. Code is separately covered by the release’s MIT license. Original ZIP bytes and their SHA256 values are supplied; neither private data nor original model pickles are needed by this study’s public reproduction.

Appliances predicts energy use in the next complete nominal hour, in Wh, by summing six available ten-minute readings. An incomplete hour has a missing target. Metro predicts next-hour traffic volume at ATR301; duplicate nominal source timestamps are averaged. Its 7,629 duplicate source rows have no disagreements in the target. Reindexing the full recorded Metro calendar gives 52,551 hourly slots, of which 11,976 have no target. Missing slots remain missing; they are not removed before constructing lags. Source timestamps retain their nominal local interpretation without an unverified daylight-saving conversion.

For Appliances, T_0 is the first midnight at or after the first source timestamp, here 12 January 2016. The four half-open windows are $[T_0, T_0 + 56d)$, $[T_0 + 56d, T_0 + 70d)$, $[T_0 + 70d, T_0 + 84d)$ and $[T_0 + 84d, T_0 + 126d)$. For Metro, the nominal windows are before 1 January 2016, January–June 2016, July–December 2016 and calendar 2017; data from 2018 are unused. The source begins on 2 October 2012, so there are no artificial training rows before that first observation. Table 22 gives observed support. The fixed eligibility checks require at least 500 training targets and at least 75% of nominal hours in each later window; failure would be retained, without replacing the task or moving a window.

Table 22: Fixed chronological windows. Dates in the displayed ranges are inclusive; windows in the implementation are half-open at the next midnight.

Task	Stage	Observed target dates	Rows
Appliances	Training	12 Jan–7 Mar 2016	1,344
	Weight calibration	8–21 Mar 2016	336
	Selection	22 Mar–4 Apr 2016	336
	Confirmation	5 Apr–16 May 2016	1,008
Metro	Training	2 Oct 2012–31 Dec 2015	17,491
	Weight calibration	1 Jan–30 Jun 2016	3,476
	Selection	1 Jul–31 Dec 2016	4,362
	Confirmation	1 Jan–31 Dec 2017	8,713

Information time. Predictions for hour t use measurements from hours strictly before t , and known calendar variables. Completed measurements are *assumed* available at the end of their measurement hour, with no additional release delay. Historical data vintages and this publication-delay assumption have not been verified. Appliances’ interpolated airport variables and source random columns are excluded; Metro’s target-hour weather and weather text are excluded. Past confirmation outcomes can enter later lag features, as in an online forecast, while fitted models and selected rules stay fixed. The source parser necessarily materializes historical confirmation labels; the restriction concerns the fixed functions that produce forecasts and selections, not physical inability to access the downloaded file. Perturbing current and later targets leaves every feature through the current forecast origin unchanged in the implemented timing checks.

25.2 Forecasts, controls and frozen augmentation

All models use exact-calendar outcome lags 1, 2, 3, 6, 12, 24, 48, 168 hours; means, population standard deviations and observed-slot fractions over the preceding 24 and 168 slots (at least one observed slot); sine/cosine hour, weekday and day-of-year features; and a weekend indicator. The annual cycle uses 365.25 days. The richer feature set adds previous-hour Appliances temperatures T1–T9, humidities RH1–RH9 and lighting consumption, or Metro temperature, one-hour rain/snow and cloud coverage. Numeric covariates are aggregated at their own measurement hour and then shifted by one hour. Missing predictors receive training medians with missingness indicators; entirely missing training columns are retained with zero imputation.

The two baselines are the previous day’s same-hour target and Ridge on lag/rolling/calendar features. Ridge is the primary baseline. The eight candidates are persistence, daily seasonality, weekly seasonality, Ridge on the richer features, histogram gradient boosting, ExtraTrees, a category-only copy and independent uniform noise. Daily and weekly forecasts fall back to lag one and then the training median if needed. Ridge uses standardized imputed features, $\alpha = 10$ and an SVD solver. Boosting uses 200 iterations, 15 leaves, learning rate .05, minimum leaf size 20, $L_2 = .01$ and no early stopping. ExtraTrees uses 256 trees, maximum depth 12, minimum leaf size 10, all features and one execution thread. Boosting and ExtraTrees use the richer imputed feature matrix. The initial models are never refitted.

Let $s = \max\{1, \hat{Q}_{.99}(Y \mid \text{training})\}$ and $[x]_s = \min\{s, \max\{0, x\}\}$. Here $s = 2477.1$ Wh for Appliances and 6729.2 traffic units for Metro; the untruncated training medians used for fallback are 360 and 3349. Model fitting uses $[Y]_s$, and delivered forecasts are clipped to $[0, s]$. The primary loss is

$$\ell_s(y, p) = \{[y]_s - [p]_s\}^2 / s^2 \in [0, 1]. \quad (29)$$

Raw MSE and MAE compare the same delivered clipped prediction with the *untruncated* outcome; normalized MAE compares the clipped outcome and prediction, divided by s . Thus raw MSE is not merely the displayed bounded MSE multiplied by s^2 when outcome clipping is active.

For each baseline b and candidate f , calibration chooses the smallest minimizer of mean (29) for $[b + t(f - b)]_s$, with $t \in \{0, .05, \dots, 2\}$. A separate convex rule uses $t \in \{0, .05, \dots, 1\}$. All weights are frozen before selection metrics. The exact complete weight tables remain in the release. The selected ExtraTrees weights for Appliances/seasonal, Appliances/Ridge, Metro/seasonal and Metro/Ridge are respectively 1.10, 1.00, 1.05, 1.10; the four corresponding convex weights are all one.

Each baseline defines 32 declared categories: 16 training-baseline forecast quantile bins crossed with weekend status. All fifteen breakpoints are retained, including ties; equality is assigned to the lower bin. Breakpoints for fitted Ridge may use its training predictions, and are fixed before calibration/selection. Empty cells remain declared. Category-copy predictions are $s(c + .5)/32$ for category $c \in \{0, \dots, 31\}$, so they have exact zero conditional-rank target. These controls are stronger than four coarse calendar/sign cells but are still not the full continuous baseline or the full predictor feature vector.

The master seed is 2026091201. With task index $j = 0$ for Appliances and $j = 1$ for Metro, boosting and ExtraTrees use seeds $2026091201 + j + 10$ and $2026091201 + j + 20$; noise uses $2026091201 + j + 100$. Its draws are uniform on $[0, s]$ over the full recorded calendar. Sampling uses a NumPy SeedSequence with seed $2026091201 + j + 1000$, spawning three independent generators for training, validation and evaluation. The same task-level indices are reused across baselines, candidates and methods.

25.3 Sampling targets, five gates and complete decisions

Condition on the fitted forecasts and complete fixed selection archive of n rows, and let P_n be the uniform law on its rows. If $R_f(f) = P_n(f' < f) + \frac{1}{2}P_n(f' = f)$ is the marginal midrank, the four association gates target

$$\vartheta_n = E_{P_n}[(R_f - E_{P_n}[R_f \mid C])(R_y - E_{P_n}[R_y \mid C])]. \quad (30)$$

The rank inputs are unaugmented candidate forecasts and original outcome values; the selected augmentation is assessed on bounded loss separately. The direct-loss gate instead targets the fixed archive mean of $\ell_s(Y, b) - \ell_s(Y, [b + t(f - b)]_s)$, in $[-1, 1]$. Neither target is a later-period population-risk parameter.

Every gate uses 32,768 independently drawn uniform row indices with replacement. Reference U (RU) and reference betting (RB) allocate 4,096 to conditional-mean fitting, 8,192 independent pairs to validation and 12,288 to evaluation. RU uses the complete order-three distinct-reference center and the signed variance-sensitive bound; its kernel-variance sample uses 4,096 disjoint triples. RB uses the same validation allowance and independent triple scores with an equal mixture of 64 fixed geometric betting fractions from 10^{-4} to .99 [27]. Its reported value is $\min\{1, \delta + 1/E\}$, not an unadjusted e-BH procedure. A test-only betting result has no borrowed U-bound interval. For both methods the prespecified validation allocation is

$$\delta = \frac{0.1 \times .05}{8H_8}, \quad H_8 = \sum_{k=1}^8 k^{-1}. \quad (31)$$

Known-forecast U (KU) and known-forecast independent pairs (KP) additionally use the complete forecast and category metadata to center forecast midranks exactly, avoiding mean-learning error. Writing $R_X = F_X(X)$, $R_Y = F_Y(Y)$ and $m_X(Z) = \mathbb{E}[R_X | Z]$, exact centering gives

$$\mathbb{E}[(R_X - m_X(Z))R_Y] = \mathbb{E}[(R_X - m_X(Z))(R_Y - \mathbb{E}[R_Y | Z])] = \vartheta.$$

The omitted conditional-outcome mean contributes zero because $\mathbb{E}[R_X - m_X(Z) | Z] = 0$. This uses complete forecast-rank and category metadata; knowing only the baseline forecast does not supply it. Their classical Gini-covariance kernel [28] and fixed-frame range use all 32,768 sampled outcomes for evaluation: KU uses the complete order-two average and KP uses 16,384 disjoint pairs. Both retain ties. The direct-loss gate (LG) uses all 32,768 bounded gain observations and the classical independent-observation empirical-Bernstein bound [17]. KU, KP and LG have no validation-failure allocation. The release specifies all constants and inversions, including zero-range conventions.

Each of the five methods applies BY at .05 to the same complete eight candidates within each task/baseline; category-copy and noise are not removed after inspection. There is no pooled selection of the best gate and no claim of one global FDR guarantee over all methods and tasks. The rule chooses the retained augmentation with the smallest selection bounded MSE if it strictly improves on the baseline; otherwise baseline wins. Ungated augmentation (UG) compares all eight fitted augmentations without a gate; convex combination (CV) compares their separately calibrated convex versions. Baseline alone (B) completes the eight rules.

Table 23: All 32 fixed decisions and confirmation losses. ET denotes ExtraTrees augmentation and B the applicable baseline; t is its frozen weight ($t = 0$ on fallback). K_r is the number of BY-retained candidates for a gate, and a dash denotes a rule without a test. Norm. losses use the clipped training scale; raw losses use untruncated outcomes. All values are rounded only for display.

Rule	Forecast	t	K_r	Norm. MSE	Raw MSE	Norm. MAE	Raw MAE
<i>Appliances / seasonal baseline</i>							
RU	B	0.00	0	0.04129840	259,901.751	0.11114419	277.022
RB	B	0.00	0	0.04129840	259,901.751	0.11114419	277.022
KU	ET	1.10	5	0.01774890	114,034.668	0.08180081	204.335
KP	ET	1.10	6	0.01774890	114,034.668	0.08180081	204.335
LG	ET	1.10	6	0.01774890	114,034.668	0.08180081	204.335
UG	ET	1.10	–	0.01774890	114,034.668	0.08180081	204.335
CV	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
B	B	0.00	–	0.04129840	259,901.751	0.11114419	277.022
<i>Appliances / Ridge baseline</i>							
RU	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
RB	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
KU	ET	1.00	5	0.01727688	111,262.587	0.08055088	201.239
KP	ET	1.00	5	0.01727688	111,262.587	0.08055088	201.239
LG	B	0.00	0	0.02080396	132,473.153	0.10627496	264.960
UG	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
CV	ET	1.00	–	0.01727688	111,262.587	0.08055088	201.239
B	B	0.00	–	0.02080396	132,473.153	0.10627496	264.960
<i>Metro / seasonal baseline</i>							
RU	B	0.00	0	0.02241721	1,016,372.888	0.08188022	552.156
RB	B	0.00	0	0.02241721	1,016,372.888	0.08188022	552.156
KU	ET	1.05	6	0.00143846	66,073.700	0.02440903	165.421
KP	ET	1.05	6	0.00143846	66,073.700	0.02440903	165.421
LG	ET	1.05	5	0.00143846	66,073.700	0.02440903	165.421
UG	ET	1.05	–	0.00143846	66,073.700	0.02440903	165.421
CV	ET	1.00	–	0.00140356	64,508.684	0.02396282	162.418
B	B	0.00	–	0.02241721	1,016,372.888	0.08188022	552.156
<i>Metro / Ridge baseline</i>							

Rule	Forecast	t	K_r	Norm. MSE	Raw MSE	Norm. MAE	Raw MAE
RU	B	0.00	0	0.00525213	238,913.748	0.05021910	339.102
RB	B	0.00	0	0.00525213	238,913.748	0.05021910	339.102
KU	ET	1.10	3	0.00137224	63,077.041	0.02370708	160.698
KP	ET	1.10	3	0.00137224	63,077.041	0.02370708	160.698
LG	ET	1.10	2	0.00137224	63,077.041	0.02370708	160.698
UG	ET	1.10	–	0.00137224	63,077.041	0.02370708	160.698
CV	ET	1.00	–	0.00140356	64,508.684	0.02396282	162.418
B	B	0.00	–	0.00525213	238,913.748	0.05021910	339.102

The reference methods fall back in all four task/baseline cases. KU and KP choose the same delivered ExtraTrees augmentation as UG in every case; LG also chooses it except for Appliances/Ridge, which falls back. For the primary Ridge comparisons, the augmentation reduces bounded MSE by 16.95% and 73.87%, and raw MSE by 16.01% and 73.60%, relative to baseline. Its *additional gain from gating is zero* because the ungated forecast is identical. Table 24 also gives the seasonal comparisons. CV differs because its calibration weights are restricted; with the seasonal baseline it slightly improves on UG in these confirmation windows. These are observed-support effects, not time-series confidence bounds.

Table 24: Confirmation MSE gains for the common KU/KP/UG forecast. Positive values favor that forecast. Its gain over UG is identically zero in all four rows. Gains against CV may have either sign.

Task	Baseline	Norm. gain vs B	Raw gain vs B	Norm. gain vs CV	Raw gain vs CV
Appliances	Seasonal	0.02354950	145,867.083	-0.00047202	-2,772.080
Appliances	Ridge	0.00352708	21,210.566	0.00000000	0.000
Metro	Seasonal	0.02097874	950,299.187	-0.00003490	-1,565.016
Metro	Ridge	0.00387988	175,836.707	0.00003132	1,431.644

The original primary criterion required RU to improve on *both* UG and CV under Ridge on *both* tasks. It is not met on either task (0/2); this criterion was not replaced by KU’s or KP’s better result. These are two new dataset tasks within the recorded exposure search, not proof that every researcher had never seen their data or that energy and transport had never appeared as broad subject domains. After the results were examined, new methods or configurations on these same windows would be exploratory rather than a new independent confirmation.

25.4 Full access cost and the learning-budget limitation

The raw source files, all observed targets, training forecasts, calibration losses and the complete selection archives are already materialized. The observed training/calibration/selection/confirmation supports total 3,024 Appliances and 34,042 Metro targets; the downloaded sources also contain unused rows. Source preparation builds forecasts over the recorded calendars, including unused periods. Model fitting and weight calibration are counted separately from rank-certificate queries. Of 32,768 replacement calls, the common stream touches all 336 Appliances selection rows and 4,361 of 4,362 Metro selection rows. The call count is respectively 97.52 and 7.51 times the selection census size. These repeated calls are not newly acquired labels. Streams are shared across methods/candidates, so multiplying calls by every method is not a physical acquisition cost either.

The known-forecast methods scan complete forecast/category metadata and allocate all sampled labels to evaluation; the reference methods have less exact forecast information and spend labels learning means and validating their errors. Equal label-call budgets therefore do not imply equal information or a one-component ablation. Computing the exact archive association uses the complete outcome census only as a diagnostic and is not supplied to a gate. Because selection already uses every selection label, no label-acquisition advantage over a census is established.

For ExtraTrees under the primary Ridge baselines, RU’s observed means are .00994791 and .00408298, its signed learning allowances .03810465 and .03704776, and its single-test .05 sampling radii .00519920 and .00505665. The exact finite-archive learning biases are only .00007855 and .00007659. All 32 unadjusted RU p-values and all 32 RB p-values are one, before BY. Holding these fitted means, validation samples and realized allowances fixed, adding evaluation observations alone cannot remove their allowance barrier. This does not prove failure of all redraws at the same budget. Substituting census bias after inspection is a diagnostic, not success of the fixed protocol or information available under its original training/validation allocation.

25.5 Four bounds allowing dependent confirmation hours

The four prespecified contrasts compare RU with UG and with CV, under Ridge on each task. Positive gain favors RU. These bounds concern a *same-period conditional availability-weighted average*; they are separate from (30) and from the descriptive MSE on observed rows. Let $T = 1008$ for Appliances and $T = 8760$ for Metro, including every nominal confirmation hour. Write I_t for target availability, and let $\mathcal{F}_{t-1}$ contain the training, calibration, selection and sampling information and all observations available before hour t . The selected prediction functions and their delivered predictions at t are $\mathcal{F}_{t-1}$ -measurable under the stated hour-end availability assumption. Set

$$D_t = I_t\{\ell_s(Y_t, p_t^{\text{comp}}) - \ell_s(Y_t, p_t^{\text{RU}})\} \in [-1, 1], \quad \mu_t = E[D_t \mid \mathcal{F}_{t-1}], \quad (32)$$

with $D_t = 0$ when the target is missing. An absent target is not imputed. The target is $\bar{\mu}_T = T^{-1} \sum_t \mu_t$, including availability inside the conditional expectation. No missing-at-random assumption, stationarity, independence of hourly observations or independence across the four contrasts is required.

For $\alpha_0 = .05/4 = .0125$, the reported lower bound is

$$L_T = \frac{1}{T} \sum_{t=1}^T D_t - \sqrt{\frac{2 \log(1/\alpha_0)}{T}}. \quad (33)$$

Conditional Hoeffding's lemma [18], applied to a variable in an interval of width two, gives for $\lambda > 0$

$$E[\exp\{\lambda(D_t - \mu_t)\} \mid \mathcal{F}_{t-1}] \leq \exp(\lambda^2/2).$$

Iterating conditional expectations and using Markov's inequality yields

$$P\left\{T^{-1} \sum_t (D_t - \mu_t) \geq r\right\} \leq \inf_{\lambda > 0} \exp\{-\lambda T r + T \lambda^2/2\} = \exp(-T r^2/2).$$

Substituting the radius in (33) gives $P\{L_T > \bar{\mu}_T\} \leq \alpha_0$. A union bound covers all four fixed conditional averages simultaneously with probability at least .95. This is a classical fixed-horizon bound, not an empirical-Bernstein time-series claim or an optional-stopping guarantee. The availability eligibility check does not license conditioning this coverage assertion on favorable observed support; failed tasks may not be silently removed.

Table 25: All four dependent confirmation bounds. Missing Metro hours contribute zero and the denominator is 8,760, not 8,713. Each bound has $\alpha_0 = .0125$; their joint coverage is at least .95 under the stated filtration. No lower bound is positive.

Task	Comparator	Nominal/observed hours	Calendar gain	Radius	Lower bound
Appliances	UG	1008/1008	-0.00352708	0.09324429	-0.09677137
Appliances	CV	1008/1008	-0.00352708	0.09324429	-0.09677137
Metro	UG	8760/8713	-0.00385907	0.03163009	-0.03548916
Metro	CV	8760/8713	-0.00382791	0.03163009	-0.03545801

The four nonpositive bounds give no positive conditional-average gain certificate; they are not a significance claim that the method harms forecasting. They do not cover an invariant future population, later years, unbounded raw squared loss or a rank-to-risk implication. The 944 weekly records in the release are descriptive summaries, not independent blocks to which an iid empirical-Bernstein formula is applied.

25.6 Exposure, mechanical correction and independent reproduction

The original local protocol and implementation were fixed at 17:02:15 UTC on 12 September 2026; source retrieval began afterward and completed at 17:02:29 UTC. This is a recorded local sequence, not an external preregistration or a trusted third-party timestamp. The forecasting/selection functions were fixed before the confirmation outcomes were inspected. Downloading and parsing all source labels does not create a claim that confirmation labels were physically inaccessible. All fixed tasks, methods and adverse outcomes are retained.

The first numerical implementation subtracted floating-point category means from forecast ranks. Category-constant scores acquired tiny nonzero residuals, which were amplified by a tiny computed kernel range, causing six erroneous category-copy rejections across KU and KP. The correction forms exact integer numerators: if $q_i = 2 \text{rank}(f_i) - 1$, category c has size n_c , and $S_c = \sum_{j: C_j=c} q_j$, then

$$A_i = \frac{n_{C_i} q_i - S_{C_i}}{2nn_{C_i}}.$$

Category-constant forecasts are now identically zero, including ties. The correction changes no model, category, split, seed, draw, coefficient or primary comparison. It removes the six false rejections; all 32 chosen rules and all four dependent confirmation bounds are unchanged. It was made after outcome exposure and is a mechanical replay of the same study, not a second independent confirmation. The defective computation is not on the public default reproduction path.

The public `results/forecast_confirmation` directory contains the original ZIPs, frozen numeric arrays, six complete result tables, protocol, source-to-export digest mapping, regression checks and separate producer and independent reconstruction code. Reproduction uses a new output directory and loads no historical model pickle. Python 3.12.11 with scikit-learn 1.7.2 rebuilds the original 70 forecast-array fields and eight sampling-index arrays exactly, and all non-runtime scientific table columns are exactly equal. The original fitting environment used scikit-learn 1.6.1; newly fitted models are saved for the installed version. Independent arithmetic additionally reconstructs all 160 gate rows, 32 choices, 32 confirmation losses, 32 candidate losses, 944 weekly rows and four dependent bounds without importing producer helpers. It checks source aggregation, scales, seasonal features, categories, weights, rank products, allowances, p-values and BY decisions. That independent arithmetic uses stated numerical tolerances, reported separately from the exact source-replay equality. No certificate is inferred from the mere agreement of two implementations.

26 Joint rank-mean constraints and an auditable numerical allowance

This section preserves the joint rank-mean allowance and records its complete exposed-archive diagnosis. The information added to the previous signed allowance is the exact mean of each population marginal midrank. Bilinear convexification, linear programming, weak duality and the sampling inequalities remain classical tools. The result is a conservative outer allowance, with no claim of sharpness or general optimality.

26.1 Original joint allowance and complete proof

The rectangular validation construction in the main paper gives intervals for each conditional midrank mean and category-mass caps u_c . Let q_c^+ be the largest endpoint product. This subsection retains the joint refinement and its proof, including its non-attainability example.

Lemma 2 (Joint rank-mean allowance). *Let (x_{ct}, y_{ct}) , $t = 1, \dots, 4$, be category c 's rectangle corners and $q_{ct} = (x_{ct} - f(c))(y_{ct} - g(c))$. Define*

$$\begin{aligned} B_{\text{joint}} &= \max_{\lambda \geq 0} \sum_{c,t} \lambda_{ct} q_{ct}, \\ \sum_t \lambda_{ct} &\leq u_c, \quad \sum_{c,t} \lambda_{ct} = 1, \\ \sum_{c,t} \lambda_{ct} x_{ct} &= \sum_{c,t} \lambda_{ct} y_{ct} = 1/2. \end{aligned} \tag{34}$$

On the same validation event, $b_H \leq B_{\text{joint}} \leq \widehat{B}_+$.

Midranks average one half, including ties, coupling the category means. This classical bilinear relaxation [29] may replace $\widehat{B}_+$ without another error budget. It is an outer bound, not generally attainable. Exact dual arithmetic verifies the numerical upper bound; empty sets or numerical failure return the separable bound.

Proof of Lemma 2. The identity $a(v, v') + a(v', v) = 1$ gives both marginal means $1/2$. On the validation event, bilinear interpolation expresses each true $(m_V(c), m_W(c), m_V(c)m_W(c))$ as a convex combination of the four lifted corners. Multiply its weights by p_c . The resulting λ is feasible in Eq. (34) and has objective b_H . Also $q_{ct} \leq q_c^+$, so every feasible objective is at most $\widehat{B}_+$. Zero-mass categories and repeated corners cause no change. The set can be a strict relaxation: one category, intervals $[0, 1]$ and $f = g = 1/2$ permit $B_{\text{joint}} = 1/4$ although the true bias is zero. No extra validation event is used.

Population and confidence event Condition on independent training information. Write $p_c = \Pr(C = c)$, $\mu_{Xc} = E[R_X | C = c]$ and $\mu_{Yc} = E[R_Y | C = c]$, where R_X, R_Y are marginal population midranks under one common law. The comparison $a(x, x') = \mathbf{1}\{x > x'\} + \frac{1}{2}\mathbf{1}\{x = x'\}$ satisfies $a(x, x') + a(x', x) = 1$. Hence, including ties,

$$\sum_c p_c \mu_{Xc} = \sum_c p_c \mu_{Yc} = \frac{1}{2}. \tag{35}$$

References must be independent draws from the same global law. This identity does not assert that category-specific reference ranks have the same target. For fixed training fits $f_c, g_c \in [0, 1]$, the distinct-reference statistic has expectation $\vartheta + b$, with $b = \sum_c p_c (\mu_{Xc} - f_c)(\mu_{Yc} - g_c)$. Conditional independence of X and Y is not required.

Let the simultaneous validation event have probability at least $1 - \delta$, and on it let $\ell_c \leq p_c \leq h_c$, $L_{Xc} \leq \mu_{Xc} \leq U_{Xc}$ and $L_{Yc} \leq \mu_{Yc} \leq U_{Yc}$. All endpoints lie in $[0, 1]$. The archived comparison keeps the previous mean rectangles and one-sided mass caps, so $\ell_c = 0$; no validation observation or error allocation changes. The argument also permits valid mass lower bounds. Empty declared categories remain in the confidence set.

Compact corner-mass formulation Let (a_{cj}, b_{cj}) , $j = 1, \dots, 4$, be the four corners of the mean rectangle for category c , allowing repeated corners. Define

$$\begin{aligned} B_{\text{MC}} &= \max_{\lambda \geq 0} \sum_{c,j} \lambda_{cj} (a_{cj} - f_c)(b_{cj} - g_c), \\ \text{subject to } \sum_{c,j} \lambda_{cj} &= 1, \quad \ell_c \leq \sum_j \lambda_{cj} \leq h_c \quad \text{for all } c, \\ \sum_{c,j} \lambda_{cj} a_{cj} &= \sum_{c,j} \lambda_{cj} b_{cj} = \frac{1}{2}. \end{aligned} \tag{36}$$

The auxiliary corner masses are not additional observed categories. On the validation event, the product of the two endpoint interpolation weights, multiplied by p_c , represents the true pair of conditional means and their product. These masses are feasible and have objective b . If q_c is the largest corner error product, any feasible objective is at most $\sum_c q_c \sum_j \lambda_{cj}$. The category masses belong to the same probability confidence box as before. Therefore

$$b \leq B_{MC} \leq B_{sep}. \quad (37)$$

Repeated endpoints use a single interpolation endpoint; zero-mass categories use zero corner masses. Omitting the two half-mean equalities recovers B_{sep} exactly. Adding them spends no additional δ because Eq. (35) is a deterministic rank identity.

Equivalence to the perspective McCormick envelope Define three lifted mass moments, retaining h_c for the probability upper bound:

$$\xi_c = p_c \mu_{Xc}, \quad \zeta_c = p_c \mu_{Yc}, \quad \omega_c = p_c \mu_{Xc} \mu_{Yc}.$$

Besides the probability box and the global equalities, the relaxation imposes

$$\begin{aligned} L_{Xc} p_c &\leq \xi_c \leq U_{Xc} p_c, & L_{Yc} p_c &\leq \zeta_c \leq U_{Yc} p_c, \\ \omega_c &\geq L_{Yc} \xi_c + L_{Xc} \zeta_c - L_{Xc} L_{Yc} p_c, \\ \omega_c &\geq U_{Yc} \xi_c + U_{Xc} \zeta_c - U_{Xc} U_{Yc} p_c, \\ \omega_c &\leq L_{Yc} \xi_c + U_{Xc} \zeta_c - U_{Xc} L_{Yc} p_c, \\ \omega_c &\leq U_{Yc} \xi_c + L_{Xc} \zeta_c - L_{Xc} U_{Yc} p_c. \end{aligned} \quad (38)$$

The objective is $\sum_c (\omega_c - g_c \xi_c - f_c \zeta_c + f_c g_c p_c)$. For positive mass, division by p_c gives the classical bilinear hull on the rectangle [29]. More explicitly, for positive interval widths $d_X = U_X - L_X$ and $d_Y = U_Y - L_Y$, the high/high corner mass is

$$\lambda_{HH} = \frac{\omega - L_Y \xi - L_X \zeta + L_X L_Y p}{d_X d_Y}.$$

The high/low and low/high masses are respectively $(\xi - L_X p)/d_X - \lambda_{HH}$ and $(\zeta - L_Y p)/d_Y - \lambda_{HH}$; the remaining mass completes their sum to p . The four inequalities are exactly the nonnegativity of these four masses. If a width is zero, the corresponding product is fixed and ordinary one-dimensional interpolation suffices. If $p = 0$, all lifted coordinates vanish. Thus Eq. (36) and Eq. (38) are equivalent in these cases as well.

What the numerical program certifies A floating-point primal optimum is not used as an upper certificate. Write the lifted LP as $\max c^T x$ subject to $Ax \leq d_0$, $Ex = e_0$ and $0 \leq x \leq 1$. For any proposed multipliers $y \geq 0$ and arbitrary z , put $r = c - A^T y - E^T z$. Weak duality with the box residual gives

$$c^T x \leq d_0^T y + e_0^T z + \sum_i \max(r_i, 0) = U_{\text{dual}}. \quad (39)$$

The software reconstructs inputs, coefficients and proposed multipliers as exact binary rational numbers and evaluates this expression exactly. Approximate dual feasibility is not assumed. Negative residuals need no absolute-value correction. The returned allowance is the upward binary64 rounding of $\min(U_{\text{dual}}, B_{\text{sep}})$. It can exceed the mathematical optimum B_{MC} and is no larger than the similarly upward-rounded separable allowance. Exact fractions and multipliers are stored, so verification requires no optimizer.

Reported joint infeasibility or numerical failure returns the separable allowance. If the probability box itself is empty, the general API returns the universal bound $b \leq 1$. Genuine emptiness cannot occur on the stated confidence event and is not used as evidence for rejection. Malformed inputs raise an error. Exact LP arithmetic is conditional on valid input confidence intervals; it does not certify the floating-point evaluation of an upstream binomial quantile. In the archived run, the largest certified-upper correction relative to the floating primal objective was 1.87×10^{-14} .

If the existing sampling radius obeys $\Pr\{\bar{D} - (\vartheta + b) > r_\epsilon \mid \text{training, validation}\} \leq \epsilon$, the union bound gives $\Pr\{\bar{D} - B - r_{\alpha-\delta} > \vartheta\} \leq \alpha$ for the returned allowance B . No new sampling law, variance floor or conditional-independence assumption is introduced. The corresponding fixed-mixture betting construction keeps its existing additive δ correction; a validation-dependent capital is not thereby an unconditional e-value.

Strict improvement and strict relaxation An attainable strict improvement has two equally likely categories, $f = g = 0$, and identical rank-mean rectangles $[1/4, 3/8]$ and $[5/8, 3/4]$. Here $B_{\text{sep}} = 45/128$ and $B_{\text{MC}} = 5/16$. At all corners, $xy \leq (x + y)/2 - 3/16$, which with the global equalities proves the joint upper value. The law $X = Y = Z$ with Z uniform on $[0, 1]$ and categories indicating its two halves attains it.

Nevertheless, one category, both rectangles $[0, 1]$, and $f = g = 1/2$ permit half the corner mass at low/low and half at high/high. The LP gives $1/4$ although the two true conditional means are known to be $1/2$ and $b = 0$. General optimized corner weights need not be products of endpoint weights for a single coherent pair of conditional means.

A nontrivial example fixes category masses $(1/4, 3/8, 3/8)$, fits $f = g = 1/2$ and all mean rectangles $[3/8, 5/8]$. The relaxed optimum is $1/64$, whereas the maximum over coherent means is $3/256$. Writing each centered mean as $t_c/8$ with $|t_c| \leq 1$ and $p^T t = 0$, Cauchy–Schwarz reduces the latter maximum to a weighted squared norm. Enumerating the vertices of that sliced box gives maximum $3/4$, attained by $t = (0, 1, -1)$. It is attainable by a real rank law: let $X = Y = Z$ be uniform and set

$$\Pr(C = 1 \mid Z) = \frac{1}{4}, \quad \Pr(C = 2 \mid Z) = \frac{3}{8} + \frac{9}{16}(Z - \frac{1}{2}), \quad \Pr(C = 3 \mid Z) = \frac{3}{8} - \frac{9}{16}(Z - \frac{1}{2}).$$

These nonnegative propensities give means $(1/2, 5/8, 3/8)$. The relaxation gap is $1/256$. Further common-rank constraints, such as $p_c/2 \leq \mu_{Xc} \leq 1 - p_c/2$, are not imposed. These facts preclude calling the LP a sharp attainable rank-law optimization.

Complete post-exposure archive diagnosis The construction and checks were frozen before its first new allowance outputs, but after exposure to the forecast study. This is explicitly an exploratory diagnosis, not a new confirmation. It retains every one of the $2 \times 2 \times 8 = 32$ task/baseline/candidate cases, their original fits, 32 declared categories, probability caps, confidence allocation and draws. Per candidate, 4,096 training draws, 8,192 disjoint validation pairs and 12,288 evaluation draws cost 32,768 row queries. No census rank or conditional-mean metadata enters the new allowance. Those values are used only afterward for diagnostics.

All 32 allowances decrease: the median absolute reduction is 0.000479251, the maximum is 0.002510925, and their relative reductions have median 1.2399% and maximum 6.5173%. Every variance and betting raw p -value remains exactly one; all BY values remain one. Both reference methods therefore retain no candidate in any family. The four category-copy controls remain nonrejected. This does not recover the earlier failed practical success criterion or demonstrate a forecast-loss gain.

Table 26: ExtraTrees allowances in the complete exploratory diagnosis. The lower bounds remain negative and both reference decisions are unchanged.

Task	Baseline	Separable	Joint	Reduction (%)	Joint lower
Appliances	Daily seasonal	.0418754	.0413886	1.1626	-.0106669
Appliances	Ridge	.0381046	.0380884	.0427	-.0333397
Metro	Daily seasonal	.0392165	.0386157	1.5320	-.0287812
Metro	Ridge	.0370478	.0361197	2.5050	-.0370934

The public directory `results/joint.bias/` contains all 32 rows, their exact dual receipts and local numeric inputs. Running `python results/joint.bias/run.py` checks the bundle, reconstructs all allowances and probabilities, and invokes a separately implemented verifier that imports neither the producer nor an optimizer. It checks 200 additional finite rank laws and 240 exact corner-formulation conversions, including ties, zero masses and degenerate rectangles. Separate producing code checks cover 120 finite frames and numerical-failure branches. The reconstruction retains the previously verified full-U means and radii; it rebuilds the training and validation quantities, disjoint-triple kernels, variances, ranges and probability inversions. It does not repeat forecast training or confirmation outcomes. Numeric equality and tolerance-based agreement are reported separately, and all old sealed results are preserved.

27 Aggregate validation with exact category probabilities

Lemma 1 in the main paper bounds the total learning bias using independent validation streams. The category probabilities are known exactly, for example from the complete control column of a finite archive. Forecast and outcome ranks are not required to obtain these probabilities. Conditioning fixes only focal validation categories; reference observations and their category labels retain their common marginal law. Estimated category frequencies alone do not meet this information condition.

Validation experiment The study crosses $C \in \{2, 8, 32, 128\}$ categories, $m \in \{512, 2048, 8192\}$ validation pairs, two category distributions and three fitting-error patterns. It has 72 settings with 2,000 repetitions each, using $\delta = 0.05$ for each method separately. Balanced masses are $p_c = 1/C$; unequal masses are $p_1 = 0.7$ and $p_c = 0.3/(C - 1)$ for $c > 1$. Let t_c be equally spaced on $[-1, 1]$, recentered to have p -weighted mean zero, and set

$$q_c = \frac{1}{2} + 0.35 t_c / \max_d |t_d|.$$

Within a category, V, W are independent Bernoulli variables with success probability q_c . Reference pairs follow the common mixture law; their two channels therefore share reference-category dependence. Since $\sum_c p_c q_c = 1/2$, both conditional marginal-midrank means are $\mu_c = 1/4 + q_c/2$. The three fixed fits are $(f_c, g_c) = (\mu_c, \mu_c)$, $(\mu_c + 0.05, \mu_c + 0.05)$, and $(\mu_c + 0.15, \mu_c - 0.15)$, with true biases 0, 0.0025 and -0.0225 . All fits lie in $[0, 1]$; these patterns probe validation accuracy rather than a particular learning algorithm.

Both methods receive the same exact p and the same m independent focal/reference pairs, costing $2m$ raw observations. The aggregate method uses forecast comparisons from the first half and outcome comparisons from the second half. The rectangle comparator uses both channels from all pairs. If $n_c > 0$, its mean radius is $\sqrt{\log(4C/\delta)/(2n_c)}$, with intervals intersected with $[0, 1]$; missing categories use $[0, 1]$. It maximizes each error product over the four endpoints, then averages those maxima with p . Thus it receives both channels on more validation units and the same category metadata. Neither bound is selected by taking an unadjusted random minimum.

The primary generator uses the exact 16-outcome binary-pair law and multinomial counts to reproduce the independent-pair sampling distribution. A separate verifier checks 288 selected replications. A second sealed seed draws the raw focal and reference categories and their four Bernoulli variables directly: 1,600 repetitions yield 4,800 fit rows and 48 method summaries. The original fixed seed is 1309202601; the second is 1309202699. These checks concern coverage and width, not forecast selection power or a new chronological confirmation.

At $C = 32$, $m = 8192$ and balanced exact fits, the median slack above true bias is 0.0185168 for rectangles and 0.00260164 for aggregation. Across all settings, the ratio of rectangle to aggregate median slack has minimum 0.5574, median 2.5000 and maximum 16.9403. Rectangles have smaller median slack in 10/72 settings: all six $C = 2$ opposite-error settings, three $C = 2$ unequal-mass exact-fit settings, and the $C = 128$, $m = 512$, unequal-mass opposite-error setting. Every one of the 144 method cells has zero observed noncoverage among 2,000 repetitions; the second-seed cells also have zero observed noncoverage. These counts do not imply zero failure probability. Negative valid allowances are retained; they are not truncated to zero.

Matched archive diagnosis After the original Appliances and Metro task outcomes were inspected, the aggregate allowance was applied to all 32 existing task/baseline/candidate cases. Fits, forecasts, draws and evaluation scores remain fixed. Each candidate uses 4096 training observations, 8192 validation pairs and 12288 evaluation observations, totaling 32768 queries. Exact category masses come from the selection archive’s control column. The original rectangle additionally estimates masses; the matched rectangle and aggregate bounds receive identical exact masses. Each construction uses the original per-candidate validation allocation and is assessed separately. BY applies within each of the four eight-candidate families.

Table 27: All 32 inspected candidates. Counts combine four separate families; they do not count independent confirmations. The exact-bias row is diagnostic.

Learning allowance	Pooled U retained	Betting retained
Original rectangles, estimated masses	0	0
Rectangles, exact masses	1	3
Aggregate, exact masses	9	13
Exact archive bias	14	16

For Appliances ExtraTrees beyond Ridge, the matched rectangle allowance is 0.0292745 and the aggregate allowance is 0.00432772. The observed association is 0.00994791 and the pooled-U sampling radius is 0.00519920. Its pooled-U

lower bound is positive at the pointwise 0.05 level, but its BY value is 0.367573, so this rule does not retain it. Classical betting with the same aggregate allowance has BY value 0.00306193 and retains it. For Metro ExtraTrees beyond Ridge, the aggregate allowance 0.00573561 exceeds observed association 0.00408298; neither rule retains it. Substituting exact archive bias makes the Metro betting rule retain it, whereas the pooled-U rule still does not. Both validation width and sampling precision therefore matter. All category-copy and independent-noise controls remain unretained. The original prespecified two-task success result remains 0/2; the post-exposure diagnostic does not establish a new gating benefit.

Reproduction and scope The public directory `results/aggregate_bias/` contains all 128 diagnostic rows, 144,000 simulation rows, 144 summaries, both seeds, source-specific provenance and separately implemented checks. Its reproduction command reconstructs all archive rows and verifies simulation arithmetic; `--regenerate-simulation` also generates every original replication. Outputs are written outside the distributed evidence. Numerical comparisons use the stated floating-point tolerances; the theorem is a real-arithmetic statement, not an outward-rounded machine certificate.

Direct product-bias estimation has classical higher-order antecedents [30,31]. Active U-statistic inference optimizes label acquisition while incorporating predictions [32]; here category probabilities are given and validation queries are fixed. The present contribution is the explicit observable allowance for this rank target and information condition. It does not claim optimal acquisition, minimax width, sharp rank-law optimization or uniform improvement over all classical procedures.

Reference design values The reference comparison in the main figure uses resampled Gaussian peers with Kendall correlation 1/3 and Spearman correlation 0.482584. Its fixed-peer counterparts have shared-score mean zero and distinct-score means -0.001423 or $+0.001423$, according to channel-effect alignment. These design-specific values are separate from the aggregate validation study.

A sufficient training rate For fixed finite categorical controls with positive cell probabilities, empirical midranks followed by cell averaging in an independent training sample of size m give $\varepsilon_f, \varepsilon_g = O_p(m^{-1/2})$. Thus $m/\sqrt{MN} \rightarrow \infty$ suffices, counting all training observations. This states a usable sufficient rate; it is not a necessary cost bound.

27.1 Expected validation width under exact fits

For exact fits, $C = C_Z$, $p_c = 1/C$ and $m_s/C \geq 1$ pairs per category in each stream ($m_{\text{val}} = 2m_s$ pairs and $4m_s$ raw observations in total),

$$\mathbb{E}[r_A + r_B + r_{AB}] \leq \frac{3\sqrt{Cx/8} + x/4}{m_s}, \quad (40)$$

where expectation conditions on these balanced counts. This describes validation precision under exact fits, not a general detection boundary.

Proof. For Eq. (40), exact fits give centered stream means with second moments at most $C/(4m_s)$. Jensen’s inequality bounds each expected linear radius by $\sqrt{Cx/8}/m_s$; the product radius equals $\sqrt{Cx/8}/m_s + x/(4m_s)$. Adding gives the result.

27.2 Explicit inversion of the reference certificate

For the main paper’s observable certificate, J counts disjoint triples, R bounds the kernel range, s is the disjoint-triple sample standard deviation, and $c_J = 2R/\sqrt{J(J-1)} + R/(3J)$. Fix validation error δ before evaluation and use the independently validated allowance $\hat{B}$. The following closed form implements monotone inversion. For explicit inversion, set $d = (\bar{D} - \hat{B})_+$, $A = s\sqrt{2/J}$, $x_H = 2J(d/R)^2$ and $x_B = \{2d/(A + \sqrt{A^2 + 4c_J d})\}^2$. For $d > 0$, $p_{\text{cert}} = \min\{1, \delta + 2e^{-\max(x_H, x_B)}\}$; set it to one for $d = 0$. Monotone inversion gives super-uniformity. The range-only version uses $\min\{1, \delta + e^{-x_H}\}$.

28 A matched fully averaged bias comparator

This retrospective comparison fixes the exact category probabilities, fitted means, validation pairs, evaluation samples and complete candidate families used by aggregate validation. It tests whether the observed improvement over rectangular bounds is specific to splitting. It is not independent confirmation.

Averaging without reusing a diagonal Condition on training and focal categories; leave reference categories unconditioned. For category c , the n_c independent pairs (A_{ci}, B_{ci}) have channel ranges of length one and means a_c, b_c . Dependence within a pair is allowed. The target is $b_H = \sum_c p_c a_c b_c$, with exact p_c . For $S = \{c : p_c > 0, n_c \geq 2\}$, put $k_c = \lfloor n_c/2 \rfloor$, $\ell_c = n_c - k_c$ and

$$U_c = \frac{(\sum_i A_{ci})(\sum_i B_{ci}) - \sum_i A_{ci} B_{ci}}{n_c(n_c - 1)}, \quad U = \sum_{c \in S} p_c U_c.$$

Write $b_{H,S} = \sum_{c \in S} p_c a_c b_c$. Then $\mathbb{E}U = b_{H,S}$. Removing the diagonal is necessary: within-pair dependence otherwise biases the product of means. Cells outside S contribute $H = \sum_{c \notin S} p_c q_c$, where q_c is the largest corner product of the two channel ranges. These omissions depend only on counts.

Let $\bar{A}_c, \bar{B}_c$ and $s_{A,c}^2, s_{B,c}^2$ be full-cell means and unbiased variances. Define

$$\begin{aligned} V_A &= \sum_{c \in S} \frac{p_c^2}{4k_c} \left(\bar{B}_c^2 + \frac{k_c s_{B,c}^2}{n_c \ell_c} \right), & V_B &= \sum_{c \in S} \frac{p_c^2}{4\ell_c} \left(\bar{A}_c^2 + \frac{\ell_c s_{A,c}^2}{n_c k_c} \right), \\ d_c &= \frac{p_c}{4\sqrt{k_c \ell_c}}, & D &= \sum_{c \in S} d_c^2, \quad d_* = \max_{c \in S} d_c. \end{aligned}$$

For nonempty S , let $x = \log(3/\delta)$ and $\lambda_0 = \sqrt{2x/D}$. An upper allowance is

$$B_{\text{pool}} = U + \frac{2x}{\lambda_0} + \frac{\lambda_0(V_A + V_B)}{2} + \sqrt{2x/D} + x d_* + H, \quad \Pr(b_H > B_{\text{pool}}) \leq \delta.$$

If S is empty, use H . The global corner bound may also cap the result. This tuning is fixed by counts and masses. It is not adaptive to arbitrary large fitting errors.

Why the observed variance proxies are valid Average uniformly over all splits into k_c and ℓ_c rows within each category. The split product averages to U_c . The squared subset means average to the bracketed expressions above. For each split, conditional Hoeffding bounds give exponential inequalities for its two linear errors. Jensen's inequality applied to these exponentials yields

$$\mathbb{E}e^{tL_A - t^2 V_A/2} \leq 1, \quad \mathbb{E}e^{tL_B - t^2 V_B/2} \leq 1,$$

where $L_A = \sum_{c \in S} p_c a_c \bar{B}_c - U$ and $L_B = \sum_{c \in S} p_c b_c \bar{A}_c - U$. The remaining centered product $R = b_{H,S} - U - L_A - L_B$ obeys

$$\log \mathbb{E}e^{tR} \leq -\frac{1}{2} \sum_{c \in S} \log(1 - t^2 d_c^2) \leq \frac{t^2 D}{2(1 - t d_*)}, \quad 0 < t < 1/d_*.$$

This follows from the product MGF of independent sub-Gaussian split errors and another application of Jensen. Markov's inequality for the linear terms and the product tail, each at failure level $\delta/3$, give the allowance. Averaging optimized bounds without this exponential argument would not justify it.

At exact fits and balanced counts, let m be the total number of validation pairs, with m/C_Z even and $p_c = 1/C_Z$. Then

$$\mathbb{E}(B_{\text{pool}} - U - H) \leq \{3\sqrt{C_Z x/2} + x/2\}/m.$$

This is the same expected-radius envelope as the split construction after matching its per-stream count to $m/2$. It establishes neither pointwise width dominance nor unique power improvement.

Table 28: Matched retrospective retentions over four families of eight candidates, at BY level 0.05 and 32,768 total raw draws per candidate. The same exact category masses are supplied to every displayed validator.

Learning allowance	Evaluation U	Betting
Rectangular	1	3
Split aggregate	9	13
Pooled, fixed tuning	9	13
Pooled, fixed tuning grid	9	12
Pooled, Gaussian mixture	9	13
Pooled, square-normalized MGF	9	13
Pooled, square-normalized and exact product MGF	10	13

The last two constructions are secondary developments after the initial comparison. The release provides every formula, all 72 simulation settings at two error levels, and all candidate-level results. These results show that improvement over rectangles is not unique to splitting. The original two-task confirmation criterion remains unmet. The public `results/pooled_bias/` directory supplies executable derivations, complete records and independent numerical checks.

29 Estimating the final target from raw observations

A stronger comparator may use all acquired observations without following the paper’s training and bias-subtraction construction. For iid observations $O_i = (C_i, V_i, W_i)$ and known positive category masses p_c , let $a_{ij} = \mathbf{1}\{V_j < V_i\} + \frac{1}{2}\mathbf{1}\{V_j = V_i\}$ and define b_{ij} from W . Independent reference positions give the exact identity

$$\vartheta = \mathbb{E}[a_{12}b_{13}] - \mathbb{E}\left[\frac{\mathbf{1}\{C_1 = C_2\}}{p_{C_1}}a_{13}b_{24}\right].$$

The first term is $\mathbb{E}[R_V R_W]$; the second is $\sum_c p_c m_V(c) m_W(c)$. Complete averaging over distinct positions gives an unbiased degree-three minus degree-four U-statistic. Repeated population IDs remain separate iid draw positions; their values may tie. Conditioning on all sampled category counts or substituting masses estimated from the inference draws would require a separate validity argument.

The release implements this identity in $O(n^2 + C_Z)$ time and $O(n + C_Z)$ working memory by subtracting position collisions from row-sum products. Exact rational enumeration independently checks small cases. For $p_* = \min_c p_c$, the symmetric four-row kernel lies in $[1/6 - 1/(4p_*), 1/3]$. The first component’s endpoints follow from the six strict relative orders; ties are averages over refinements. The unweighted second component averages to $1/4$ by pairing complementary comparisons, so its weighted value lies in $[0, 1/(4p_*)]$. Thus a valid range for the kernel has width $1/6 + 1/(4p_*)$. The constant-category specialization has the sharper width $1/6$.

Two preselected conservative calibrations were examined: full-U Hoeffding concentration using $\lfloor n/4 \rfloor$ blocks, and empirical Bernstein concentration for the separate mean of disjoint four-row kernels. The latter uses its own mean and variance; its radius is not attached to the full-U center. The release provides the confidence formulas and their p -value inversions.

The matched retrospective archive comparison pools all 32,768 independent replacement draws previously allocated to training, validation and evaluation. The target, predictor and category law were fixed before those draws. Across the same four eight-candidate BY families, both calibrations retain 0/32 candidates. The smallest raw p -values are 0.98443 and 0.18484. All 64 lower bounds are below the exact archive target in these saved draws; this is not an empirical coverage-rate estimate. The sparse categories make these bounds conservative. Their failure to retain candidates does not prove superiority over other direct-target calibrations.

The release supplies all results, exact small-sample checks, an independent reconstruction of 262,144 raw four-row blocks and every BY decision, and a command for repeating the larger quadratic calculations. No new confirmation data are used, and the original two-task criterion is unchanged.

Table 29: Matched sampling comparison.

Sampling bound	Detections	Null rejections
Signed variance bound	130/300	0/300
Classical betting mixture	296/300	0/300

Eight categories, 30,976 observations and 300 paired replications per condition, with the same signed allowance. Betting’s paired detection gain is 55.3 percentage points with pointwise 95% interval [47.2, 61.8] points. Each null rate has interval [0, 0.0123].

Table 30: Complete-family error and power.

Fitting / moment	Gaussian FDR	Calibrated FDR	Calibrated power
Complementary	0.000	0.008	0.844
Gapped complementary	0.928	0.015	0.694
Directional blocks	0.011	0.009	0.666
Classical FOD	0.012	0.015	0.752

400 independent entities, 11 models, 1,000 evaluation families. FDR columns use the all-null law; power uses three effects of 0.06. Gaussian denotes a normal reference law. All rules use nominal 0.05. Calibration uses a separate 5,000-draw simulator bank.

30 Additional sampling and family comparisons

Local alternatives For common-law arrays with $\sigma_{\psi, M, N}^2 \rightarrow \sigma_{\psi}^2 > 0$ and $\sqrt{MN} \vartheta_{M, N} \rightarrow d$, the main-paper argument gives $T_D \Rightarrow \mathcal{N}(d/\sigma_{\psi}, 1)$.

Table 31: Temporal feedback under the null.

Mean fitting	$\mathbb{E}[S_g]$	Mean T	Rejections
Complementary	-0.045052	-2.677	0/500
Gapped complementary	0.065533	3.148	458/500
Directional	0	-0.002	27/500

400 independent entities, 20 observations, five folds and lookback five. Equal entity weights and identical middle rows; 500 paired replications.

31 Reference overlap and target preservation

Weighted, partially overlapping references. Condition on training and design information H . Suppose the focal row O_0 and peers $O_1, \dots, O_k$ are mutually independent given H ; the peer laws may differ. Let $\mathcal{F} = \sigma(H, O_0)$, and define

$$\begin{aligned} A_j &= a(V_0, V_j), & B_j &= a(W_0, W_j), \\ p_j &= \mathbb{E}[A_j \mid \mathcal{F}], & q_j &= \mathbb{E}[B_j \mid \mathcal{F}], \\ \kappa_j &= \text{Cov}(A_j, B_j \mid \mathcal{F}). \end{aligned}$$

Take nonnegative H -measurable weights u_j, v_j with $\sum_j u_j = \sum_j v_j = 1$; zeros permit partial overlap. Set $U_V = \sum_j u_j A_j$, $U_W = \sum_j v_j B_j$, $p = \sum_j u_j p_j$, $q = \sum_j v_j q_j$, and let $f(Z_0), g(Z_0)$ be fixed given H . With $m_p = \mathbb{E}[p \mid Z_0, H]$ and m_q defined analogously,

$$\begin{aligned} \mathbb{E}[(U_V - f)(U_W - g) \mid H] &= \mathbb{E}[(p - m_p)(q - m_q) \mid H] \\ &\quad + \mathbb{E}[(m_p - f)(m_q - g) \mid H] \\ &\quad + \mathbb{E}\left[\sum_j u_j v_j \kappa_j \mid H\right]. \end{aligned} \tag{41}$$

This concerns the target induced by the specified weighted peer laws. For an externally fixed target $T_V(V_0), T_W(W_0)$, set $m_V = \mathbb{E}[T_V \mid Z_0, H]$, $r_V = T_V - m_V$, $e_V = m_V - f$, $\Delta_V = p - T_V$, and analogously on the outcome side. The right side is then $\mathbb{E}[r_V r_W \mid H] + \mathbb{E}[e_V e_W \mid H]$ plus the overlap term and the additional reference-target shift

$$\mathbb{E}[r_V \Delta_W + r_W \Delta_V + e_V \Delta_W + e_W \Delta_V + \Delta_V \Delta_W \mid H]. \tag{42}$$

For $d = \sum_j u_j v_j < 1$, define the marginally centered off-diagonal statistic

$$D_{\text{off}} = \frac{U_V U_W - \sum_j u_j v_j A_j B_j}{1 - d} - f U_W - g U_V + f g.$$

Its expectation is

$$\begin{aligned} \mathbb{E}[D_{\text{off}} \mid H] &= \mathbb{E}[(p - f)(q - g) \mid H] \\ &\quad + \mathbb{E}\left[\frac{dpq - \sum_j u_j v_j p_j q_j}{1 - d} \mid H\right]. \end{aligned} \tag{43}$$

The last term vanishes for common-law peers. Uniform weights reduce it to $-\mathbb{E}[\overline{p_j q_j} - \bar{p} \bar{q} \mid H]/(k - 1)$, the fixed-peer distortion. For common-law peers, the uncorrected overlap coefficient is d ; uniform reference sets give $d = |\mathcal{R}_V \cap \mathcal{R}_W|/(|\mathcal{R}_V||\mathcal{R}_W|)$.

Proof. For $j \neq l$, conditional independence gives $\mathbb{E}[A_j B_l \mid \mathcal{F}] = p_j q_l$; on the diagonal the expectation is $p_j q_j + \kappa_j$. Expanding $U_V U_W$ and then decomposing p, q into their conditional means and residuals proves Eq. (41). Expanding $(r_V + e_V + \Delta_V)(r_W + e_W + \Delta_W)$ proves Eq. (42), since only $\mathbb{E}[r_V e_W \mid H]$ and $\mathbb{E}[r_W e_V \mid H]$ vanish automatically. Removing the diagonal leaves conditional expectation $pq - \sum_j u_j v_j p_j q_j$; division by $1 - d$ proves Eq. (43).

A limit of index deletion. Removing overlap must also preserve the declared reference mixture. A universal correction from one observation per heterogeneous peer is impossible when some $u_j v_j > 0$. To see this, fix the focal row and all other peers, and let the shared peer's comparison pair be $A_j = B_j \sim \text{Bernoulli}(t)$. The product pq has nonzero quadratic coefficient $u_j v_j t^2$, whereas the expectation of any statistic of this single random observation is affine in t . Thus it cannot estimate pq unbiasedly for every t . Common-law peers, known peer laws, or independent replicate rows change this information condition. Independent reference samples from the same declared mixtures remove overlap without changing the target; duplicating an observed row does not create an independent replicate. These statements describe sampling requirements, not an optimality or empirical superiority result.

Following the original observations Different rows can reuse the same earlier sale. Let $X_{is} = \sum_{b \in D_s} Y_{ib}$, using either one lag for every nonconstant map or two equally weighted lags for every nonconstant map. Constant maps contribute zero. Define $P_{sj} = a(X_{is}, X_{js}) - 1/2$, $Q_{tj} = a(Y_{it}, Y_{jt}) - 1/2$. Expanding fixed mean fits and reference weights gives $A = \sum_{s,j} c_{sj} P_{sj}$, $B = \sum_{t,j} d_{tj} Q_{tj}$. Write $c_{s+} = \sum_j c_{sj}$ and $d_{t+} = \sum_j d_{tj}$. Then

$$\mathcal{T} = \sum_s \sum_{t \in D_s} c_{s+} d_{t+}, \quad \Omega = \sum_s \sum_{t \in D_s} c_s^T d_t. \quad (44)$$

Lemma 3 (Source windows and reference reuse). *Let all Y_{it} be i.i.d. from F , with fixed coefficients and peers $j \neq i$. Let $M_F(y) = \mathbb{E}[a(y, Y')]$, $Y' \sim F$ independent. Replace comparisons by their expectations over references given focal outcomes to obtain A_0, B_0 . Then*

$$\mathbb{E}[A_0 B_0] = \kappa_F \mathcal{T}, \quad \mathbb{E}[AB] = \kappa_F \mathcal{T} + \lambda_F \Omega. \quad (45)$$

For one-lag copies, $\kappa_F = \text{Var } M_F(Y)$ and $\lambda_F = \mathbb{E} \text{Var}\{a(Y, Y') \mid Y\}$. For two-lag sums, writing $F * F$ for the sum law and $\pi_{\text{tie}} = \Pr(Y = Y')$,

$$\begin{aligned} \kappa_F &= \text{Cov}\{M_{F*F}(Y + Y'), M_F(Y)\}, \\ \lambda_F &= (1 - \pi_{\text{tie}}^2)/8 - \kappa_F. \end{aligned} \quad (46)$$

Both constants are nonnegative. Preservation for all F is equivalent to $\Omega = 0$; empirical centering for all F is equivalent to $\mathcal{T} = \Omega = 0$.

Table 32 fixes $\mathcal{T} = 0$ and gives two-lag means $-3/128, 0, 3/128$. Signed overlap can cancel; for one fixed law, centering can also occur with nonzero design coefficients.

This specializes classical rank covariance [5, 6] to temporal provenance, including ties. Uniform complete peers, fixed N and no ties in either ranked vector give variance $(N+1)/\{12(N-1)\}$ with divisor N , so standardization rescales the identity. The main trajectory result handles externally learned maps and dependent coordinates before random rank standardization. Reference separation preserves $\kappa_F \mathcal{T}$ and does not make rank-coupled entities independent.

Proof of Lemma 3. Outside the source window, covariance vanishes. Inside, conditioning and averaging over focal outcomes gives κ_F for distinct peers; a shared peer adds λ_F , the conditional comparison variance for copies. For sums, let D_1, D_2 be independent differences of two F draws. Exchangeability and symmetry give the same-peer covariance $\mathbb{E}[\text{sgn}(D_1 + D_2) \text{sgn } D_1]/4 = (1 - \pi_{\text{tie}}^2)/8$. Expansion proves (45). Monotonicity gives $\kappa_F \geq 0$; conditioning on focal outcomes and averaging the other peer component leaves a covariance of monotone functions, giving $\lambda_F \geq 0$. Bernoulli(1/2) and uniform $\{0, 1, 2\}$ give $(\kappa_F, \lambda_F) = (1/16, 1/16), (2/27, 5/54)$ for copies and $(3/64, 3/64), (13/243, 14/243)$ for sums. Nonproportionality proves necessity; the identity proves sufficiency.

32 Temporal fitting with weighted rank references

This section develops the one-lag specialization of Lemma 3. Section 33 derives the two-lag sum extension. A general forecast need not admit either fixed comparison representation. A sufficient forecast class is $X_{is} = h_s(Y_{i, \ell(s)})$, where $\ell(s) < s$ is deterministic and each h_s is strictly increasing and common to all entities at time s . Its period rank equals a rank of the source panel at $\ell(s)$.

From temporal fits to comparison coefficients. For evaluation row r , write $\alpha_{rs} = \mathbf{1}\{s = r\} - f_{rs}$ and $\beta_{rt} = \mathbf{1}\{t = r\} - g_{rt}$, where f_{rs}, g_{rt} are fixed linear mean-fit weights with $\sum_s f_{rs} = \sum_t g_{rt} = 1$. If the forecast and outcome reference vectors are u_s, v_t , respectively, then the main lemma uses

$$c_{rt} = \sum_{s: \ell(s)=t} \alpha_{rs} u_s, \quad d_{rt} = \beta_{rt} v_t.$$

Table 32: Two-lag source and peer coefficients.

Design	d_1	d_2	d_3	Ω	$\mathbb{E}[AB]$
Positive	$(\frac{1}{2}, 0)$	$(-\frac{1}{2}, -\frac{1}{2})$	$(0, \frac{1}{2})$	$\frac{1}{2}$	$\frac{3}{128}$
Canceling	$(\frac{1}{2}, 0)$	$(-\frac{1}{2}, -\frac{1}{2})$	$(\frac{1}{2}, 0)$	0	0
Negative	$(0, \frac{1}{2})$	$(-\frac{1}{2}, -\frac{1}{2})$	$(\frac{1}{2}, 0)$	$-\frac{1}{2}$	$-\frac{3}{128}$

$X_a = Y_1 + Y_2$, $X_b = Y_2 + Y_3$; rank each sum after addition. $A = P_{a1} - P_{b1}$ fixes $c_a = (1, 0)$, $c_b = (-1, 0)$. All designs have $\mathcal{T} = 0$. Lemma 3 gives $\mathbb{E}[AB] = 3\Omega/64$ for independent Bernoulli(1/2) sources.

Thus the design coefficient is

$$\Omega_r = \sum_s \alpha_{rs} \beta_{r,\ell(s)} u_s^T v_{\ell(s)}.$$

Constant initial forecast ranks contribute zero after centering at $1/2$. Normalized mean-fit weights cancel the rank constants; with unnormalized weights, retain the same deterministic offsets in the empirical and population-reference scores. Uniform complete-peer ranks with fixed N and no ties have deterministic variance $(N+1)/\{12(N-1)\}$ using divisor N . Their common standardization rescales the identity. Ties or nonuniform reference weights can make the scale random and require another argument. For any fixed weighted aggregate of row scores, aggregate the corresponding Ω_r with those weights. This expectation statement does not require the row scores themselves to be independent.

If forecast effects use earlier rows and outcome effects use later rows, their innovation source times are disjoint, so the population-reference score and the reference interaction both have expectation zero in this model. Disjoint peer observations along each matched source time remove the reference interaction alone. With fixed peer vectors u, v across time, $\Omega_r = (u^T v) T_r$, where $T_r = \sum_t (\mathbf{1}^T c_{rt}) (\mathbf{1}^T d_{rt})$. Consequently $T_r = 0$ then implies $\Omega_r = 0$. Different peer vectors across periods need not preserve this implication.

The covariance operator of the reference graph. Write $M_F(x) = \mathbb{E}a(x, Y)$, $\psi_F = M_F - 1/2$, and

$$r_F(x, y) = a(x, y) - 1/2 - \psi_F(x) + \psi_F(y).$$

The kernel r_F is antisymmetric and has zero conditional mean given either argument. Let $v_F = \mathbb{E}\psi_F(Y)^2$ and $\tau_F = \mathbb{E}r_F(Y, Y')^2$. For a row-stochastic reference matrix U with zero diagonal, set $L_U = I - U$. Orient every edge $e = (p, q)$ with $p < q$ and let $(B_U)_{pe} = U_{pq}$, $(B_U)_{qe} = -U_{qp}$, with other entries zero. The vector of centered ranks is $q^U = L_U \psi + B_U r$. Hoeffding projection orthogonality [5] gives

$$\text{Cov}(q^U, q^V) = v_F L_U L_V^T + \tau_F B_U B_V^T. \quad (47)$$

Indeed, distinct node variables are independent. Distinct unordered-edge residuals are uncorrelated even when they share one endpoint, by conditional degeneracy; node and edge terms are also orthogonal. This proves the formula. In particular, the same-focal covariance is $v_F + (v_F + \tau_F) \sum_j U_{ij} V_{ij}$, so the main lemma's $\omega_F = v_F + \tau_F$. For different focals i, l it is

$$v_F(e_i - U_i)^T(e_l - V_l) - \tau_F U_{il} V_{li}.$$

This shows why independence of the raw entities does not imply independence of the rank scores.

For atom masses p_x , put $s_2 = \sum_x p_x^2$ and $s_3 = \sum_x p_x^3$. The sums ignore the continuous part. Then

$$v_F = \frac{1 - s_3}{12}, \quad \tau_F = \frac{1 - 3s_2 + 2s_3}{12}, \quad \omega_F = \frac{2 - 3s_2 + s_3}{12}.$$

The randomized probability integral transform has conditional mean $M_F(Y)$ and conditional variance $p_Y^2/12$, proving the first formula. Also $\mathbb{E}\{a(Y, Y') - 1/2\}^2 = (1 - s_2)/4$; projection orthogonality gives the other two. The quantity $12\tau_F$ is the probability that three independent observations are all distinct. It is zero for binary laws, but positive for any law supported on at least three distinct values. Continuous laws have $(v_F, \tau_F, \omega_F) = (1/12, 1/12, 1/6)$.

Heterogeneous peer laws. Suppose instead that source observations remain independent and each entity j has its own law F_j , fixed across time. For a fixed focal i , define

$$\begin{aligned} p_j(y) &= \mathbb{E}_{F_j} a(y, Y_j), & \mu_j &= \mathbb{E}_{F_i} p_j(Y_i), \\ K_{jl} &= \text{Cov}_{F_i}(p_j(Y_i), p_l(Y_i)), & \omega_j &= \mathbb{E}_{F_i} \text{Var}_{F_j}(a(Y_i, Y_j) \mid Y_i). \end{aligned}$$

For $Z_{tj} = a(Y_{it}, Y_{jt}) - \mu_j$, $\text{Cov}(Z_t) = K + \text{diag}(\omega_j)$. If A_0, B_0 replace each centered comparison by $p_j(Y_{it}) - \mu_j$, the exact interaction is

$$\mathbb{E}[AB] - \mathbb{E}[A_0 B_0] = \sum_j \omega_j \Lambda_j, \quad \Lambda_j = \sum_t c_{tj} d_{tj}.$$

Conditioning on the focal observation proves this covariance identity. The interaction vanishes for every collection of independent entity laws if and only if every $\Lambda_j = 0$. To prove necessity, make the focal and all peers except j deterministically zero, and let peer j be equally likely -1 or 1 . Only $\omega_j = 1/4$ is nonzero. If laws may also change with source time, the corresponding uniform criterion is $c_{tj} d_{tj} = 0$ separately for every t, j .

This targets the transform induced by each specified weighted reference mixture. For uncentered ranks, retain the deterministic residual means m_A, m_B : the expectation is $m_A m_B + \sum_t c_t^T K d_t + \sum_j \omega_j \Lambda_j$. An externally declared reference law adds the target-shift terms discussed in the weighted-reference result. Independence of references alone does not preserve that law.

Positive, negative and zero interactions. Let three entities have independent outcomes uniform on $\{-2, 0, 5\}$ and $X_{it} = Y_{i,t-1}$. At time 2, the focal-0 score is

$$S = (P_{02} - P_{04})\{Q_{02} - (Q_{01} + Q_{03})/2\}.$$

Let P_{02}, P_{04}, Q_{03} use peer 1 and let Q_{01} use peer 2; Q_{02} may use any fixed mixture. The population-reference score has expectation zero, while $\Omega = 1/2$ and $\omega_F = 5/54$, giving $\mathbb{E}S = 5/108$. Swapping the references of Q_{01}, Q_{03} gives $-5/108$; using peer 1 for both gives zero. All preserve the same common-law reference target. These are expectation statements, not rejection rates.

The original three-time construction has $T = -1$ and $\Omega = -1/(N - 1)$, hence mean $-v_F - \omega_F/(N - 1)$. At $N = 3$ this is $-3/32$ for Bernoulli(1/2) and $-13/108$ for uniform ternary outcomes, versus population-reference values $-1/16$ and $-2/27$.

Scope. These formulas use fixed coefficients and independent source observations. The rank of a sum of raw lags need not be a linear combination of their ranks: focal-minus-peer lag differences $(2, -1)$ and $(1, -2)$ have the same component comparison signs but opposite signs after summation. The results therefore do not calibrate general learned forecasts, dependent panels or data-selected projections. The Gram calculation is a specialization of classical rank projection machinery, and the design consequence is the explicit zero criterion, not a new general covariance or concentration theorem.

32.1 A finite bound for independent source panels

The identity can be used with classical bounded-mean inference when independent whole-panel replicates are available. Let $S_g = A_g B_g$, $g = 1, \dots, M$, use the same fixed coefficients and independent common-law source observations. Write $\theta_0 = \mathbb{E}[A_0 B_0]$ for the population-reference temporal target. Then $\mathbb{E}S_g = \theta_0 + \Omega \omega_F$. The individual entities within a rank panel are not independent score replicates.

Independently of these panels, take n independent triples (Z_0, Z_1, Z_2) from the same F and form

$$W = \frac{1}{2}\{a(Z_0, Z_1) - a(Z_0, Z_2)\}^2.$$

Conditional independence of Z_1, Z_2 given Z_0 gives $\mathbb{E}W = \omega_F$, and $0 \leq W \leq 1/2$. For $0 < \delta < \alpha < 1$, define

$$\epsilon = \frac{1}{2}\sqrt{\frac{\log(2/\delta)}{2n}}, \quad [l, u] = [\bar{W} - \epsilon, \bar{W} + \epsilon] \cap [0, 1/2].$$

This uses the kernel range; the preceding sharper bound $\omega_F \leq 1/6$ is not required by the implementation. Since $|Z_{bj}| \leq 1/2$, a valid range width for S_g is $R = \frac{1}{2}(\sum_b \|c_b\|_1)(\sum_b \|d_b\|_1)$. The implemented lower bound is

$$L = \bar{S} - R\sqrt{\frac{\log\{1/(\alpha - \delta)\}}{2M}} - \Omega \begin{cases} u, & \Omega \geq 0, \\ l, & \Omega < 0. \end{cases} \quad (48)$$

Hoeffding's inequality gives a validation failure probability at most δ and a one-sided evaluation failure probability at most $\alpha - \delta$. On their joint success event, the subtracted interaction is at least $\Omega \omega_F$, including when $\Omega < 0$. Thus $\Pr\{\theta_0 \geq L\} \geq 1 - \alpha$ by a union bound. This is a pointwise guarantee for fixed coefficients under the stated data design. The implementation preserves integer ordering before computing comparisons; casting large integers to floating point could create ties.

The target retains the same temporal fits and may itself contain temporal fitting bias. This bound does not certify future forecasting gain, a category-adjusted covariance, or a dependent observational panel. Coefficients and error budgets are fixed before evaluation. Independent validation from a different law does not estimate the required ω_F .

32.2 An implementation demonstration of the developed mechanism

The executable files `reference_time_audit.py` and `reproduce_interaction.py` implement Eq. (48) and a synthetic check. This checks a mechanism developed in this revision; it is not independent confirmation of predictive utility. The protocol records seed 73190527 and 400 replications for each of four laws: Bernoulli(0.1), Bernoulli(0.5), uniform on $\{-2, 0, 5\}$, and uniform on $(0, 1)$. Each replication draws $M = 8,192$ independent panels with three source times and three entities, together with $n = 2,048$ fresh independent validation triples. All observations in a replication have the same law.

The paired designs use positive, negative and zero interactions: $\Omega = 1/2, -1/2, 0$, respectively. In all three executed designs, Q_{02} uses peer 1. All have $\theta_0 = 0$ and score range width $R = 2$. The validated bound uses total error budget $\alpha = 0.05$ and validation budget $\delta = 0.01$. Two diagnostic comparators use the same score arrays: the uncorrected classical

bound $\bar{S} - 2\sqrt{\log(20)/(2M)}$ and that bound minus the known quantity $\Omega\omega_F$. The former bounds the empirical-reference mean, which is generally the wrong mean for the declared θ_0 ; the latter is an oracle diagnostic supplied with the exact law constant. They are not interchangeable procedures with the same target and available information.

Table 33 reports every positive-interaction cell. In all negative- and zero-interaction cells, all three methods have exactly 0/400 positive lower bounds. These statements cover all 36 law–design–method cells and 4,800 paired design replications in the stored results. No power alternative is included.

Table 33: Positive lower-bound counts in the positive-interaction design. The declared population-reference target is zero in every row; the empirical mean is $\omega_F/2$. Each count has denominator 400.

Law	Empirical mean	Uncorrected	Known ω_F	Validated ω_F
Bernoulli(0.1)	9/800	0/400	0/400	0/400
Bernoulli(0.5)	1/32	388/400	0/400	0/400
Uniform ternary	5/108	400/400	0/400	0/400
Continuous uniform	1/12	400/400	0/400	0/400

For a count of 0/400, the stored pointwise exact 95% binomial interval is $[0, 0.009180]$; zero observed failures does not imply zero error probability. The intervals for 388/400 and 400/400 are $[0.948183, 0.984404]$ and $[0.990820, 1]$, respectively. These are pointwise simulation intervals, not simultaneous uncertainty statements across all cells.

Each paired replication draws $9M + 3n = 79,872$ observations: 73,728 source observations and 6,144 validation observations. The uncorrected and oracle formulas do not use validation triples; the oracle additionally assumes the law constant. Results are therefore a check of centering and a conservative finite bound, not evidence of a cost advantage, unique capability, or superiority to an equally informed valid comparator. The release includes the protocol, all 4,800 replication rows, 36 summary rows and the exact-check scripts. Exact enumeration separately covers 171,808 raw temporal arrays, including ties, unequal peer laws, partial overlap and repeated source times; those checks concern expectations and covariance geometry rather than observational-panel calibration.

33 Ranks of raw two-lag sums

This section develops the two-lag part of Lemma 3 and Eq. (45). The forecast is ranked after its raw components are added. Its comparison need not be a fixed linear combination of the component comparisons. Nevertheless, equal weights and a common source law give an exact scalar description of the reference interaction. The underlying bivariate rank covariances and projection machinery are classical [5, 6]; the calculation below identifies their constants and the temporal source coefficients for this forecast class, including ties.

33.1 Objects, reference laws, and the design map

Let all raw observations Y_{it} be independent with common law F , and put

$$D_s = \{s - 1, s - 2\}, \quad X_{is} = Y_{i,s-1} + Y_{i,s-2}, \quad H = F * F.$$

No moment assumption on F is needed. Define the bounded mid-comparison and the mid-distribution transform by

$$a(x, y) = \mathbf{1}\{y < x\} + \frac{1}{2}\mathbf{1}\{y = x\}, \quad M_G(x) = \mathbb{E}_{Z \sim G} a(x, Z).$$

Fix a focal entity i and peer indices different from i . Write

$$P_{sj} = a(X_{is}, X_{js}) - \frac{1}{2}, \quad Q_{tj} = a(Y_{it}, Y_{jt}) - \frac{1}{2},$$

$$A = \sum_{s,j} c_{sj} P_{sj}, \quad B = \sum_{t,j} d_{tj} Q_{tj}.$$

All sums are finite and all coefficients are fixed independently of the evaluated raw array. For evaluation row r , normalized temporal mean fits and reference vectors give the explicit map

$$\alpha_s = \mathbf{1}\{s = r\} - f_{rs}, \quad \beta_t = \mathbf{1}\{t = r\} - g_{rt},$$

$$c_{sj} = \alpha_s u_{sj}, \quad d_{tj} = \beta_t v_{tj},$$

where $\sum_s f_{rs} = \sum_t g_{rt} = 1$ and every reference vector sums to one. Normalization cancels the deterministic rank constants. Constant forecast rows have centered comparison zero. Random learned residual coefficients and random sample-scale denominators are not covered by this map.

The population-reference counterparts replace P_{sj} by $p_s = M_H(X_{is}) - 1/2$ and Q_{tj} by $q_t = M_F(Y_{it}) - 1/2$; denote the resulting scores by A_0, B_0 . In particular, the forecast reference law is $F * F$, not F . Both transforms have mean zero, also when F or H has atoms. Define the two fixed source-window coefficients

$$\begin{aligned} T_2 &= \sum_s \sum_{t \in D_s} \left(\sum_j c_{sj} \right) \left(\sum_j d_{tj} \right), \\ \Omega_2 &= \sum_s \sum_{t \in D_s} \sum_j c_{sj} d_{tj}. \end{aligned} \tag{49}$$

The design map must retain the two actual source times of each sum. Replacing its rank by a sum of component ranks is incorrect: focal-minus-peer component differences $(2, -1)$ and $(1, -2)$ have the same comparison signs but opposite signs after addition.

33.2 Exact pair identity, including ties

For independent $Y_1, Y_2 \sim F$, let

$$\begin{aligned} \kappa_F &= \text{Cov}\{M_H(Y_1 + Y_2), M_F(Y_1)\}, \\ s_2 &= \text{Pr}(Y_1 = Y_2), \quad \eta_F = (1 - s_2^2)/8, \quad \lambda_F = \eta_F - \kappa_F. \end{aligned} \tag{50}$$

The complete same-focal pair identity is

$$\mathbb{E}[P_{sj}Q_{tk}] = \mathbf{1}\{t \in D_s\} \{\kappa_F + \mathbf{1}\{j = k\}\lambda_F\}. \tag{51}$$

Consequently the main identity takes the form

$$\mathbb{E}[A_0B_0] = \kappa_F T_2, \quad \mathbb{E}[AB] = \kappa_F T_2 + \lambda_F \Omega_2. \tag{52}$$

Pair calculation. If $t \notin D_s$, all raw source times in the two comparisons are disjoint, so their centered product has expectation zero. If $t \in D_s$ and $j \neq k$, condition on the two focal components. The two peers are independent, giving $\mathbb{E}[p_s q_t] = \kappa_F$ by exchangeability of the equal-weight components. For a shared peer, introduce independent symmetric differences $D_1 = Y_1 - Y'_1$ and $D_2 = Y_2 - Y'_2$, with all four variables drawn from F . The centered comparisons are one half of the corresponding signs, so their product expectation is

$$\eta_F = \frac{1}{4} \mathbb{E}\{\text{sgn}(D_1 + D_2) \text{sgn} D_1\}.$$

Here $\text{sgn} 0 = 0$ implements the midrank convention. Exchanging the independent components gives

$$2\mathbb{E}\{\text{sgn}(D_1 + D_2) \text{sgn} D_1\} = \mathbb{E}[\text{sgn}(D_1 + D_2)\{\text{sgn} D_1 + \text{sgn} D_2\}].$$

The integrand is two when both differences are nonzero with the same sign, one when exactly one difference is zero, and zero otherwise. Symmetry gives probability $(1 - s_2)^2/2$ for the first case and $2s_2(1 - s_2)$ for the second. Its expectation is therefore $(1 - s_2)^2 + 2s_2(1 - s_2) = 1 - s_2^2$, proving the stated η_F without restrictions on atom spacing or tie sizes. Expansion of the finite score sums proves (52).

Signs of the constants. Conditioning $M_H(Y_1 + Y_2)$ on Y_1 leaves a nondecreasing function of Y_1 , as is $M_F(Y_1)$. Their covariance is nonnegative, hence $\kappa_F \geq 0$. For λ_F , condition on focal values (x_1, x_2) and average over the second peer component. The shared-peer conditional covariance is the covariance, over $U \sim F$, of

$$U \mapsto \mathbb{E}_{V \sim F} a(x_1 + x_2, U + V), \quad U \mapsto a(x_1, U).$$

Both functions are nonincreasing. For an independent copy U' ,

$$\text{Cov}\{f(U), g(U)\} = \frac{1}{2} \mathbb{E}[\{f(U) - f(U')\}\{g(U) - g(U')\}] \geq 0.$$

Averaging this conditional covariance gives $\lambda_F \geq 0$. Thus $0 \leq \kappa_F \leq \eta_F \leq 1/8$ and $0 \leq \lambda_F \leq \eta_F \leq 1/8$.

33.3 Computable distributional constants

For a finite law with m ordered atoms y_a and probabilities p_a , form the convolution masses, sort the at most $m(m+1)/2$ distinct sums, and calculate their mid-distribution values by prefix sums. Then

$$\kappa_F = \sum_{a,b} p_a p_b \{M_H(y_a + y_b) - \tfrac{1}{2}\} \{M_F(y_a) - \tfrac{1}{2}\}, \quad \lambda_F = \frac{1 - (\sum_a p_a^2)^2}{8} - \kappa_F. \quad (53)$$

This requires $O(m^2 \log m)$ sorting and arithmetic operations and $O(m^2)$ storage. Rational atom values and probabilities permit exact rational calculation; bit complexity additionally depends on their encoding lengths. No conditional-covariance oracle is required.

Table 34: Constants for ranks of equal-weight two-lag sums. In the first row, $q = 1 - p$; the Gaussian law is nondegenerate.

Source law	κ_F	λ_F	η_F
Bernoulli(p)	$pq(1 - pq)/4$	$pq(1 - pq)/4$	$pq(1 - pq)/2$
Bernoulli($1/2$)	$3/64$	$3/64$	$3/32$
Uniform $\{0, 1, 2\}$	$13/243$	$14/243$	$1/9$
Uniform $\{-2, 0, 5\}$	$13/243$	$14/243$	$1/9$
Uniform $(0, 1)$	$7/120$	$1/15$	$1/8$
Gaussian	$\arcsin\{1/(2\sqrt{2})\}/(2\pi)$	$1/8 - \kappa_F$	$1/8$

For binary support, mapping the two ordered atoms to $\{0, 1\}$ preserves all comparisons and gives $a(Y, Y') - 1/2 = (Y - Y')/2$. The node projections therefore account for the entire cross-moment against this raw comparison: $\eta_F = 2\kappa_F$. Evaluating the convolution mid-distribution yields the Bernoulli row in Table 34. For uniform $(0, 1)$, the triangular sum distribution gives

$$\mathbb{E}\{M_H(x + Y) - \tfrac{1}{2}\} = -x^3/3 + x^2/2 + x/2 - 1/3, \quad 0 \leq x \leq 1.$$

Multiplying by $x - 1/2$ and integrating over $[0, 1]$ gives $\kappa_F = 7/120$. For Gaussian sources, $(Y_1 + Y_2, Y_1)$ has correlation $\rho = 1/\sqrt{2}$. The classical bivariate rank calculation gives $\eta_F = \arcsin(\rho)/(2\pi)$; two independent reference observations give $\kappa_F = \arcsin(\rho/2)/(2\pi)$. These are familiar Gaussian rank moments, not new distributional formulas.¹

33.4 Sharp uniform design criteria and examples

For a fixed design, the following statements hold uniformly over source laws:

$$\begin{aligned} \mathbb{E}[AB] &= \mathbb{E}[A_0 B_0] \text{ for every } F &\iff \Omega_2 &= 0, \\ \mathbb{E}[A_0 B_0] &= 0 \text{ for every } F &\iff T_2 &= 0, \\ \mathbb{E}[AB] &= 0 \text{ for every } F &\iff T_2 = \Omega_2 &= 0. \end{aligned}$$

Sufficiency follows from (52). A nondegenerate binary law proves necessity for the first two statements. For the third, binary sources require $T_2 + \Omega_2 = 0$, whereas uniform ternary sources require $13T_2 + 14\Omega_2 = 0$; hence both coefficients vanish. For one specified law, $\kappa_F T_2 + \lambda_F \Omega_2 = 0$ can instead hold by cancellation with nonzero coefficients. Uniform centering is a stronger design property.

Separating all forecast-residual source windows from outcome-residual source times forces both coefficients to zero. Using disjoint peer sets whenever source times match forces $\Omega_2 = 0$ only: it preserves a possibly nonzero population-reference target. Separation is sufficient, not necessary, because signed cancellation can also make either coefficient zero.

For a concrete construction, suppress the fixed focal index and write $\tilde{P}_s, \tilde{Q}_t$ for comparisons using the stated peers. Take

$$S = (\tilde{P}_2 - \tilde{P}_5)\{\tilde{Q}_2 - (\tilde{Q}_0 + \tilde{Q}_4)/2\}. \quad (54)$$

The forecast sources are $\{1, 0\}$ and $\{4, 3\}$, so $T_2 = -1/2 + 1/2 = 0$. Use peer 1 for both forecast comparisons, peer 2 for $\tilde{Q}_0$, and peer 1 for $\tilde{Q}_4$; the peer for $\tilde{Q}_2$ is immaterial. Then $\Omega_2 = 1/2$ and $\mathbb{E}S = \lambda_F/2$. Swapping the two relevant outcome peers gives $-\lambda_F/2$; using peer 1 for both gives zero despite source reuse. All three have the same common-law population-reference target zero. For Bernoulli($1/2$) their means are $3/128, -3/128, 0$.

¹P. A. P. Moran, *Rank correlation and product-moment correlation*, Biometrika, 35, 203–206, 1948, <https://doi.org/10.1093/biomet/35.1-2.203>. For the classical Gaussian identities and appendix proofs, see C. Croux and C. Dehon, *Influence functions of the Spearman and Kendall correlation measures*, Statistical Methods & Applications, 19, 497–515, 2010, Eqs. (1)–(5), <https://doi.org/10.1007/s10260-010-0142-z>.

The directional construction $A = \tilde{P}_3 - \tilde{P}_2$, $B = \tilde{Q}_3 - \tilde{Q}_4$ uses forecast sources $\{0, 1, 2\}$ and outcome sources $\{3, 4\}$, so both expectations vanish. In contrast, $A = \tilde{P}_2 - \tilde{P}_3$ and $B = \tilde{Q}_2 - (\tilde{Q}_0 + \tilde{Q}_1)/2$, with disjoint forecast and outcome peer sets, has $\Omega_2 = 0$ and $T_2 = -3/2$. For uniform ternary sources its empirical and population-reference means both equal $-13/162$. Removing the reference interaction does not remove this temporal target.

33.5 Unequal lags require indexed overlap

One scalar interaction coefficient suffices for the equal-weight sum against a raw one-period outcome because either matched component has the same distributional constant. It does not suffice for all nonlinear forecast classes. For fixed positive unequal weights, let

$$X_{is} = w_1 Y_{i,s-1} + w_2 Y_{i,s-2}, \quad H_w = \mathcal{L}(w_1 Y_1 + w_2 Y_2),$$

and define

$$\begin{aligned} \kappa_{F,\ell} &= \text{Cov}\{M_{H_w}(w_1 Y_1 + w_2 Y_2), M_F(Y_\ell)\}, \\ \eta_{F,\ell} &= \frac{1}{4} \mathbb{E}[\text{sgn}(w_1 D_1 + w_2 D_2) \text{sgn } D_\ell], \\ \lambda_{F,\ell} &= \eta_{F,\ell} - \kappa_{F,\ell}, \\ T^{(\ell)} &= \sum_s \left(\sum_j c_{sj} \right) \left(\sum_j d_{s-\ell,j} \right), \quad \Omega^{(\ell)} = \sum_s \sum_j c_{sj} d_{s-\ell,j}. \end{aligned}$$

Absent coefficients are zero. The same pair proof gives

$$\mathbb{E}[A_0 B_0] = \sum_{\ell=1}^2 \kappa_{F,\ell} T^{(\ell)}, \quad \mathbb{E}[AB] - \mathbb{E}[A_0 B_0] = \sum_{\ell=1}^2 \lambda_{F,\ell} \Omega^{(\ell)}. \quad (55)$$

For Gaussian sources, put $\rho_\ell = w_\ell / \sqrt{w_1^2 + w_2^2}$. Then

$$\kappa_{F,\ell} = \frac{\arcsin(\rho_\ell/2)}{2\pi}, \quad \lambda_{F,\ell} = \frac{\arcsin \rho_\ell - \arcsin(\rho_\ell/2)}{2\pi}.$$

Thus the unweighted sum $\Omega^{(1)} + \Omega^{(2)}$ discards relevant information. For an exact counterexample, use (54) with one common peer, Bernoulli(1/2) sources, and $(w_1, w_2) = (2, 1)$. Then

$$(T^{(1)}, T^{(2)}) = (\Omega^{(1)}, \Omega^{(2)}) = (1/2, -1/2), \quad (\kappa_{F,1}, \kappa_{F,2}) = (\lambda_{F,1}, \lambda_{F,2}) = (1/16, 1/32).$$

The naive total overlap is zero, yet the reference interaction is $1/64$, the population-reference expectation is $1/64$, and the empirical expectation is $1/32$. The superscripted coefficients index individual lags; their sums recover the equal-weight totals in (49). Heterogeneous laws across source times or peers likewise require indexed constants. Ranking raw sums on both score sides introduces additional overlap patterns and generally requires more than one constant.

33.6 Cross-focal covariance and classical scaling

At a matched component, view the pairs (X_i, Y_i) as independent bivariate entity vectors with the induced common law. Let U, V be fixed row-stochastic reference matrices with zero diagonal, and set $L_U = I - U$, $L_V = I - V$. Orient each unordered edge $e = (p, q)$ with $p < q$ and put $(B_U)_{pe} = U_{pq}$, $(B_U)_{qe} = -U_{qp}$, with other entries zero; define B_V in the same way. For the vectors of centered weighted comparisons p_s^U and q_t^V , the two-term rank projection gives

$$\text{Cov}(p_s^U, q_t^V) = \mathbf{1}\{t \in D_s\} \{\kappa_F L_U L_V^T + (\lambda_F - \kappa_F) B_U B_V^T\}. \quad (56)$$

Indeed, the node transforms are $M_H(X_i) - 1/2$ and $M_F(Y_i) - 1/2$, with cross-covariance κ_F . Subtracting the two node projections from an edge comparison leaves an antisymmetric kernel with zero conditional mean at either endpoint. Its same-edge cross-moment is $\eta_F - 2\kappa_F = \lambda_F - \kappa_F$. Different edges are uncorrelated even when they share an endpoint, by conditional degeneracy, and all node-edge cross-moments vanish. This proves (56) by classical projection orthogonality [5]. For distinct focals i, l the matched-source entry is

$$\kappa_F(e_i - U_i)^T(e_l - V_l) - (\lambda_F - \kappa_F)U_{il}V_{li}.$$

Independent raw entities therefore need not have independent rank scores.

With fixed N , uniform complete peers, and no ties in either ranked vector, the empirical rank variance with divisor N is deterministically $(N+1)/\{12(N-1)\}$. The standardized same-focal matched-source expectation is

$$\frac{12\{(N-1)\kappa_F + \lambda_F\}}{N+1} = \frac{(N-2)\rho_S + 3\tau}{N+1}, \quad (57)$$

where $\rho_S = 12\kappa_F$ and $\tau = 4\eta_F$ are the continuous-margin population Spearman and Kendall correlations of $(Y_1 + Y_2, Y_1)$. This is the classical expected sample Spearman identity, not a new standardization theorem [5, 6]. Tie-dependent random scale denominators remain outside the fixed-coefficient expectation identity.

33.7 Finite inference, validation, and raw cost

Let $S_g = A_g B_g$, $g = 1, \dots, M$, be independent whole-panel replicates using the same source law and fixed design. The estimand is $\theta_0 = \mathbb{E}[A_0 B_0]$, and $\mathbb{E}S_g = \theta_0 + \Omega_2 \lambda_F$. Individual entities within a rank panel are not independent score replicates. The target can itself retain temporal fitting bias and is not a forecast-loss gain or a general category-adjusted covariance.

Independently of these panels, take three two-component trajectories $Z = (Z_1, Z_2)$, $Z' = (Z'_1, Z'_2)$ and $Z'' = (Z''_1, Z''_2)$ whose six coordinates are independent draws from F . Form the validation kernel

$$R = \frac{1}{2}\{a(Z_1 + Z_2, Z'_1 + Z'_2) - a(Z_1 + Z_2, Z''_1 + Z''_2)\}\{a(Z_1, Z'_1) - a(Z_1, Z''_1)\}. \quad (58)$$

Conditional on Z , the primed and double-primed trajectories are independent copies. The independent-copy covariance identity therefore gives $\mathbb{E}[R \mid Z]$ equal to the shared-peer conditional comparison covariance, and $\mathbb{E}R = \lambda_F$. Each peer's two-component trajectory must be reused consistently across the two factors. The kernel lies in $[-1/2, 1/2]$, while its expectation lies in $[0, 1/8]$.

For n independent fresh validation replicates and fixed $0 < \delta < \xi < 1$, let

$$\epsilon_\lambda = \sqrt{\frac{\log(2/\delta)}{2n}}, \quad [l, u] = [\bar{R} - \epsilon_\lambda, \bar{R} + \epsilon_\lambda] \cap [0, 1/8].$$

If the intersection is empty, return $[0, 1/8]$. On the original coverage event the intersection contains λ_F , so this fallback preserves coverage and defines the procedure for every sample. A valid score-range width is

$$R_S = \frac{1}{2} \left(\sum_{s,j} |c_{sj}| \right) \left(\sum_{t,j} |d_{tj}| \right).$$

This follows from $|A| \leq \sum |c_{sj}|/2$ and $|B| \leq \sum |d_{tj}|/2$. For nonnegative normalized fit and peer weights, $R_S \leq 2$; this displayed bound is two when the evaluation row is excluded from both mean fits. It is a valid enclosing range width, not a claim that both endpoints of the actual score support are attained. Dependence or cancellation among the comparison terms can yield a tighter range. The lower bound is

$$L = \bar{S} - R_S \sqrt{\frac{\log\{1/(\xi - \delta)\}}{2M}} - \Omega_2 \begin{cases} u, & \Omega_2 \geq 0, \\ l, & \Omega_2 < 0. \end{cases} \quad (59)$$

Hoeffding's inequality [18] gives a two-sided validation failure probability at most δ and a one-sided panel-mean failure probability at most $\xi - \delta$. On their joint success event, $\bar{S} - R_S \sqrt{\log\{1/(\xi - \delta)\}/(2M)} \leq \mathbb{E}S_g$, and the interaction subtracted in (59) is at least $\Omega_2 \lambda_F$, including when $\Omega_2 < 0$. Thus

$$\Pr\{\theta_0 \geq L\} \geq 1 - \xi.$$

This is a pointwise finite-sample bound for the declared independent-panel experiment. It does not calibrate a dependent observational panel, and validation observations from another law do not identify λ_F .

When $\Omega_2 = 0$, validation is unnecessary and the full error budget ξ can be assigned to the panel mean. More generally, the proved parameter range gives the validation-free conservative bound

$$\bar{S} - R_S \sqrt{\frac{\log(1/\xi)}{2M}} - \frac{\max(\Omega_2, 0)}{8}.$$

For complete peers, $\Omega_2 = T_2/(N-1)$, so the maximum positive correction decreases with peer count at fixed temporal design. This fact alone does not establish a raw-cost advantage or sensitivity to arbitrary small targets.

To display the validation cost, let one independent panel consume c_M raw observations and one validation replicate consume six. With fixed budgets δ, ξ , and ignoring interval clipping, the two radius contributions are $a/\sqrt{M} + b/\sqrt{n}$, where

$$a = R_S \sqrt{\log\{1/(\xi - \delta)\}/2}, \quad b = |\Omega_2| \sqrt{\log(2/\delta)/2}.$$

For positive a, b and a continuous allocation satisfying $c_M M + 6n = C$, Lagrange differentiation gives

$$-\frac{a}{2M^{3/2}} + \nu c_M = 0, \quad -\frac{b}{2n^{3/2}} + 6\nu = 0, \quad \frac{n}{M} = \left(\frac{bc_M}{6a}\right)^{2/3}.$$

The objective is strictly convex on positive allocations, so the stationary point is its unique constrained minimum. Integer sample sizes require a feasible predetermined rounding. If $b = 0$, omit validation and allocate the budget to panels. This optimizes only the displayed conservative radius at a fixed error split, not statistical power or an unknown optimal procedure.

33.8 Verification and scope

The exact raw-array checker evaluates 183,523 complete temporal arrays, covering positive, negative and canceling interactions, source-separated directional fits, peer separation with a nonzero population target, and the unequal-lag counterexample. Independent exact calculations cover four-draw law constants, all nine entries of unequal-reference three-entity covariance matrices, and six-draw validation means for five tied laws: balanced and skew binary, uniform ternary, gapped uniform ternary, and skew ternary. Integer-scaled accumulation and rational arithmetic preserve ordering and prevent rounding-created ties. The checker does not import the estimator implementation. Its enumeration counts are checks of identities, not independent forecast tasks, error-rate simulations, or evidence of power.

Nine separate bound-API tests cover the validation moments and kernel endpoints, large-integer comparisons, sign-correct interaction subtraction, fixed error allocation, and malformed or overflowing observations. These are deterministic implementation checks; they do not establish observational-panel coverage. The supplied implementation and receipts state the tested numerical contract.

The additive-source calculation is a derived specialization of classical rank moments, Hoeffding projections, and bounded-mean inference. Its concrete content is the tie-aware formula $\eta_F = (1 - s_2^2)/8$, computable convolution constants, the source-window map beyond lag-copy ranks, sharp uniform cancellation criteria, and the failure of unweighted overlap for unequal lags. It is not a claim of priority for a general rank theorem, a new concentration principle, or unique advantage over equally informed valid procedures. Fixed coefficients, independent raw sources, and the declared reference law remain essential; general learned forecasts, dependent sources, and random tied standardization require additional analysis.

34 Equal-fold temporal feedback calculation

For $K \geq 2$ contiguous folds of $h \geq p$ rows, consider the complementary-fold entity-mean score of Proposition 4. A forecast equals a times the sum of its previous p available outcomes; at the beginning of the sequence, only outcomes within that sequence are available. Under the proposition's orthogonality assumptions, the total expected numerator is $a\sigma^2 Kp(h - p - 1)/\{(K - 1)h\}$.

Proof. For the closed form, let $S = a\{Khp - p(p + 1)/2\}$ be the sum of all forecast weights and $B = a(K - 1)p(p + 1)/2$ the sum crossing fold boundaries. Summing Eq. (17) yields $\sigma^2\{(K - 1)S - (2K - 1)B\}/\{(K - 1)^2h\}$, which simplifies to the stated value.

35 Conditional designs for learned trajectory maps

This section develops Proposition 1. Conditioning on external training fixes arbitrary measurable transformations, rather than requiring a linear or equal-lag forecast. The comparison identities, independent copy covariance, and rank U-statistics are classical [5, 6]. The construction below specifies the target, feasible reference assignments, observable interaction, and complete sampling costs. It does not assert a new concentration inequality or an efficiency advantage over the classical comparator.

35.1 Information, maps, and the exact operator

Let $\mathcal{H}$ fix maps ϕ_s, ψ_t , finite coefficient matrices $\mathbf{C} \in \mathbb{R}^{S \times J_{\text{ref}}}$, $\mathbf{D} \in \mathbb{R}^{T \times J_{\text{ref}}}$, and all sample sizes and error allocations. Independent external training can learn the maps and both coefficient matrices before the evaluated trajectories are drawn. Conditional on $\mathcal{H}$, complete trajectories $W_0, W_1, \dots$ are independent with a common law $P_{\mathcal{H}}$. Independence of external training and fresh trajectories is a sufficient special case. Within-trajectory dependence and unequal coordinate marginals are unrestricted. In a finite-archive experiment $\mathcal{H}$ also contains the archive, and independently sampled indices give the conditional product law. Repeated sampled archive records are permitted. This does not assert physical independence of historical days or justify a forecasting availability claim.

With $a(x, y) = \mathbf{1}\{y < x\} + \frac{1}{2}\mathbf{1}\{y = x\}$, write

$$p_{sj} = a(\phi_s(W_0), \phi_s(W_j)) - \frac{1}{2}, \quad q_{tj} = a(\psi_t(W_0), \psi_t(W_j)) - \frac{1}{2}, \quad A = \sum_{s,j} C_{sj} p_{sj}, \quad B = \sum_{t,j} D_{tj} q_{tj}.$$

For a normalized temporal fit, $C_{sj} = (\mathbf{1}\{s = r\} - w_{rs})u_{sj}$, with $\sum_s w_{rs} = 1$ and $\sum_j u_{sj} = 1$; the other channel is analogous. Signed fits give fixed comparison coefficients for each evaluated row. Several rows require their stated aggregate of products, as below. Constant maps contribute zero. A forecast learned on separate data is allowed as a whole map; ranking a nonlinear forecast is not replaced by a linear combination of component ranks.

Put $c = \mathbf{C}\mathbf{1}$, $d = \mathbf{D}\mathbf{1}$, $\mathbf{\Omega} = \mathbf{C}\mathbf{D}^T$ and $m_s(W_0) = \mathbb{E}[p_{s1} \mid W_0, \mathcal{H}]$, with n_t analogous. Define

$$K_{st} = \mathbb{E}[m_s n_t \mid \mathcal{H}], \quad \Lambda_{st} = \mathbb{E}[\text{Cov}(p_{s1}, q_{t1} \mid W_0, \mathcal{H}) \mid \mathcal{H}].$$

Conditioning on the focal trajectory and then averaging gives exactly, including ties,

$$\mathbb{E}[p_{sj} q_{tk} \mid \mathcal{H}] = K_{st} + \mathbf{1}\{j = k\} \Lambda_{st}, \quad \mathbb{E}[AB \mid \mathcal{H}] = \underbrace{c^T K d}_{\theta_{\mathcal{H}}} + \underbrace{\langle \mathbf{\Omega}, \Lambda \rangle}_{\gamma_{\mathcal{H}}}. \quad (60)$$

The target is the fitted population-reference residual association for this trajectory law. It is not automatically true conditional covariance, an earlier target under different reference laws, or future loss improvement. Before training, $\theta_{\mathcal{H}}$ is random; conditional coverage concerns this trained target, not its average over training samples. External fitting alone does not establish nuisance consistency or score orthogonality of the kind used in double machine learning [33].

The entries need not be nonnegative: $\phi(Y) = Y$, $\psi(Y) = -Y$, with $Y \sim \text{Bernoulli}(1/2)$, give $K = \Lambda = -1/16$. The common-law bound $|\Lambda_{st}| \leq 1/6$ follows by Cauchy–Schwarz from

$$\mathbb{E} \text{Var}\{a(X, X') \mid X\} = \{2 - 3 \sum_z p_z^2 + \sum_z p_z^3\}/12 \leq 1/6,$$

where the sums cover the atoms of the marginal law. If primitive coordinates are conditionally independent, maps depending on disjoint coordinate sets have $K_{st} = \Lambda_{st} = 0$. Only under this additional assumption may a source-overlap mask remove matrix entries. Distinct coordinate names do not establish independence: $W = (Y, Y)$ gives $K = \Lambda = 1/16$ for the two coordinate maps.

35.2 Sharp design conditions and a heterogeneous counterexample

For fixed $\mathbf{C}, \mathbf{D}$, preservation over every common trajectory law and every measurable map holds if and only if $\mathbf{\Omega} = 0$ entrywise. Necessity isolates one pair $\phi_s = \psi_t = Y \sim \text{Bernoulli}(1/2)$ and makes all other maps constant; its only nonzero interaction is $1/16$. Sufficiency is (60). For fixed maps and a fixed law, a nonzero matrix may still satisfy $\langle \mathbf{\Omega}, \Lambda \rangle = 0$; nonzero $\mathbf{\Omega}$ alone does not establish a nonzero interaction.

The stronger requirement $\mathbb{E}[AB \mid \mathcal{H}] = 0$ for every law and map holds if and only if

$$cd^T = 0 \quad \text{and} \quad \mathbf{\Omega} = 0.$$

For each isolated entry, binary and uniform ternary maps have $(K, \Lambda) = (1/16, 1/16)$ and $(2/27, 5/54)$; the two pairs are not proportional, so both coefficients must vanish. Since $cd^T = 0$ implies $c = 0$ or $d = 0$, one population residual is then identically zero. Uniform preservation is a useful goal; demanding centering without a substantive null removes every nontrivial target. For fixed windows under independent primitive coordinates, the corresponding conditions are needed only on the active overlap graph.

Here is an exact failure of signed scalar cancellation. Let $W = (Y_1, Y_2, Y_3)$ have independent Bernoulli coordinates with probabilities $(1/2, 1/2, 1/10)$, and take

$$\phi = (Y_1, Y_3), \quad \psi = (Y_1, Y_2, Y_3), \quad \mathbf{C} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1/2 & 0 \\ -1/2 & -1/2 \\ 1/2 & 0 \end{pmatrix}.$$

Both temporal coefficient sums are zero. The active entries $\Omega_{11} = 1/2$ and $\Omega_{23} = -1/2$ sum to zero, but $\Lambda_{11} = 1/16$, $\Lambda_{23} = 9/400$. Consequently

$$\gamma_{\mathcal{H}} = \theta_{\mathcal{H}} = 1/50, \quad \mathbb{E}[AB \mid \mathcal{H}] = 1/25.$$

The scalar condition from common-law equal-weight source windows therefore does not extend unchanged to heterogeneous maps. Disjoint same-law reference pools provide a feasible matrix-zero design while preserving c, d and the target. Under ordinary nonnegative peer weights, each relevant zero overlap requires disjoint peer supports.

For several evaluated rows, use a common dictionary of trajectory maps and let $S = \sum_r v_r A_r B_r$, with fixed row weights v_r . Then $T_{\star} = \sum_r v_r c_r d_r^T$ and $\Omega_{\star} = \sum_r v_r \mathbf{C}_r \mathbf{D}_r^T$ give $\mathbb{E}[S \mid \mathcal{H}] = \langle T_{\star}, K \rangle + \langle \Omega_{\star}, \Lambda \rangle$. Thus preservation over all maps and laws is equivalent to $\Omega_{\star} = 0$; universal centering additionally requires $T_{\star} = 0$. The aggregate target need not equal the product of averaged residuals.

35.3 One aggregate validator and finite block bounds

Three fresh trajectories suffice to estimate the full signed interaction:

$$\Delta p_s = a(\phi_s(W_0), \phi_s(W_1)) - a(\phi_s(W_0), \phi_s(W_2)), \quad Q = \frac{1}{2} \Delta p^T \Omega \Delta q, \quad \mathbb{E}[Q \mid \mathcal{H}] = \gamma_{\mathcal{H}}.$$

The same two references must occur in both differences. Four independent references across the factors would instead give expectation zero. Since centered comparisons lie in $[-1/2, 1/2]$, $|A| \leq \|\mathbf{C}\|_1/2$, $|B| \leq \|\mathbf{D}\|_1/2$ and $|Q| \leq \|\Omega\|_1/2$. Enclosing range widths are

$$W_S = \frac{1}{2} \|\mathbf{C}\|_1 \|\mathbf{D}\|_1, \quad W_Q = \|\Omega\|_1,$$

where matrix norms sum absolute entries. A tighter validation width can use $\max_{x \in \{-1,1\}^S, y \in \{-1,1\}^T} |x^T \Omega y|$. For the multiple-row aggregate replace Ω by $\Omega_{\star}$; the same three trajectories suffice. A score width is

$$\frac{1}{2} \sum_r |v_r| \|\mathbf{C}_r\|_1 \|\mathbf{D}_r\|_1.$$

Shared trajectories across rows form one block, not additional independent observations.

For m independent bounded observations X in a common interval of width W , set

$$h(W, m, \epsilon) = W \sqrt{\frac{\log(1/\epsilon)}{2m}}, \quad e(X, W, \epsilon) = \sqrt{\frac{2s_X^2 \log(2/\epsilon)}{m}} + \frac{7W \log(2/\epsilon)}{3(m-1)},$$

where the second formula requires $m \geq 2$ and s_X^2 is the ordinary unbiased sample variance. These are classical Hoeffding and empirical Bernstein bounds [17, 18]; Maurer and Pontil's Theorem 11 allows independent nonidentical observations. A valid predeclared hybrid is $r(X, W, \epsilon) = \min\{h(W, m, \epsilon/2), e(X, W, \epsilon/2)\}$; for $m = 1$ use $h(W, m, \epsilon)$. Zero known width has zero radius.

Let $M \geq 1$ evaluation blocks supply $S_b = A_b B_b$, and $n \geq 1$ independent validation triples supply Q_b . Explicitly, $\bar{S} = M^{-1} \sum_{b=1}^M A_b B_b$ and $\bar{Q} = n^{-1} \sum_{b=1}^n Q_b$. For fixed $0 < \delta < \alpha < 1$,

$$L = \bar{S} - r(S, W_S, \alpha - \delta) - \bar{Q} - r(Q, W_Q, \delta) \implies \Pr\{L > \theta_{\mathcal{H}} \mid \mathcal{H}\} \leq \alpha.$$

This follows by intersecting the score lower event and interaction upper event, and handles negative interactions without truncating them to zero. If $\Omega = 0$, omit validation and use the full α . Different known block widths permit Hoeffding radius $\{\log(1/\epsilon) \sum_b W_b^2 / (2M^2)\}^{1/2}$ for the average mean. Heterogeneous block laws require matched interactions, not a validator from an unrelated law.

The guarantee conditions on training and any fixed archive, not on realized validation outcomes. Evaluation concentration may additionally condition on independent validation, but its own coverage event generally cannot: ten mean-zero Rademacher validation observations all equal to -1 give a negative Hoeffding upper bound at $\delta = .05$. This positive-probability event has conditional coverage zero. Validation-selected designs therefore need a simultaneous family guarantee or separate selection data.

35.4 The complete classical comparator

For $J_{\text{ref}} \geq 2$, let $a_j = \sum_s c_s p_{sj}$ and $b_j = \sum_t d_t q_{tj}$. The same focal and peers give the unbiased direct statistic

$$D_0 = \frac{(\sum_j a_j)(\sum_j b_j) - \sum_j a_j b_j}{J_{\text{ref}}(J_{\text{ref}} - 1)}, \quad \mathbb{E}[D_0 \mid \mathcal{H}] = \theta_{\mathcal{H}}.$$

Using every trajectory as focal produces the classical complete degree-three rank U-statistic [5, 6]:

$$U_N = \frac{\sum_i [(\sum_{j \neq i} a_{ij})(\sum_{k \neq i} b_{ik}) - \sum_{j \neq i} a_{ij} b_{ij}]}{N(N-1)(N-2)}, \quad a_{ij} = \sum_s c_s \{a(\phi_s(W_i), \phi_s(W_j)) - \tfrac{1}{2}\}. \quad (61)$$

The fixed-focal width is $\|c\|_1 \|d\|_1 / 2$, but the full symmetric kernel has range width at most $W_U = \|c\|_1 \|d\|_1 / 6$. For one map pair, put $u_i = \phi_s(W_i)$, $v_i = \psi_t(W_i)$, $s_{ij} = \text{sgn}(u_i - u_j)$ and $t_{ij} = \text{sgn}(v_i - v_j)$. Its three-trajectory kernel is $\sum_{i,j,k \text{ distinct}} s_{ij} t_{ik} / 24$. Order $u_1 \leq u_2 \leq u_3$. The numerator is $-2t_{13}$ with no u ties, $-(t_{13} + t_{23})$ when $u_1 = u_2 < u_3$, $-(t_{12} + t_{13})$ when $u_1 < u_2 = u_3$, and zero when all are tied. Its absolute value is at most $1/12$. Bilinearity and averaging over triples prove the stated width for every N . The scalar bound is attained; the coefficient bound may remain loose for particular fitted maps.

At the same observed cost, this comparator receives all evaluation blocks and validation triples. For $J_{\text{ref}} \geq 2$ each supplies an independent $U_{J_{\text{ref}}+1}$ or U_3 value with the same mean and range, so their pooled block average can use the preceding independent, nonidentical bound. The overlapping triples inside a block are not independent observations.

If all raw draws are conditionally IID, the stronger complete statistic can instead use every draw in (61). Its center is computable without quadratic storage. Put $A_{is} = \sum_j p_{sij}$, the centered sample midrank, and let $K_{st}^{\pm}$ count concordant minus discordant unordered pairs of map columns; ties in either coordinate contribute zero. Antisymmetry gives

$$U_N = \frac{\sum_i (c^T A_i)(d^T B_i) - \frac{1}{2} \sum_{s,t} c_s d_t K_{st}^{\pm}}{N(N-1)(N-2)}.$$

Sorting midranks and accumulating concordance counts with a Fenwick tree costs $O((S+T+ST)N \log N)$ time and $O((S+T)N)$ storage. Querying each tied-first-coordinate group before insertion preserves both types of ties.

For finite calibration let $q = \lfloor N/3 \rfloor \geq 2$, and let s^2 be the sample variance of the first q prespecified disjoint symmetric triple kernels. With $x = \log(2/\alpha)$, a valid complete-U radius is

$$r_U = \min \left\{ W_U \sqrt{\frac{x}{2q}}, s \sqrt{\frac{2x}{q}} + \left[\frac{2W_U}{\sqrt{q(q-1)}} + \frac{W_U}{3q} \right] x \right\}. \quad (62)$$

To prove it, write the complete U-statistic as the average, over permutations, of means of q disjoint kernels. Jensen's inequality transfers the classical Hoeffding and Bernstein moment bounds [18]. If the kernel variance is σ^2 , the deterministic radii $W_U \sqrt{x/(2q)}$ and $\sigma \sqrt{2x/q} + W_U x/(3q)$ each have failure at most e^{-x} ; their smaller deterministic value has the same guarantee. Theorem 10 of Maurer–Pontil [17] gives $\sigma \leq s + W_U \sqrt{2x/(q-1)}$ except on another event of probability e^{-x} . Substitution and a union bound yield (62). The variance and complete center may share the same draws. For $q = 1$ use pure Hoeffding with $\log(1/\alpha)$; for $W_U = 0$ use zero radius. This pooled construction requires IID raw trajectories: the nonidentical block argument alone does not authorize arbitrary pooling of raw laws.

35.5 Target-preserving construction and executable checks

A reference redesign keeps the fitted target when it retains the maps, the reference law and both row sums c, d . One explicit construction is

$$\mathbf{C}_{\text{new}} = [c, 0], \quad \mathbf{D}_{\text{new}} = [0, d].$$

Two independent same-law reference roles therefore suffice for universal preservation. If both c and d are nonzero, one role cannot suffice: with one column, $\mathbf{C} = c$, $\mathbf{D} = d$ and $\mathbf{CD}^T = cd^T \neq 0$. This is an elementary feasibility boundary for the specified design, not a minimax efficiency bound. The new roles require actual independent draws; relabeling a reused observation does not create them.

The released `design_contract.py` builds matrices from declared map IDs, reference draw-role IDs and weights. Repeated terms with one reference ID add in the same column. The target-preservation check retains map signatures and reference-law declarations and compares row sums in exact rational arithmetic. It also checks the exact row sums of the

generated binary floating coefficients: three separately rounded $1/3$ weights need not equal one stored unit weight. A re-assignment with that discrepancy is rejected instead of being reported as numerically target preserving. Distinct coordinate names or reference IDs are not evidence of stochastic independence.

For a concrete target distinction, let the forecast, outcome and named baseline all equal $Y \sim \text{Bernoulli}(1/2)$. The centered population midrank $r(Y)$ takes values $-1/4, 1/4$. If each fitted baseline coefficient is $1/2$, both fitted residuals equal $r(Y)/2$, giving

$$\theta_{\mathcal{H}} = 1/64 \quad \text{with disjoint references,} \quad \theta_{\infty} = 0 \quad \text{when fully conditioning on the baseline.}$$

This example makes reference correction and baseline adjustment separate, checkable operations. With the baseline itself defining the categories, the category-adjusted population-rank target is also zero.

The distinction also occurs with the unit baseline coefficients used in the application. Let the baseline U be uniform on $\{-1, 0, 1\}$ and let both forecast and outcome equal U^2 . With row sums $c = d = (1, -1)$, the centered population-rank contrasts at $U = -1, 0, 1$ are $1/2, -1/3, -1/6$. Their mean squared value is

$$\theta_{\mathcal{H}} = \frac{1}{3} \left(\frac{1}{4} + \frac{1}{9} + \frac{1}{36} \right) = \frac{7}{54}.$$

Both fully baseline-adjusted targets are zero because forecast and outcome are deterministic given U . Disjoint references remove their interaction without changing this positive fitted target. Thus the distinction is not confined to half-weighted fits or an exact identity copy.

A second example uses only nonconstant maps. Let U, V be independent fair Bernoulli variables, $\phi = \psi = (U, V)$ and $\mathbf{C} = \mathbf{D} = (1, -1)^T$. Then

$$\mathbf{\Omega} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \Lambda = I_2/16, \quad \sum_{s,t} \Omega_{st} = 0, \quad \gamma_{\mathcal{H}} = \theta_{\mathcal{H}} = 1/8.$$

The empirical product has mean $1/4$. This refutes summing all matrix entries; its active-overlap diagonal sum is nonzero. The heterogeneous example above provides the stronger failure in which active scalar overlap also cancels. Conversely, independent U, V with $\phi = U, \psi = V$ and one shared reference have $\mathbf{\Omega} = \mathbf{1}$ but $K = \Lambda = 0$: nonzero design interaction is not itself evidence of bias for a particular map and law.

35.6 A matched-budget range comparison

Assume positive integers $M, n, J_{\text{ref}}, 0 < \delta < \alpha < 1$ and $W_S > 0$. Give the complete U-statistic all $N_{\text{tot}} = M(J_{\text{ref}} + 1) + 3n$ conditionally IID trajectory draws, and put $q = \lfloor N_{\text{tot}}/3 \rfloor$. Its pure-Hoeffding radius and the split evaluation radius satisfy

$$\frac{r_{\text{pool}}}{r_S} = \frac{\|c\|_1 \|d\|_1}{3\|\mathbf{C}\|_1 \|\mathbf{D}\|_1} \sqrt{\frac{M \log(1/\alpha)}{q \log\{1/(\alpha - \delta)\}}} < \frac{1}{\sqrt{6}}. \quad (63)$$

Indeed, $q \geq \lfloor 2M/3 \rfloor + 1 > 2M/3$, row-sum norms do not exceed entrywise matrix norms, and the logarithm ratio is below one. If $c = 0$ or $d = 0$, the pooled radius is zero. If $W_S = 0$, both radii are zero and the ratio is undefined. The split rule additionally pays a nonnegative correction radius. Equation (63) concerns these two range-only radii; it does not compare different realized centers, empirical-Bernstein alternatives, rejection decisions or power.

Thus complete U is a direct estimator to use when the same raw trajectories can be recombined, while the signed correction explains interference in an existing construction. When the complete finite archive is already available, its exact target census is another legitimate comparator. Any efficiency claim for split correction requires an actual information or computation constraint, with its cost stated separately.

35.7 Fixed-budget inversion and numerical precision

For the pure-Hoeffding rule, fix δ before outcomes and put $G = \bar{S} - \bar{Q} - W_Q \sqrt{\log(1/\delta)/(2n)}$. For $W_S > 0$, the one-sided inversion for $H_0 : \theta_{\mathcal{H}} \leq 0$ is

$$p_{\delta} = \begin{cases} \min\{1, \delta + \exp(-2MG^2/W_S^2)\}, & G > 0, \\ 1, & G \leq 0. \end{cases} \quad (64)$$

This is the infimum of levels $\alpha \in (\delta, 1)$ at which the corresponding lower bound is positive, with value one if none exists. The fixed correction event and score concentration prove validity at each level $\alpha > \delta$; smaller levels cannot reject. When $\mathbf{\Omega} = 0$,

omit validation and set effective $\delta = 0$. For zero score width, the generic inversion is δ if $G > 0$ and one otherwise; valid inputs for a zero-width design here force $G = 0$.

The released `fixed.delta.inference.py` implements (64). It evaluates exact rational input means and coefficient widths using directed high-precision decimal arithmetic, then rounds the final floating p value upward. Tiny positive tails do not silently become zero. Declared whole-family Bonferroni thresholds round downward; unrepresentable positive thresholds trigger an explicit error. Inputs outside the asserted symmetric kernel support are rejected. Justified numerical mean-error allowances can be subtracted from G . The outward-arithmetic guarantee covers inversion of supplied values; floating kernel calculations require their own error allowance for a complete numerical guarantee. No outcome-dependent optimization over δ is used.

This API implements the range-only inversion. The Beijing study uses the preselected hybrid bound at fixed marginal and family levels, with separate union allocations for the two concentration rules. Its reported decisions are therefore not numerical outputs of the pure-Hoeffding inversion.

35.8 Learning and information available at forecast issue

Independent external training may learn both maps and coefficient matrices; all these objects are then fixed by $\mathcal{H}$. A valid forecast additionally requires its inputs and fitted model to be available at issue time. The released availability validator checks declared, timezone-aware issue, target, feature observation, feature release and model-availability times. It rejects a future feature, a delayed fitted model, or known training-source reuse. These checks verify supplied declarations, not actual measurement latency or the conditional IID assumption. Outcome maps may contain later realized outcomes; forecast maps may not use those outcomes as unavailable features.

Foldwise training separation alone does not prove independent fold scores. For independent Rademacher X_1, X_2 , let fitted rules be $m_1 = X_2, m_2 = X_1$ and evaluation outputs $T_1 = X_1 m_1, T_2 = X_2 m_2$. Each fit is independent of its evaluation variable, but $T_1 = T_2$ exactly. Combining individually valid fold bounds by a predeclared union allocation can remain valid; treating fold outputs as independent observations requires another argument. This distinguishes the stated conditional trajectory experiment from an unproved extension to arbitrary cross-fitting.

35.9 Access, cost, and limitations

The validated design uses exactly $M(J_{\text{ref}} + 1) + 3n$ trajectory draws. For maps requiring L distinct primitive coordinates per trajectory, the coordinate draw count is $L\{M(J_{\text{ref}} + 1) + 3n\}$; irregular windows require their actual coordinate union. Repeated draws, unique archive records, full ingestion, training, map evaluation and inference computation remain separate reported costs. This is the access needed by the construction, not a universal lower bound. If the full archive is already read, its exact rank-target census is also an available alternative; resampling does not create label acquisition savings.

For a Hoeffding-only validated bound with fixed δ , let $a = W_S \sqrt{\log(1/(\alpha - \delta))/2}$, $b = W_Q \sqrt{\log(1/\delta)/2}$. Minimizing $a/\sqrt{M} + b/\sqrt{n}$ under $c_M M + c_R n = B$, for positive a, b , gives $n/M = (bc_M/(ac_R))^{2/3}$. The exact integer minimum follows by enumerating feasible M and taking the largest feasible n for each. This optimizes the displayed bound only and must precede outcomes.

Three draws are the minimal generic order of an unbiased fixed kernel without additional law information: for $\phi = \psi = Y$ with ordered masses $(t, (1-t)/2, (1-t)/2)$, $\theta = \{1 - t^3 - (1-t)^3/4\}/12$ has a nonzero cubic coefficient. Any fixed statistic of at most two IID draws has expectation of degree at most two in t . Equation (61) attains order three. This is not a minimax testing bound, and binary support or known transforms can supply extra information.

Finally, if peer j has a different law $P_{j,\mathcal{H}}$, preserve its indexed conditional rank means in the target. Two independent copies of each such peer law yield the explicit validator

$$Q_{\text{het}} = \frac{1}{2} \sum_j \left\{ \sum_s C_{sj}(p'_{sj} - p''_{sj}) \right\} \left\{ \sum_t D_{tj}(q'_{tj} - q''_{tj}) \right\}.$$

Its mean is the indexed interaction, its width is at most $\sum_j \|C_{:j}\|_1 \|D_{:j}\|_1$, and its cost is one focal plus $2J_{\text{ref}}$ matched-law draws. One historical record per heterogeneous source does not provide these independent copies. Grouping or pooling references changes the target unless the declared reference laws and mixture masses are preserved. For example, a Bernoulli(1/2) focal has squared centered midrank expectation 1/16 under its own reference law, but 5/64 under a half mixture of that law and its shift by ten. Even mean recentring gives variance 1/64. External fitting therefore fixes an auditable design; it does not remove reference-law changes, nuisance errors, or the need for separate chronological decision validation.

35.10 Additional examples for the main reference results

For $X_{i1} = 0$, $X_{it} = Y_{i,t-1}$, fit times 1, 3 and evaluate time 2. Uniform peers give $\mathcal{T} = -1$, $\Omega = -1/(N-1)$; Bernoulli(1/2) with $N = 3$ gives population/empirical expectations $-1/16$ and $-3/32$.

For the estimated-mean corollary, with training size m , for fixed finite controls with positive masses, empirical-midrank cell means are ratios of bounded order-two U-statistics and cell proportions. Their root- m rate yields the product condition when $m/\sqrt{MN} \rightarrow \infty$.

Other conditional association tests include kernel [9] and partial-copula [34] methods. Recent calibration work [35] studies further practical departures. These broader antecedents complement the main paper’s closest-target comparisons.

36 Learned-trajectory forecasting study

The Beijing PM2.5 source [36] contains 43,824 hourly records from 2010 to 2014. A protocol was committed locally before downloading its raw observations. Only provider schema and licensing metadata had been consulted. This was local prespecification, not public preregistration. The released source ZIP is unchanged under CC BY 4.0, with creator attribution.

36.1 Chronology, models, and execution correction

All models train on 23,006 complete training records from 2010–2012. Inputs comprise PM2.5 lags 1, 2, 3, 6, 12, 24; previous-hour temperature, pressure, dew point, cumulative wind speed, snow, rain and wind direction; and the known target hour, weekday and month. Missing target or PM2.5 lag rows are excluded. Meteorological imputation and numeric scaling use training statistics. There are no future-weather inputs. Timestamps establish the lag order; the source does not document publication latency, so availability of previous-hour measurements at the forecast origin is an assumption.

The eight candidates are persistence, Ridge with penalties 1 and 100, histogram gradient boosting on persistence residuals with 7 or 31 leaves, and equal persistence blends with Ridge-1 or either boosting model. Boosting uses 100 iterations, learning rate 0.05, L2 penalty 1, no early stopping and seed 20260914. All predictions are clipped below zero; primary MSE and MAE are otherwise in raw measurement units, without upper clipping.

The protocol specifies origin 14:00 and next-hour target 15:00. The initial implementation filtered target 14:00, giving 339 selection records in 2013 and 349 confirmation records in 2014. Independent code review detected this one-hour mismatch after initial losses had been viewed. Re-execution at the specified target hour gives 342 and 350 records, respectively, with unchanged models and selection rules. Both executions are preserved. The correction is not a new untouched confirmation, and its result is not chosen between hours.

36.2 Reference experiment and strong comparators

A trajectory W holds the frozen forecast’s input history, its prediction and the corresponding target. The issue time is 14:00 and target time is 15:00. The used hourly rows have offsets 0, -1 , -2 , -3 , -6 , -12 , -24 from the target: seven hourly records spanning 24 hours. Weather and wind direction are at offset -1 ; target-hour calendar fields are known at issue. These comprise 16 logical input features and one target; 15 numeric input features are stored in the forecast arrays, while wind direction is retained in the attributed raw source. The selection trajectories involve 2,069 distinct raw hourly records. The conditional law is uniform on the fixed selection archive. Independent replacement index draws implement the conditional independence assumption; original days need not be independent. For candidate h , persistence baseline b and outcome Y , define $\phi = (h, b)$, $\psi = (Y, b)$ and

$$\mathbf{C} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{D}_{\text{shared}} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{D}_{\text{separate}} = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{D}_{\text{crossed}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

All designs keep $c = d = (1, -1)^T$. The target is the product of the two population-midrank differences. The maps are frozen before archive index draws; conditioning fixes this archive experiment, not the law of future days. It is not full adjustment for the baseline or a theorem of future MSE improvement. Three two-peer designs fix the same target: shared, separate and crossed reference assignments. One three-trajectory kernel estimates their signed reference interactions. Exact archive midranks provide a census target.

There are 300 fixed-seed replications per candidate and design: 1,024 evaluation triples and 512 validation triples at marginal level 0.05, with validation budget 0.025. A fixed union-budget minimum of Hoeffding and empirical Bernstein radii is used. The classical triple comparator uses all 1,536 triples and its sharper six-role support width. Each cohort has 9,600 rows in 32 cells. In the corrected cohort, four cells have target zero and 28 have positive target, up to 0.000948129. All cells record zero noncoverage and zero detections at this small budget. For 0/300, the two-sided exact 95% upper endpoint

is approximately 0.01222. Shared seeds pair designs; cells are not pooled as independent replications. These results show conservative coverage accompanied by no detection at the diagnostic budget. Absence of observed failures does not establish zero error probability. All 32 corrected cells are reported below; a one-sided interval is unbounded above, so the finite quantities reported are kernel range widths and lower-bound radii, not total confidence-interval widths.

Selection uses 16,384 evaluation and 8,192 validation triples, or 73,728 trajectory draws per candidate. Bonferroni assigns 0.05/8 to each of the eight candidates. The shared-score certificate assigns half its marginal budget to validation. Each gate chooses the smallest selection MSE among positive lower bounds, otherwise persistence; ungated selection minimizes the same MSE. The same index arrays are shared across candidates for paired comparison. Multiplying the per-candidate draw count by eight does not count independent new labels: all 342 existing selection records are accessed, and archive resampling acquires no new labels. The selection seed also occurs in diagnostic replication 99, as fixed in the protocol. It is not an independent additional simulation stream.

The initial classical comparator symmetrizes within each triple. A subsequent stronger analysis recombines every already sampled evaluation and validation trajectory in the complete U-statistic using the same target and total draws. Its disjoint-triple variance and pooled center share observations legally under the joint bound above. A further census comparison uses all accessible selection outcomes. On the corrected archive, per-candidate census computation takes 0.010–0.013 seconds; complete U takes 0.430–0.439 seconds. These are single-run wall times, not acquisition-cost evidence. Only 342 distinct selection records exist. The nominal 17-coordinate count denotes 16 inputs plus target per record, not measured array access operations; training records and computation are additional.

36.3 Selection evidence and chronological loss

The following table uses the corrected hour. Values are rounded for display; all unrounded arrays, exact targets and replication records are released.

Table 35: Fitted-rank target and simultaneous lower bounds.

Candidate	Census target	Signed	Triple U	Pooled U
Persistence	0.000000	-0.010190	-0.000409	-0.000365
Ridge 1	0.000948	-0.015048	0.000141	0.000146
Ridge 100	0.000887	-0.014660	0.000165	0.000105
Boosting 7	0.000572	-0.015389	-0.000106	-0.000080
Boosting 31	0.000852	-0.015367	0.000114	0.000155
Blend Ridge 1	0.000456	-0.013258	-0.000171	-0.000207
Blend boosting 7	0.000281	-0.013927	-0.000341	-0.000262
Blend boosting 31	0.000374	-0.013822	-0.000288	-0.000202

At the corrected hour, triple U, pooled U, census and ungated selection choose boosting with 31 leaves. Its confirmation MSE is 295.9390; the signed-reference gate reverts to persistence with MSE 288.3200. Thus the latter has a 2.5745% point improvement over ungated selection. A subsequent reporting check gives the matching target-weighted absolute MSE gain 7.6190 with descriptive 95% interval $[-36.4804, 51.3807]$; its relative gain interval is $[-18.8887\%, 11.9981\%]$. The originally specified equal-week paired-loss difference is 6.5342 with interval $[-35.0218, 48.6297]$. The gate and persistence baseline are identical, with zero gain and a degenerate zero contrast interval. Stable incremental benefit is not established; the intervals spanning zero also do not establish an absence of benefit. In the initial target-14 cohort, ungated boosting has MSE 317.7244 against persistence 455.5874, a 30.2605% improvement; triple U selects Ridge-1 with MSE 380.2824, and the signed gate selects persistence. The contrast across hours reinforces the need to report the implementation correction.

Intervals use paired, noncircular four-week contiguous blocks, 2,000 draws and seed 2026091455. The target-weighted interval resamples the paired squared-loss sums and record counts, then recomputes their ratio; the relative interval recomputes the paired MSE improvement divided by comparator MSE in each draw. It therefore matches the record-weighted primary MSE, rather than treating each week equally. This additional interval was calculated after inspection of the corrected outcomes; it is not a new prespecified or untouched confirmation. The original equal-week endpoint remains separately reported in squared PM2.5 units, not percentage points. The 53 weekly groups are not asserted independent, and the intervals are descriptive, without finite time-series coverage claims. Every candidate's MAE, MSE, weekly losses and both rule-comparator contrasts is released. No unfavorable run or candidate is removed. Independent checks reconstructed features from raw timestamps, reproduced the initial preparation exactly, recomputed pooled-U centers by finite-support multiplicities, and reproduced all frozen decisions and bootstrap values. A separate reconstruction of all 9,600 corrected diagnostic rows has maximum absolute discrepancy below 1.12×10^{-16} ; final bounds and complete-U centers agree within 1.0×10^{-16} .

Checks of all 692 selection and evaluation records match target values, persistence and stored numeric features to the raw timestamps. These are independent recomputations of inspected results, not new observations.

36.4 Correction and uncertainty decomposition

Table 36 separates the shared-score mean, the signed interaction estimate and both uncertainty radii. The unchanged fixed-level hybrid is used. For boosting with 31 leaves, the pure-Hoeffding split lower bound is -0.101208 . The hybrid’s Hoeffding component, which allocates half of each failure budget, gives -0.107139 , compared with the reported hybrid value -0.015367 . This numerical distinction matters when interpreting the range-only theoretical comparison.

Table 36: Corrected archive selection: shared-score decomposition and pooled radius.

Candidate	$\bar{S}$	$\bar{Q}$	Score radius	Correction radius	Split lower	Pooled radius
Persistence	0.000000	0.000000	0.002038	0.008152	-0.010190	0.000365
Ridge 1	0.014435	0.012344	0.005588	0.011550	-0.015048	0.000792
Ridge 100	0.014496	0.012039	0.005599	0.011517	-0.014660	0.000776
Boosting 7	0.006439	0.006744	0.004460	0.010623	-0.015389	0.000653
Boosting 31	0.008530	0.008148	0.004805	0.010943	-0.015367	0.000692
Blend Ridge 1	0.008026	0.005997	0.004734	0.010553	-0.013258	0.000661
Blend boosting 7	0.002594	0.003052	0.003648	0.009821	-0.013927	0.000543
Blend boosting 31	0.003647	0.003571	0.003908	0.009990	-0.013822	0.000576

36.5 All corrected diagnostic cells

For the shared, separate and crossed designs, score width is 2 and interaction widths are 4, 0, 2, respectively. The triple-U width is $2/3$. These are enclosing kernel widths; exact cancellation in a particular candidate can make the attained support smaller. In Table 37, every row contains 300 replications with 0 noncoverage and 0 detections. The correction-radius column is zero for designs requiring no correction. Targets are multiplied by 10^3 for readability; radii and bounds use the unscaled rank-product units.

Table 37: All 32 corrected diagnostic cells at $M = 1,024$, $n = 512$ and marginal level 0.05.

Candidate	Design	$10^3\theta_{\mathcal{H}}$	$10^3\gamma_{\mathcal{H}}$	Mean score radius	Mean correction radius	Mean lower
Persistence	Shared	0.000000	0.000000	0.023152	0.092697	-0.115849
Persistence	Separate	0.000000	0.000000	0.019990	0.000000	-0.019990
Persistence	Crossed	0.000000	0.000000	0.023152	0.046349	-0.069500
Persistence	Triple U	0.000000	0.000000	0.004441	0.000000	-0.004441
Ridge 1	Shared	0.948129	14.496794	0.035408	0.104791	-0.138973
Ridge 1	Separate	0.948129	0.000000	0.024690	0.000000	-0.023778
Ridge 1	Crossed	0.948129	13.786931	0.035232	0.057988	-0.092210
Ridge 1	Triple U	0.948129	0.000000	0.005823	0.000000	-0.004902
Ridge 100	Shared	0.887306	14.155784	0.035223	0.104690	-0.138838
Ridge 100	Separate	0.887306	0.000000	0.024452	0.000000	-0.023657
Ridge 100	Crossed	0.887306	12.860932	0.034721	0.057622	-0.091711
Ridge 100	Triple U	0.887306	0.000000	0.005758	0.000000	-0.004894
Boosting 7	Shared	0.571738	7.157135	0.031967	0.101216	-0.132337
Boosting 7	Separate	0.571738	0.000000	0.022946	0.000000	-0.022478
Boosting 7	Crossed	0.571738	5.960398	0.031288	0.054109	-0.084763
Boosting 7	Triple U	0.571738	0.000000	0.005392	0.000000	-0.004803
Boosting 31	Shared	0.852133	9.056899	0.033088	0.102279	-0.134132
Boosting 31	Separate	0.852133	0.000000	0.023759	0.000000	-0.022935
Boosting 31	Crossed	0.852133	8.360385	0.032785	0.055481	-0.087239
Boosting 31	Triple U	0.852133	0.000000	0.005582	0.000000	-0.004720
Blend Ridge 1	Shared	0.455562	7.858962	0.032183	0.101638	-0.133195
Blend Ridge 1	Separate	0.455562	0.000000	0.023061	0.000000	-0.022620
Blend Ridge 1	Crossed	0.455562	6.498026	0.031398	0.054431	-0.085533

Candidate	Design	$10^3\theta_{\mathcal{H}}$	$10^3\gamma_{\mathcal{H}}$	Mean score radius	Mean correction radius	Mean lower
Blend Ridge 1	Triple U	0.455562	0.000000	0.005373	0.000000	-0.004916
Blend boosting 7	Shared	0.281438	3.266662	0.029115	0.098541	-0.127293
Blend boosting 7	Separate	0.281438	0.000000	0.021722	0.000000	-0.021489
Blend boosting 7	Crossed	0.281438	2.279329	0.028346	0.051359	-0.079435
Blend boosting 7	Triple U	0.281438	0.000000	0.005056	0.000000	-0.004777
Blend boosting 31	Shared	0.374003	4.507841	0.030126	0.099430	-0.128965
Blend boosting 31	Separate	0.374003	0.000000	0.022177	0.000000	-0.021807
Blend boosting 31	Crossed	0.374003	3.323847	0.029387	0.052268	-0.081165
Blend boosting 31	Triple U	0.374003	0.000000	0.005171	0.000000	-0.004796

36.6 Active references and the limit of sequential correction

The following consequences clarify costs and sampling scope; they are not new concentration or minimax results.

Let $I = \{j : \mathbf{C}_{:,j} \neq 0 \text{ or } \mathbf{D}_{:,j} \neq 0\}$. Deleting columns outside I preserves both row sums, $\mathbf{CD}^T$, and the scores pointwise when the retained roles use the same draws. It also preserves the entrywise norms, hence the declared range widths. Each evaluation block then costs $g = 1 + |I|$ used trajectories. In particular, $\mathbf{C} = \mathbf{D} = ((1, 0), (-1, 0))$ requires two used draws per evaluation block, not three. This follows by deleting zero summands; the retained roles keep their conditional IID law. Columns used by either channel cannot be deleted by this argument.

For a fixed range rule write its radius as $a/\sqrt{M} + b/\sqrt{n}$, where $a = W_S \sqrt{\log\{1/(\alpha - \delta)\}/2}$ and $b = W_Q \sqrt{\log(1/\delta)/2}$. For $a, b > 0$, under $gM + 3n \leq N$, the continuous optimum obeys

$$\frac{M}{n} = \left(\frac{3a}{gb}\right)^{2/3}.$$

The Lagrange equations are $aM^{-3/2}/2 = \lambda g$ and $bn^{-3/2}/2 = 3\lambda$; strict convexity gives uniqueness. For integer budgets, enumerate all admissible n and take $M = \lfloor (N - 3n)/g \rfloor$. This allocation uses prespecified widths and levels, without inspecting outcomes. It does not optimize a hybrid radius chosen from the observed sample, nor establish a power advantage over complete U.

Unbiased correction is insufficient for sequential validity. Let five latent pairs (X_i, Y_i) , $i = 0, \dots, 4$, be IID, each with independent fair Bernoulli coordinates, and repeat these five pairs periodically. The resulting process is strictly stationary. Every consecutive five-observation tuple has marginally IID roles; aligned disjoint five-observation rounds coincide pointwise. Use the first two roles for the score and the last three for its independent validation correction, with $\phi = X$, $\psi = Y$ and $C = D = 1$:

$$S = \frac{(X_0 - X_1)(Y_0 - Y_1)}{4}, \quad Q = \frac{(X_4 - X_3)(Y_4 - Y_3)}{8}.$$

The validation focal cancels in this binary comparison difference. The marginal fitted target is $\theta_{\mathcal{H}} = 0$, and $R = S - Q$ has mean zero and variance $5/256$. Yet $R_b = R$ in every block. Thus the putative betting process $E_t = \prod_{b \leq t} (1 + R_b)$ satisfies $\mathbb{E}E_2 = 261/256 > 1$ and is not an e-process for that target. Its eventual threshold-crossing probability is $\Pr(R > 0) = 7/32$, which exceeds 0.05. These values follow by enumerating 2^{10} binary configurations, or by convolving the independent three-point laws of S, Q .

Valid sequential inference requires a filtration-specific assertion such as $\mathbb{E}[R_t \mid \mathcal{F}_{t-1}] = \theta_t$, with predictable maps, bounds and acquisition rules. Fresh focal and reference trajectories that are conditionally IID under the same declared law are sufficient; the unconditional Q identity is not. Classical methods can instead infer the average of $\mathbb{E}[R_t \mid \mathcal{F}_{t-1}]$ for bounded adapted scores, but that changes the target unless the required identity is established. Sequential forecast-score comparisons [37] therefore provide a distinct alternative, not an automatic extension of the trajectory certificate.

36.7 Using every acquired reference draw

The diagnostic in `results/trajectory_budget/` keeps the corrected Beijing archive, all eight fits and all 73,728 original draw positions. Both coefficient matrices are $((1, 0), (-1, 0))$. The unused second column accounts for 16,384 draw positions: deleting it at the original block counts preserves every score and bound, reducing used positions from 73,728 to 57,344. Both access all 342 selection records and 2,069 original hourly rows. There are no newly acquired labels.

For a same-budget comparison, integer minimization of the declared range radius with widths $W_S = 2, W_Q = 4, \alpha = 0.05/8$ and $\delta = 0.05/16$ gives $M = 13,086, n = 15,852$. Flattening the original evaluation then validation indices in

recorded order and repartitioning them into pairs and triples uses exactly the same independent draw positions. The allocation depends only on widths and levels, not on observed scores or variances. Its design followed inspection of earlier results; it is a diagnostic comparison rather than a new independent confirmation.

For boosting with 31 leaves, the original hybrid radius 0.01574802 falls to 0.01212612, a 23.00% reduction. Its new score mean is 0.01035458, interaction estimate 0.00847685, score radius 0.00588669 and validation radius 0.00623944, giving lower bound -0.01024839 . The center changes because the draw roles change. The corresponding pure range radius falls from 0.10158980 to 0.08364628, a 17.66% reduction. All shared-reference gates still retain zero candidates. Every pure range gate, including triple and complete U, also retains zero; the three positive triple/complete-U bounds use their hybrid/variance rules.

All eight targets, certificates, costs and subsequent losses are retained. Exact census finds positive fitted targets for all seven augmentations, but four have higher observed future MSE than persistence. Blends of boosting 7 and boosting 31 have future MSE 283.4075 and 282.8837, respectively, below persistence's 288.3200; these inspected rankings were not used to select another rule. Triple U, complete U, census and ungated selection all choose boosting 31, with MSE 295.9390. The shared rule chooses persistence and contributes zero improvement over choosing persistence directly.

The paired bootstrap uses 53 Monday–Sunday weeks, 2,000 draws of noncircular four-week blocks and seed 2026091455. Each concatenation is truncated to 53 weeks. Weekly counts range from three to seven. The primary loss contrast divides resampled loss totals by resampled record counts; the original equal-week endpoint remains separate. These pointwise descriptive intervals provide no finite-coverage guarantee for the physical series. The package contains the full bootstrap indices and all future loss comparisons, as well as an independent finite-support reconstruction of the reference kernels.

36.8 Using the common baseline to sharpen uncertainty bounds

For $\phi = (h, b)$, $\psi = (Y, b)$ and one active shared reference, the main paper proves the sharp intervals $S \in [-1/4, 1]$, $Q \in [-1/2, 2]$ and $U \in [-1/6, 1/3]$, including ties. All endpoint pairs are attained by three-point maps. For example, uncentered comparison values $(x, y, z) = (0, 1, 1/2)$ and $(1, 1, 0)$ attain the S endpoints. Difference values $(u, v, w) = (1, -1, 0)$ and $(1, 1, -1)$ attain the Q endpoints. For U , the triples of map values $(h, Y, b) = ((0, 1, 2), (1, 2, 0), (0, 2, 1))$ and $((2, 1, 0), (2, 1, 0), (0, 1, 2))$ attain $-1/6$ and $1/3$, respectively. The public exact checker enumerates all $13^3 = 2,197$ weak-order combinations and compares them with the generic reference kernels.

These widths use known relations among maps, not observed sample ranges. Persistence is the baseline map by construction, so its kernels and widths are exactly zero. This algebraic identity is declared separately from an observed equality of arrays. Generic coefficient-based widths remain valid when these map identities are unavailable.

The pure-range shared radii shrink by 37.5% at either fixed allocation; complete U's pure-range radius shrinks by 25%. Scaling both shared widths by $5/8$ leaves the width-optimal integer allocation (13, 086, 15, 852) unchanged. For Boosting 31 at this allocation, the hybrid radius falls from 0.01212612 to 0.00958948 (20.9%); the score center is unchanged. Complete U's variance bound additionally certifies Boosting 7, with lower bound 0.00001128 instead of -0.00007999 . Its total retention becomes four; triple hybrid retains three and shared rules retain none. Each rule's selected forecast remains unchanged, so this calculation supplies no new forecast-loss benefit. Every comparison uses the same existing draw arrays; it is a diagnosis after inspecting the archive, not independent confirmation.

Table 38: All fitted archive targets and reference interactions. Scores use units 10^{-3} ; the share divides the exact interaction by the exact expected shared score.

Forecast	Expected score	Target	Interaction	Share (%)
Persistence	0.0000	0.0000	0.0000	undefined
Ridge 1	15.4449	0.9481	14.4968	93.86
Ridge 100	15.0431	0.8873	14.1558	94.10
Boosting 7	7.7289	0.5717	7.1571	92.60
Boosting 31	9.9090	0.8521	9.0569	91.40
Blend Ridge 1	8.3145	0.4556	7.8590	94.52
Blend Boosting 7	3.5481	0.2814	3.2667	92.07
Blend Boosting 31	4.8818	0.3740	4.5078	92.34

The population share differs from the ratio of a sampled correction to a sampled score. For Boosting 31 it is

$$0.00905689905/0.00990903184 = 91.40044\%.$$

The original sample ratio is about 95.53% and is not used for that claim. Full-precision results, reconstruction checks and

the map declaration are in `results/shared_baseline_ranges/`. The earlier general-width comparison and loss endpoints remain in `results/trajectory_budget/`.

36.9 Why the unbiased correction can require three draws

The number of roles has an elementary explanation through the classical polynomial-degree argument for unbiased estimability. Count all observations that supply information about the unknown law. Fix maps independently of that law and allow auxiliary randomness independent of it. Over the unrestricted class, no integrable statistic using at most two IID draws almost surely can estimate the shared-reference interaction unbiasedly for every law. This is an order statement, not a confidence-width or decision-cost bound.

For a proof, take an issue-time covariate $X \in \{0, 1, 2\}$ with probabilities $(t, t, 1 - 2t)$, $0 < t < 1/2$, outcome $Y = X$, forecast $h = X$ and a constant baseline. The forecast uses the covariate, without reading the future outcome. Writing a for the midrank comparison, direct enumeration gives

$$\mathbb{E}S = t - \frac{3}{2}t^2, \quad \theta_{\mathcal{H}} = \frac{1}{2}(t - 2t^2 + t^3), \quad \Gamma = \frac{1}{2}(t - t^2 - t^3).$$

The expectation of any fixed two-draw statistic is $\sum_{i,j} p_i(t)p_j(t)g(i,j)$, a polynomial of degree at most two. Randomization independent of the law is absorbed into g . A rule that stops after one draw can be supplied an unused second draw. Neither case can produce the nonzero cubic coefficient of Γ . In contrast, the three-draw kernel $\{a(X_0, X_1) - a(X_0, X_2)\}^2/2$ has expectation Γ .

Restricted laws can require fewer draws. For binary X with probabilities $(t, 1 - t)$, the same maps give $\Gamma = t(1 - t)/4 = \mathbb{E}[\mathbf{1}\{X_1 \neq X_2\}/8]$. For the declared identity $h = b$, no observations are needed to establish zero interaction. Extra training information, known probabilities or an oracle can also change the information restriction. The argument does not rule out conservative tests based on fewer draws or address sample counts bounded only in expectation.

36.10 Prespecified checks of the final common-baseline rules

We check the final sharp-width implementations on six prescribed finite laws, with ties and near-degenerate cases. The protocol fixes four maps $(b, h, -h, 0)$, the family level 0.05, two budgets of 3,072 and 12,288 IID trajectory draws, and 300 replications per law and budget before simulation. Candidate levels are 0.05/4; the shared construction splits this level equally between its score and interaction bounds. The baseline candidate is declared to equal b and has zero kernels and widths. All other candidates use the same common-baseline widths for every comparator, without selecting tighter law-specific ranges after observing results.

The tied laws have uniform marginals on $\{-1, 0, 1\}$ and baseline zero. The joint law mixes independent coordinates with an equal-coordinate draw, with mixture weights 0, 1/10 and 1. Near-degenerate laws use marginal probabilities $(0.01, 0.98, 0.01)$ and weights 0 and 1. The informative baseline law sets b uniform on $\{-1, 0, 1\}$ and $h = Y = b^2$. Its positive target 7/54 does not represent information beyond b .

Each comparator receives the same full draw stream. Shared allocation minimizes the declared range radius; complete U can recombine every draw, using $\lfloor N/3 \rfloor$ for its concentration bound. Exact rational population midranks provide the targets independently of the simulation kernels. We evaluate both the range and preselected hybrid or variance-aware bounds, retaining all outcomes. No forecast fitting or prediction-loss experiment is performed.

Table 39: Certifications for h in 300 replications, using the hybrid or variance-aware rule.

Law	Target	Draws	Shared	Triple U	Complete U
Tied null	0	3,072	0	0	0
		12,288	0	0	0
Tied weak	1/135	3,072	0	0	0
		12,288	0	300	300
Tied strong	2/27	3,072	0	300	300
		12,288	300	300	300
Near-degenerate null	0	3,072	0	0	0
		12,288	0	0	0
Near-degenerate positive	0.0049005	3,072	0	0	0
		12,288	0	300	300
Informative baseline	7/54	3,072	4	300	300
		12,288	300	300	300

All 72 law/budget/procedure cells have 0/300 family noncoverage events and false certifications. The pointwise two-sided 95% Clopper–Pearson upper limit is 1.222%, not zero; these intervals are not simultaneous across cells. For 300/300 certifications, the corresponding lower limit is 98.778%. Both direct U constructions detect the weak and near-degenerate positive targets at the larger budget. Shared correction remains conservative in those cases.

All zero-target cells in this grid have zero expected reference interaction. The study therefore does not probe a zero target with positive shared-reference bias and is not an exhaustive biased-null stress test. It checks the implemented final rules on these finite IID laws, not a dependent physical-panel sampling model or independent application benefit. Eight independent formula and kernel tests, byte-identical deterministic replay, and a separate recount of all 86,400 candidate/procedure records accompany the protocol in `results/shared_baseline_calibration/`.

A separate null with positive reference interaction After inspecting the initial grid, we construct a separate law with B uniform on $\{0, 1, 2, 3\}$, $h = \mathbf{1}\{B = 3\}$ and $Y = \mathbf{1}\{B \geq 2\}$. Here $\theta_h = 0$ but $\mathbb{E}[S] = \Gamma = 1/32 > 0$, and h is not the baseline identity. The four candidate targets are $(0, 0, 1/32, 1/64)$. Thus this example probes shared-reference bias at a genuine zero fitted rank-contrast target. Its separate protocol is frozen before its own simulation, while retaining the original rules, levels, widths and budgets.

The additional 600 replications give 0/300 family noncoverage events and false certifications in each of 12 procedure cells. The nonidentity zero-target candidate is never certified. Shared hybrid certifies the positive reverse candidate only 1/300 times at the larger budget; both direct U hybrid or variance rules certify both positive candidates 300/300 times at both budgets. All outcomes remain available. An independent exact oracle and separate implementation reconstruct all 14,400 candidate records and 3,600 family records; deterministic replay preserves every result CSV byte. These results address one biased-null construction, not exhaustive calibration. The original 72 cells are unchanged. The additional protocol and outcomes remain separate in `results/shared_baseline_calibration/biased_null/`.

37 Final-target precision and a separate chronological task

The preceding comparisons separate a learning allowance from sampling uncertainty. This section evaluates confidence bounds for the final target itself. The synthetic designs and news protocol were fixed before their respective runs; none supplies a universal superiority claim. All code, method outputs and unsuccessful settings accompany the release.

37.1 Complete averaging, variance and classical betting

For one category with binary forecast and outcome values, the marginal-midrank target is $\vartheta = \text{Cov}(V, W)/4$. The complete four-observation target estimator reduces exactly to the unbiased sample covariance divided by four:

$$U_n = \frac{nN_{11} - (N_{10} + N_{11})(N_{01} + N_{11})}{4n(n-1)}.$$

Here N_{ab} counts observations equal to (a, b) . This is a classical second-order U-statistic; the reduction is not a new covariance estimator. Comparing its confidence bounds with betting on independent pairs or disjoint four-row kernels separates estimator averaging from the choice of inference rule.

The assessment crosses four laws, four raw-observation budgets and thirteen methods, with 500 paired repetitions per setting: 8,000 repetitions and 208 method cells. Every method receives the same binary support information and raw observations. In addition to the rare positive law below, the grid includes balanced independence, balanced anticorrelation and balanced perfect correlation. The first Hoeffding projection vanishes for balanced perfect correlation, while a higher-order projection remains nonzero.

Table 40: Rare perfect correlation: $V = W \sim \text{Bernoulli}(0.01)$, $\vartheta = 0.002475$. Rejections out of 500 at marginal level 0.05. This is a small absolute covariance with correlation one. All thirteen methods and paired intervals are retained in the code release.

Method	512	2,048	8,192	32,768
Complete U, range bound	0	0	0	478
Complete U, variance bound	0	0	0	500
Stratified hybrid bound	0	0	0	500
Stratified profiled betting	0	0	488	500
Classical four-row betting	4	486	500	500
Classical binary-pair betting	15	498	500	500

At 2,048 rows, classical four-row betting exceeds the complete-U joint bound by 0.972 in rejection rate, with pointwise paired 95% empirical Bernstein interval $[0.9092, 1]$. Binary-pair betting exceeds it by 0.996, interval $[0.9467, 1]$. The pair-versus-four-row difference 0.024 has interval $[-0.0373, 0.0853]$ and is unresolved. The four-row method uses the generic one-category symmetric kernel range $[-1/12, 1/12]$; the pair method additionally uses the binary target identity. These outcomes establish neither a general weak-correlation advantage nor superiority of a new sampling construction.

For $V = W \sim \text{Bernoulli}(p)$ the exact first-projection variance is $\sigma_\psi^2 = p(1-p)(1-2p)^2/16$. At $p = 0.01$ it is 0.0005942475 and $\sigma_\psi^2/\vartheta^2 = 97.0101$. Small projection variance alone does not make the implemented confidence radius small: at $n = 2,048$, the two deterministic remainder terms of the generic complete-U variance bound are 0.01453236 and 0.00420458, already larger than ϑ . The binary-specialized remainders are 0.00743181 and 0.00240249. The executable derivation retains every term; it does not substitute a leading-order standard error for the proved radius.

Different submodels have different sample requirements. For balanced binary margins with correlation $\rho = 16\vartheta$, the KL/Pinsker argument gives $N \geq (1 - \alpha - \eta)^2/(128\vartheta^2)$ for level α and power $1 - \eta$. This is a nondegenerate submodel lower bound. In the rare perfectly correlated law, compare with the all-zero null: on the event that no rare observation occurs, a level- α test rejects with probability at most α . Power is therefore at most $1 - (1 - \alpha)(1 - p)^N$, requiring $N \geq \log[\eta/(1 - \alpha)]/\log(1 - p)$. Neither statement is a universal variance-adaptive optimality result.

37.2 News popularity: protocol and observed decisions

The News Popularity dataset [38] supplies Facebook trajectories for economy articles. The target is the popularity count after 48 hours; features use the first hour and calendar information. Their historical availability is assumed from nominal timestamps. The delivered outcome is $\text{clip}(\log(1 + Y)/s, 0, 1)$, where s is the larger of one and the training 99th percentile of $\log(1 + Y)$. Primary loss is squared error on this delivered scale.

The four disjoint chronological stages have 2,706 training, 931 calibration, 1,324 selection and 1,110 confirmation articles, with 48-hour embargoes. Training runs from 10 November 2015 to 30 January 2016; calibration from 1 to 28 February; selection from 1 March to 29 April; confirmation from 1 May to 29 June, all as half-open nominal intervals. The entire 1,416-hour confirmation grid is retained, including 1,069 empty hours. Eligible labels occupy 347 hours; the last is 20 June. No later truncation changes this grid.

Six candidates comprise Ridge, histogram gradient boosting, ExtraTrees, persistence, a category-only copy and deterministic hash noise. Four training baseline quartiles define the categories. Nine gates share a 32,768-draw budget; BY includes all six candidates. The primary stratified rule, profiled betting, pooled-reference betting and ungated selection choose identical boosting augmentation predictions with weight 0.45. Their added gain over ungated selection is exactly zero; the lower gain bound is -5.33904×10^{-8} , reflecting the fixed betting-grid penalty.

Table 41: Confirmation losses for the identical selected boosting augmentation. Only the first column is the fixed primary loss. Differences on other scales are descriptive and are not new success criteria.

Prediction	Clipped scaled MSE	Unclipped log MSE	Share-count MSE
Ridge baseline	0.0201528374	0.912768786	29,677.3677
Selected augmentation	0.0201209783	0.913848929	33,481.1495

The observed primary improvement is 0.158087%; raw share-count MSE worsens by 12.8171%. Ridge and ExtraTrees augmentations are harmful on the primary confirmation loss. Persistence has zero augmentation weight and equals the baseline. The category copy has exact archive target zero; the realized hash noise has target 0.00226977 and is not a finite-archive null. All six candidates, thirteen selection rules and fifty-four gate outputs are reported, including zero, negative and non-retained results.

The execution order was recorded locally: protocol and code were fixed before source download; selection predictions were fixed before confirmation losses were evaluated. Ingestion nevertheless materialized all labels. This was not third-party blind custody, and reproducing these data is not a new independent confirmation. Gain guarantees concern the conditional expected average at settled hours, not issuance-time future risk. A later declared census check computes all six selection targets in about 0.0044 seconds and chooses the same augmentation. All selection outcomes had already been acquired; random query counts do not establish label savings. Exact array equality and stable paired-loss arithmetic preserve zero gains without rounding thresholds.

37.3 Additional retained comparisons

On the earlier thirty-two candidates, stratified hybrid and pooled-variance bounds retain fourteen, and a classical profiled rule retains sixteen; these are development diagnostics on exposed tasks. They do not change the original confirmation results. The earlier joint rank-mean refinement reduced allowances by median 1.24% without changing decisions. For Seoul rental demand, Ridge augmentation gives 22,431 raw-count MSE gain and equals ungated selection. Excluding a useful seasonal candidate incurs 299,790 opportunity loss. A weekly interval for gain over convex blending is positive; a fortnight interval includes zero. Every strategy, interval and stored prediction remains in the release.

References

- [1] “Seoul Bike Sharing Demand,” UCI Machine Learning Repository, 2020.
- [2] W. K. Newey and K. D. West, “A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix,” *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.
- [3] R. D. Shah and J. Peters, “The hardness of conditional independence testing and the generalised covariance measure,” *The Annals of Statistics*, vol. 48, no. 3, pp. 1514–1538, 2020.
- [4] H. D. Chiang, K. Kato, Y. Ma, and Y. Sasaki, “Multiway cluster robust double/debiased machine learning,” *Journal of Business & Economic Statistics*, vol. 40, no. 3, pp. 1046–1056, 2022.
- [5] W. Hoeffding, “A class of statistics with asymptotically normal distribution,” *The Annals of Mathematical Statistics*, vol. 19, no. 3, pp. 293–325, 1948.
- [6] D. Leung and M. Drton, “Testing independence in high dimensions with sums of rank correlations,” *The Annals of Statistics*, vol. 46, no. 1, pp. 280–307, 2018.
- [7] E. F. Fama and K. R. French, “A five-factor asset pricing model,” *Journal of Financial Economics*, vol. 116, no. 1, pp. 1–22, 2015.
- [8] M. M. Carhart, “On persistence in mutual fund performance,” *The Journal of Finance*, vol. 52, no. 1, pp. 57–82, 1997.
- [9] K. Zhang, J. Peters, D. Janzing, and B. Schölkopf, “Kernel-based conditional independence test and application in causal discovery,” in *Uncertainty in Artificial Intelligence*, 2011, pp. 804–813.
- [10] G. J. Székely and M. L. Rizzo, “Partial distance correlation with methods for dissimilarities,” *The Annals of Statistics*, vol. 42, no. 6, pp. 2382–2412, 2014.
- [11] A. Trindade, “ElectricityLoadDiagrams20112014,” UCI Machine Learning Repository, 2015, doi: 10.24432/C58C86.
- [12] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “M5 accuracy competition: Results, findings, and conclusions,” *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.
- [13] F. M. Harper and J. A. Konstan, “The MovieLens datasets: History and context,” *ACM Transactions on Interactive Intelligent Systems*, vol. 5, no. 4, pp. 19:1–19:19, 2015.
- [14] A. Y. Chen and T. Zimmermann, “Open source cross-sectional asset pricing,” *Critical Finance Review*, vol. 11, no. 2, pp. 207–264, 2022.
- [15] M. Arellano and O. Bover, “Another look at the instrumental variable estimation of error-components models,” *Journal of Econometrics*, vol. 68, no. 1, pp. 29–51, 1995.
- [16] R. M. Gray and D. L. Neuhoff, “Quantization,” *IEEE Transactions on Information Theory*, vol. 44, no. 6, pp. 2325–2383, 1998.
- [17] A. Maurer and M. Pontil, “Empirical Bernstein bounds and sample variance penalization,” in *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009.
- [18] W. Hoeffding, “Probability inequalities for sums of bounded random variables,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.

- [19] W. R. Schucany, H. L. Gray, and D. B. Owen, “On bias reduction in estimation,” *Journal of the American Statistical Association*, vol. 66, no. 335, pp. 524–533, 1971.
- [20] B. Phipson and G. K. Smyth, “Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn,” *Statistical Applications in Genetics and Molecular Biology*, vol. 9, no. 1, p. Article 39, 2010.
- [21] Ø. Langsrud, “Rotation tests,” *Statistics and Computing*, vol. 15, pp. 53–60, 2005.
- [22] P. O. Perry and A. B. Owen, “A rotation test to verify latent structure,” *Journal of Machine Learning Research*, vol. 11, no. 18, pp. 603–624, 2010.
- [23] P. Kline, R. Saggio, and M. Sølvesten, “Leave-out estimation of variance components,” *Econometrica*, vol. 88, no. 5, pp. 1859–1898, 2020.
- [24] N. Liu, C. Li, Y. Gu, and L. Liu, “Stabilized higher-order influence functions: Statistical theory of a class of bilinear forms,” arXiv:2607.04743v2, 2026.
- [25] L. Candanedo, “Appliances energy prediction,” UCI Machine Learning Repository, 2017.
- [26] J. Hogue, “Metro interstate traffic volume,” UCI Machine Learning Repository, 2019.
- [27] I. Waudby-Smith and A. Ramdas, “Estimating means of bounded random variables by betting,” *Journal of the Royal Statistical Society Series B*, vol. 86, no. 1, pp. 1–27, 2024.
- [28] E. Schechtman and S. Yitzhaki, “A measure of association based on Gini’s mean difference,” *Communications in Statistics: Theory and Methods*, vol. 16, no. 1, pp. 207–231, 1987.
- [29] G. P. McCormick, “Computability of global solutions to factorable nonconvex programs: Part i: Convex underestimating problems,” *Mathematical Programming*, vol. 10, pp. 147–175, 1976.
- [30] J. M. Robins, L. Li, R. Mukherjee, E. Tchetgen Tchetgen, and A. van der Vaart, “Minimax estimation of a functional on a structured high-dimensional model,” *The Annals of Statistics*, vol. 45, no. 5, pp. 1951–1987, 2017.
- [31] L. Liu, R. Mukherjee, and J. M. Robins, “Assumption-lean falsification tests of rate double-robustness of double-machine-learning estimators,” *Journal of Econometrics*, vol. 240, no. 2, p. 105500, 2024.
- [32] X. Wang, Y. Huo, L. Peng, and C. Zou, “Learning U-statistics with active inference,” arXiv:2605.11638, 2026. [Online]. Available: <https://arxiv.org/abs/2605.11638>
- [33] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, “Double/debiased machine learning for treatment and structural parameters,” *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.
- [34] L. Petersen and N. R. Hansen, “Testing conditional independence via quantile regression based partial copulas,” *Journal of Machine Learning Research*, vol. 22, no. 70, pp. 1–47, 2021.
- [35] Z. He, R. Pogodin, Y. Li, N. Deka, A. Gretton, and D. J. Sutherland, “On the hardness of conditional independence testing in practice,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [36] S. Chen, “Beijing PM2.5,” UCI Machine Learning Repository, 2015.
- [37] Y. J. Choe and A. Ramdas, “Comparing sequential forecasters,” *Operations Research*, vol. 72, no. 4, pp. 1368–1387, 2024.
- [38] L. Torgo and N. Moniz, “News popularity in multiple social media platforms,” UCI Machine Learning Repository, 2018.